Sharp embezzlement laws for quantum chains and finite Bell tests

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Abstract

For the finite open critical XX and Ising chains, we prove a sharp entanglement-embezzlement law from their exact discrete Schmidt spectra, including all degeneracies. For a maximally entangled target of fixed dimension d, the optimal squared local-unitary error is $\mathcal{E}_d = (\ln d)^2/(4S) + O_d(S^{-2})$, where S is the half-chain entanglement entropy. A counting criterion gives a Gaussian crossover for growing d and the same sharp law for the XXZ corner-transfer-matrix spectrum as $\Delta \downarrow 1$. We compare the optimal loss in a fixed Bell game with 243 question pairs to the Bell-pair embezzlement error with explicit constants, and prove the loss floor $(3-2\sqrt{2})/243$ for maximally entangled states of every dimension. With a shared nine-dimensional maximally entangled register, the open XX ground state at $N = 2^{19}$ beats this floor in the original game. At $N = 2^{84}$ the chain alone suffices. A tagged test gives an exact tilted CHSH separation for the four-spin XX ground state, with classical play in the original game branch. For the infinite critical XX ground state, the half-chain factors are type III_1 and the game value is one without attainment.

1 Introduction and main results

Entanglement embezzlement uses local unitaries to extract entanglement from a shared state while disturbing it as little as possible [12]. We determine the optimal squared error from the exact half-chain Schmidt spectra of the finite open XX and critical Ising chains. For a pure bipartite state ψ with decreasing Schmidt probabilities p_j, define the squared error for a Bell-pair target by

$$\mathcal{E}(\psi) = \inf_{U_A,U_B} \left\| (U_A \otimes U_B)(\psi \otimes |00\rangle) - \psi \otimes \Phi_2 \right\|^2 = 2 - \sqrt{2} \sum_{j\geq 1} \sqrt{p_j p_{\lceil j/2 \rceil}}. \tag{1}$$

Here $\Phi_d = d^{-1/2} \sum_{j=1}^{d} |jj\rangle$ is a maximally entangled pair. We use $\mathcal{E}_d$ for the analogous error with target Φ_d, and $\mathcal{E} = \mathcal{E}_2$. Entropies are in nats, and log and ln denote natural logarithms.

For the finite chains we use $H_N^{\mathrm{XX}} = \frac{1}{2} \sum_{j=1}^{N-1} (X_j X_{j+1} + Y_j Y_{j+1})$ and $H_L^{\mathrm{I}} = -\sum_{j=1}^{L-1} X_j X_{j+1} - \sum_{j=1}^{L} Z_j$, with open boundaries.

Theorem 1 (Finite open XX chain)**.** *Let ψ_N be the ground state of the half-filled open XX chain on an even number N of sites, cut in the middle. Write S_N for its entanglement entropy and V_N for its information variance, the variance of $-\log p_j$ when j has the Schmidt probability law (p_j). For every fixed integer $d \geq 2$,*

$$S_N = \frac{\log N}{6} + O(1), \qquad V_N = \frac{\log N}{6} + O(1), \tag{2}$$

$$\mathcal{E}_d(\psi_N) = \frac{(\log d)^2}{4V_N} + O_d((\log N)^{-2}) = \frac{3(\log d)^2}{2\log N} + O_d((\log N)^{-2}), \tag{3}$$

$$S_N \mathcal{E}_d(\psi_N) = \frac{(\log d)^2}{4} + O_d((\log N)^{-1}). \tag{4}$$

Corollary 2 (Finite open critical Ising chain)**.** *For the ground state of the open critical transverse-field Ising chain on an even number L of spins, cut in the middle,*

$$S_L = \frac{\log L}{12} + O(1), \qquad V_L = \frac{\log L}{12} + O(1), \qquad \mathcal{E}_d = \frac{3(\log d)^2}{\log L} + O_d((\log L)^{-2}).$$

Consequently $S_L \mathcal{E}_d = (\log d)^2/4 + O_d((\log L)^{-1})$.

The proof and the common spectral criterion appear in Section 3. Under that criterion, as the information variance V tends to infinity, $\mathcal{E}_d = 2(1 - e^{-(\log d)^2/(8V)}) + o(1)$ uniformly over integers $d \geq 2$. The error vanishes precisely when $\log d = o(\sqrt{V})$. The corresponding root fidelity $1 - \mathcal{E}_d/2$ has the same limiting curve

as the optimal LOCC root fidelity obtained by Zhang, Liu and Chen under a Gaussian logarithmic-rank hypothesis, with fixed-charge random states as an application [68, Eq. (6) and Supplement, Theorem 1]. Here it holds for local unitaries on the chain spectra, uniformly over the target dimension.

A Bell game gives an operational test of the Bell-pair error $\mathcal{E}$. A referee sends questions to two separated players, receives their answers, and applies a fixed winning rule [15–17]. We use the game G of Ref. [21], built from nine linked flagged magic squares with 243 equally likely question pairs. Write $\omega(\psi)$ for the supremal winning probability when the players share ψ.

Theorem 3 (The game measures the embezzlement error)**.** *Every pure bipartite state satisfies the following bounds with the numerical constant $C = 2{,}706{,}291/2$:*

$$\frac{\mathcal{E}(\psi)}{C} \le 1 - \omega(\psi) \le \min\left\{\frac{1}{27}, \mathcal{B}(p)\right\}, \qquad \mathcal{B}(p) \le \frac{7}{3}\mathcal{E}(\psi).$$

The loss $\mathcal{B}(p)$ is achieved by the explicit measurements in Section 4.1, without an additional shared state. If a shared Φ_9 is included, their loss is exactly $2\mathcal{E}(\psi)/243$. The classical value of G is $26/27$.

The lower comparison follows from the robust principle of Ref. [21]. Appendix B gives a self-contained proof with the displayed numerical constant. The upper comparison, its exact state-dependent loss, and the classical value are proved from the complete rule table. We also obtain the following numerical bound independent of C.

$$1 - \omega(\Phi_d) \ge \gamma = \frac{3 - 2\sqrt{2}}{243} = 7.060612\ldots \times 10^{-4} \quad \text{for every } d. \tag{5}$$

An explicit state on nine qubits per party has a strategy with loss below 6.90×10^{-4} (Section 4.2).

With an additional shared Φ_9, the open XX ground state at $N = 2^{19} = 524{,}288$ has loss below 0.000703 in the original game, and at $N = 2^{20}$ its loss is below 0.000663. Both bounds are certified, including the physical-spectrum and arithmetic errors (Section 5.1). The same auxiliary register preserves the comparison class, since $\Phi_D \otimes \Phi_9$ is a maximally entangled state of rank $9D$.

The open XX ground state at $N = 2^{84}$ admits a chain-only strategy in the original game with loss below 0.000461, beating the floor γ. The certificate uses a covariance calculation of dimension 1418 and exact integer rank counts (Section 5.2).

The four-spin open XX ground state gives an exact separation in a tagged extension of G with a tilted CHSH branch of five question pairs. The strategy plays the original game branch classically and uses only the four-spin ground state and explicitly specified local measurements. Its total loss is below 0.177, and every maximally entangled state has loss at least 0.180 (Section 5.3).

When a quantum system is divided into two infinite halves, the observables of one half generate a von Neumann algebra M_A. Gapped one-dimensional ground states with bounded entanglement entropy have type I half-chains [3, 4]. The type-III_1 property of the critical XX chain goes back to Matsui [5], as recorded in Ref. [72, Proposition 6.9]; the general critical-free-fermion result is due to van Luijk, Stottmeister and Wilming [11]. Type III_1 is precisely the factor type for which every normal state can embezzle any finite entangled state with arbitrarily small disturbance to the original state [13].

Theorem 4 (Critical chains approach perfect play)**.** *The infinite XX ground state, divided into its two physical half-chains, has $\omega(\Omega) = 1$. The same value holds at every unit vector in its ground-state sector and for the critical free-fermion chains covered by Ref. [11], with Alice and Bob assigned the complementary physical half-chains. For the ground state ψ_N of the even-length open XX chain, cut in the middle, $\mathcal{E}(\psi_N) \to 0$ and $\omega(\psi_N) \to 1$ along the full sequence.*

Theorem 5 (Nonattainment at the infinite ground state)**.** *Every strategy at the infinite XX ground state loses with positive probability. The same holds for the vacuum of a free massive scalar field when the players measure in complementary wedges.*

The reason for nonattainment is spectral. Perfect play would create a modular eigenvector of eigenvalue two, whereas the XX ground state and the free-field vacuum have no modular eigenvectors outside the vacuum line. This applies the modular obstruction to exact embezzlement of van Luijk, Stottmeister, Werner and Wilming [13, Proposition 35 and Corollary 36], through the game's bridge, to two concrete states of matter.

For the XXZ corner-transfer-matrix spectrum, the same sharp product law holds as $\Delta \downarrow 1$, with a remainder of order $\arccosh \Delta$. The spectral formula for a pure Néel sector is an input to this result, as specified in Section 8. At fixed critical coupling, explicit finite-depth circuit Hamiltonians retain the type-III_1 half-chain structure, and one preserves the entire finite Schmidt spectrum at the chosen cut (Section 8).

Relation to earlier work. The critical-free-fermion result is due to van Luijk, Stottmeister and Wilming [11]; the general embezzlement classification is due to van Luijk, Stottmeister, Werner and Wilming [13]. Qualitative finite-size embezzling families were also established in Ref. [11, Section 6.3]. Kera, van Luijk, Stottmeister and Wilming bound Gaussian conversion through the density of the covariance spectrum: if the embezzling covariance has ε-dense spectrum, the trace-distance error of converting the embezzler together with one Gaussian state into the embezzler together with another, minimized over Gaussian unitaries, is at most $11\varepsilon^{1/4}$ times the number of modes of these states [24, Theorem 1]. In a smoothed Calabrese–Lefevre model, Sierant derives inverse-square-root logarithmic decay of the rank-window statistic that controls universal LOCC embezzlement [25]. These quantities differ from the optimal squared local-unitary error studied here, so their rates and ours are not directly comparable; our law determines its leading coefficient and remainder for the exact open-chain spectra. The Schmidt-overlap formula in (1) is standard [12, 22]; LOCC embezzlement and its rank-window criterion were characterized by Zanoni, Theurer and Gour [23]. The game and its robust lower estimate come from Ref. [21], using Mančinska's flagged magic square [28]. The modular obstruction of Ref. [13] and later exact-embezzlement self-tests [26, 27] precede the present applications.

Junge and Palazuelos constructed Bell inequalities whose violation on maximally entangled states is bounded uniformly over all finite dimensions, with growing violations on suitable nonmaximally entangled states [18, Theorem 1.4]. Vidick and Wehner gave an explicit separation for the I_{3322} inequality [19]. Liang, Vértesi and Brunner also give a Bell inequality whose maximal quantum violation cannot be attained by any finite-dimensional maximally entangled state [20]. For the flagged magic square underlying G, Mančinska proves a dimension-independent gap for maximally entangled states and gives a rank-five state that wins perfectly [28, Corollary 4 and Proposition 1]. The existence of a positive floor for G follows already from that result; the deficit bound in Ref. [21, Section 6.6] gives another route. Here we obtain the explicit constant $(3-2\sqrt{2})/243$, relate the optimal loss to Bell-pair embezzlement, and certify strategies on finite XX ground states. The four-spin result realizes the known tilted CHSH separation [74, 75] on a chain ground state.

The tower law builds on the Calabrese–Lefevre distribution and conformal level counting [48, 49, 52]. The physical finite-chain proof instead starts from the exact covariance and derives microscopic Fourier counting estimates. Both control the ordered, discrete spectrum at the scale needed for the optimal local-unitary error. Related logarithmic-rank mechanisms occur in nonlocal magic and diffusive embezzlement [25, 68, 69], with different observables or allowed operations. Entropy scaling alone is already known not to determine half-chain type: Liu and Yi construct translation-invariant pure states of types II_∞, III_λ and III_1 with logarithmic-order interval entropy [63]. Our singlet family supplies the exact law $S_N = (c/6)\ln N + O_c(1)$ for any $c > 0$, together with a direct game-loss bound and an exponentially decaying inhomogeneous parent interaction.

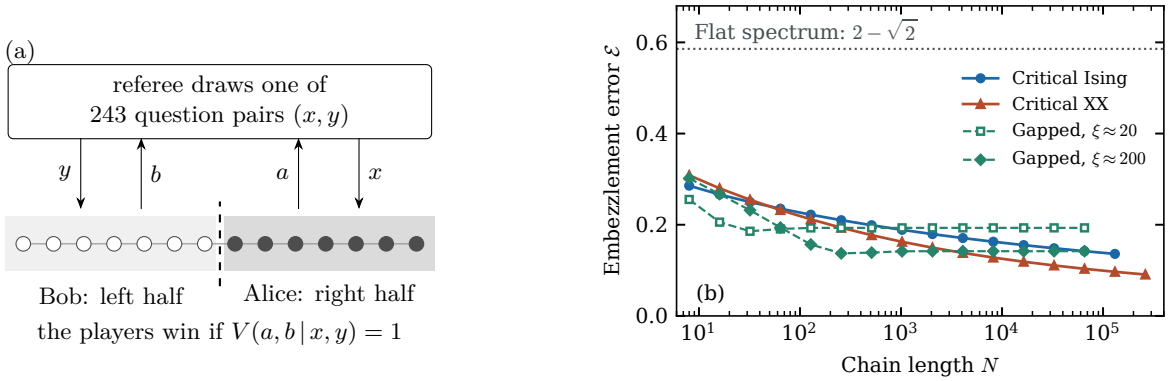


Figure 1: (a) The game. A chain in its ground state is cut in the middle, and each player measures one half without communicating with the other. (b) Embezzlement error $\mathcal{E}$ across the cut. By Theorem 3 the smallest losing probability lies between $\mathcal{E}/C$ and $(7/3)\,\mathcal{E}$. For critical chains $\mathcal{E}$ keeps decreasing as the chain grows, for gapped chains it levels off after a few correlation lengths ξ, and for product states and Bell pairs it stays at $2-\sqrt{2}$.

2 Setting

2.1 Nonlocal games, bipartite systems and types

A nonlocal game consists of finite sets of questions x for Alice and y for Bob, finite sets of answers a and b, a probability distribution $\pi(x,y)$ and a rule $V(a,b\,|\,x,y)\in\{0,1\}$. The referee draws (x,y) from π; the players, who cannot communicate, answer a and b and win if $V(a,b\,|\,x,y)=1$ [16, 17].

Finite systems. Let $\psi\in\mathcal{H}_A\otimes\mathcal{H}_B$ be a unit vector. A strategy assigns to each question a positive operator-valued measure (POVM), $\{A_a^x\}_a$ on $\mathcal{H}_A$ and $\{B_b^y\}_b$ on $\mathcal{H}_B$. Its losing probability is $\langle\psi,L\psi\rangle$ with the loss operator

$$L=\sum_{x,y}\pi(x,y)\sum_{a,b\,:\,V(a,b|x,y)=0}A_a^x\otimes B_b^y,\qquad 0\le L\le 1,\tag{6}$$

and the value of the game at ψ is $\omega(\psi)=1-\inf\langle\psi,L\psi\rangle$, the infimum being over all strategies. Local ancillas in fixed pure states are included, since a measurement on system plus ancilla is a POVM on the system.

Infinite systems. The Hilbert space of an infinite system need not factorize. A bipartite system is then a pair of von Neumann algebras M_A, M_B on a separable Hilbert space $\mathcal{H}$ that commute, $M_B\subseteq M_A'$, together with a unit vector $\Omega\in\mathcal{H}$. Haag duality is the stronger equality $M_B=M_A'$. Alice's POVMs are taken in M_A and Bob's in M_B, the product $A_a^x\otimes B_b^y$ in (6) is replaced by $A_a^xB_b^y$, and $\omega(\Omega)$ is defined as before. For $M_A=\mathcal{B}(\mathcal{H}_A)\otimes 1$ and $M_B=M_A'$ this is the tensor-product definition. For a pure state of a spin chain with quasi-local algebra $\mathfrak{A}=\mathfrak{A}_L\otimes\mathfrak{A}_R$ (left and right half-chains) and GNS representation $(\pi,\mathcal{H},\Omega)$ we put $M_A=\pi(\mathfrak{A}_R)''$ and $M_B=\pi(\mathfrak{A}_L)''$.

Types. A von Neumann algebra with trivial centre is a factor. A factor has type I if it is isomorphic to all bounded operators on a Hilbert space, type II if it is not of type I but has a faithful normal semifinite trace, and type III otherwise. For a faithful normal state $\varphi=\langle\Omega,\,\cdot\,\Omega\rangle$ on M, use the cyclic GNS space $\overline{M\Omega}$, where Ω is also separating. Tomita–Takesaki theory provides the modular operator Δ_φ on this space and the modular group $\sigma_t^\varphi(x)=\Delta_\varphi^{it}x\Delta_\varphi^{-it}$; the fixed points of σ^φ form the centralizer $M_\varphi=\{x:\varphi(xy)=\varphi(yx)\ \forall y\}$. For $M=\mathcal{B}(\mathcal{H}_A)$ with a faithful density matrix of eigenvalues $p_j>0$, the modular point spectrum consists of the ratios p_i/p_j; the full spectrum is their closure. Connes's invariant $S(M)$ is the intersection of modular spectra over faithful normal semifinite weights. It is $\{1\}$ for semifinite factors, $\{0,1\}$ for III_0, $\{0\}\cup\lambda^{\mathbb{Z}}$ for III_λ, $0<\lambda<1$, and $[0,\infty)$ for III_1 [1, 2].

2.2 The game

The game G is the game called G_{III} in Ref. [21], and Appendix A reproduces its definition. It is built from nine copies of the flagged magic-square game of Mančinska [28], which extends the Mermin–Peres magic-square game [29, 30] by one extra answer, called a flag, and two extra questions per player. One further question to Bob, whose answer is a label in $\mathbb{Z}/9\mathbb{Z}$, links the nine copies. Alice has 45 possible questions and Bob 46, each with at most nine answers, and the 243 question pairs that occur are asked with equal probability.

If either local algebra admits a faithful normal semifinite trace, then every strategy loses with positive probability [21, Theorem 1.1]. For a factor M_A with separable predual in standard form, the same theorem characterizes the existence of a normal state with a perfect strategy on (M_A,M_A') by an invariant normal state on the flow of weights. Every type-III_1 factor meets that criterion. The robust lower comparison in Theorem 3 uses Ref. [21, Theorem 1.4], with an explicit constant proved in Appendix B. For CHSH on a standard pair, van Luijk, Stottmeister, Werner and Wilming obtained $\beta\ge 2\sqrt{2}-8\sqrt{\kappa(\varphi)}$, where β is the CHSH correlation coefficient and κ is their worst-case marginal error in predual norm [13, Theorem J and Proposition 100].

Lemma 6 (Linked modular bridge)**.** *Let M_A,M_B be commuting von Neumann algebras on $\mathcal{H}$ and Ω a unit vector whose marginal φ on M_A is faithful. Suppose that POVMs in these algebras win G with probability one at Ω. For each $j\in\mathbb{Z}/9\mathbb{Z}$, let F_j be Alice's flag effect on R_1^j. The F_j are orthogonal projections in the centralizer of φ, with $\sum_j F_j=1$ and $\varphi(F_j)=1/9$. There are partial isometries $w_j\in M_A$ such that, for all real t,*

$$w_j^*w_j=F_j,\qquad w_jw_j^*=F_{j+1}+F_{j+2},\qquad \sigma_t^\varphi(w_j)=2^{it}w_j.\tag{7}$$

Proof. Let e project onto $\overline{M_A\Omega}$. This space reduces M_A, and the representation there is faithful and normal. Compress Bob's effects to it by $B\mapsto eBe|_{e\mathcal{H}}$. They still form POVMs in the commutant, and every

expectation $\langle \Omega, AB\Omega \rangle$ is preserved because $e\Omega = \Omega$ and $[e, A] = 0$. We work in this cyclic and separating representation of the faithful normal state φ [21, Lemma 1.6].

A perfect binary match of commuting effects P, Q has zero expectation of $(P-Q)^2 + P(1-P) + Q(1-Q)$. Positivity gives $P\Omega = Q\Omega$, and faithfulness makes Alice's effect P a projection. Replacing $P\Omega$ by $Q\Omega$ and commuting Q past $X \in M_A$ gives $\varphi(PX) = \varphi(XP)$. The first label test therefore gives centralizer projections F_j with $F_j\Omega = B_j^{\mathrm{Lab}}\Omega$. The label effects sum to one, so separation gives $\sum_j F_j = 1$ and the F_j are orthogonal. The second label test matches Alice's R_2^j outcome $+++$ to $B_{j+1}^{\mathrm{Lab}} + B_{j+2}^{\mathrm{Lab}}$. On Ω this sum equals $F_{j+1} + F_{j+2}$, so separation identifies Alice's outcome with that projection [21, Section 2.4].

Each copy is a perfect flagged magic-square strategy on this faithful normal support. The checked square generates a centralizer copy of M_4 with unit $1 - F_j$, and the four nonflag outcomes of R_2^j are rank-one projections in it [21, Lemma 2.1]. The calibrated sign tests give a partial isometry β_j from F_j onto a rank-one projection q_j of that copy, with $\sigma_t^{\varphi}(\beta_j) = 2^{it}\beta_j$ [21, Lemma 2.2]. Choose a partial isometry v_j in that matrix algebra with initial projection q_j and final projection $F_{j+1} + F_{j+2}$. Then $w_j = v_j\beta_j$ has the required initial and final projections, and $\sigma_t^{\varphi}(w_j) = v_j\sigma_t^{\varphi}(\beta_j)$ gives the last identity in (7). The KMS identity gives $2\varphi(F_j) = \varphi(F_{j+1}) + \varphi(F_{j+2})$. A largest weight forces both successors to be largest, so all nine weights are $1/9$. □

For a nonfaithful marginal with support $p \in M_A$, the bridge applies in the corner pM_Ap on $p\mathcal{H}$. The equality $p\Omega = \Omega$ and commutation of p with Bob's effects preserve the correlations in this reduction. On the standard pair of a faithful marginal, the maps $s_r = \sum_j w_j^* F_{j+r}$ for $r = 1, 2$ satisfy $s_r^* s_s = \delta_{rs}1$, $\sum_r s_r s_r^* = 1$ and $\sigma_t^{\varphi}(s_r) = 2^{-it}s_r$. The local column isometries with entries s_r^* and Js_r^*J, where J is the modular conjugation, extract a Bell pair exactly. Indeed, Tomita's identity gives $s_r^*(Js_s^*J)\Omega = 2^{-1/2}s_r^*s_s\Omega = 2^{-1/2}\delta_{rs}\Omega$.

2.3 The embezzlement error

For a unit vector $\psi \in \mathcal{H}_A \otimes \mathcal{H}_B$ the embezzlement error $\mathcal{E}(\psi)$ is defined by the infimum in (1), taken over unitaries U_A on $\mathcal{H}_A \otimes \mathbb{C}^2$ and U_B on $\mathcal{H}_B \otimes \mathbb{C}^2$; $\Phi_2 = (|00\rangle + |11\rangle)/\sqrt{2}$ is a Bell pair on the two added qubits. This is embezzlement in the sense of van Dam and Hayden with ψ as the catalyst, measured by the squared distance; the maximal overlap, or root fidelity, is $1 - \mathcal{E}/2$. Let $\lambda = (\lambda_j)_{j\geq 1}$, $\lambda_j = \sqrt{p_j}$, be the decreasing sequence of Schmidt coefficients of ψ, and define an isometry D on $\ell^2(\mathbb{N})$ by

$$(D\lambda)_{2j-1} = (D\lambda)_{2j} = \lambda_j/\sqrt{2}. \tag{8}$$

Lemma 7 (Formula for the embezzlement error)**.** $\mathcal{E}(\psi) = \|\lambda - D\lambda\|^2 = 2 - \sqrt{2}\sum_{m\geq 1}\sqrt{p_m\, p_{\lceil m/2 \rceil}}$.

Proof. The Schmidt coefficients of $\psi \otimes |00\rangle$ are λ, and those of $\psi \otimes \Phi_2$ are $D\lambda$, which is again decreasing. For two unit vectors with decreasing Schmidt coefficients α and β, the supremum of the overlap $|\langle \chi, (U_A \otimes U_B)\chi' \rangle|$ over local unitaries is $\sum_j \alpha_j\beta_j$. This is von Neumann's trace inequality [31]; the bound is attained by aligning the Schmidt bases in finite dimensions and approached in infinite dimensions. Hence the infimum in (1) equals $2 - 2\langle \lambda, D\lambda \rangle = \|\lambda - D\lambda\|^2$, because D is an isometry. □

For a target of dimension d the same argument gives

$$\mathcal{E}_d(p) = \|(1 - D_d)\lambda\|_2^2, \qquad (D_d\lambda)_j = d^{-1/2}\lambda_{\lceil j/d \rceil}. \tag{9}$$

The distinction between local-unitary and LOCC error matters for rates; we use (9) throughout.

3 The sharp chain law

Theorems 1 and 2 concern the exact finite spectra. Their common mechanism is a smooth limit of logarithmic Schmidt rank, with quantitative control of every spectral plateau. We first prove a criterion that isolates this mechanism.

Energy counts and logarithmic rank. Consider a family of spectra written as $p_\nu = e^{-L(1)-A_\nu}$, where $A_\nu \geq 0$ are energies listed with multiplicity and $L(z) = \log \sum_\nu e^{-zA_\nu}$ is the logarithm of the partition function. Under the tilted law $p_{\beta,\nu} = e^{-L(\beta)-\beta A_\nu}$, the mean and variance of A are $m_\beta = -L'(\beta)$ and $v_\beta = L''(\beta)$. In particular $V = v_1$ is the information variance of the original spectrum. Let $W \to \infty$ index the family.

Proposition 8 (A counting criterion for the embezzlement law)**.** *Suppose L is analytic on a fixed complex neighborhood of a fixed real interval $[1-\eta, 1+\eta] \subset (0,\infty)$, with $|L^{(k)}(z)| \leq C_k W$ for each fixed $k \geq 1$, and $v_\beta \geq cW$ on the real interval. Suppose the tilted characteristic functions $\varphi_\beta(t) = \mathbb{E}_\beta e^{itA}$ obey, uniformly in that interval,*

$$|\varphi_\beta(t)| \leq \begin{cases} e^{-cWt^2}, & |t| \leq \delta, \\ e^{-cW}, & \delta \leq |t| \leq \tau W, \end{cases} \tag{10}$$

for positive constants c, δ, τ independent of W. Then, for all integers $d \geq 2$, with $h = \log d$,

$$\left| \mathcal{E}_d - \frac{h^2}{4V} \right| \leq \frac{C(1+h^2+h^4)}{W^2}. \tag{11}$$

Here C and the large-W threshold are independent of d.

Proof. Write $R(E) = \#\{\nu : A_\nu \leq E\}$ and $R(E^-) = \#\{\nu : A_\nu < E\}$. Set $\mu = m_1$ and $B = \sqrt{W}\log W$. For $|E-\mu| \leq 2B$, let $\beta(E)$ solve $m_\beta = E$. Since $m'_\beta = -v_\beta$ has magnitude of order W, this solution is unique and lies in the given interval for large W. Define

$$F(E) = L(\beta(E)) + \beta(E)E - \log \beta(E) - \tfrac{1}{2}\log(2\pi v_{\beta(E)}). \tag{12}$$

Lemma 44 in Appendix F gives the uniform estimate for $|E-\mu| \leq 2B$,

$$\log R(E) = F(E) + O(W^{-1}), \qquad \log R(E^-) = F(E) + O(W^{-1}). \tag{13}$$

Indeed, with $\beta = \beta(E)$ the exact tilted count is

$$e^{-L(\beta)-\beta E} R(E) = \mathbb{E}_\beta\big[e^{-\beta(E-A)}\mathbf{1}_{\{A \leq E\}}\big].$$

The lemma uses both frequency bounds in (10) and an even smoothing kernel with Fourier support in $[-\tau W, \tau W]$. The tilted interval-probability estimate (67) bounds the change under smoothing and the mass of any atom at E by $O(W^{-3/2})$. The analytic derivative bounds and $v_\beta \asymp W$ give the saddle-point value $(\beta\sqrt{2\pi v_\beta})^{-1}$ with relative error $O(W^{-1})$ in the Fourier integral (68).

Put $p(x) = p_{\lceil x \rceil}$ for the decreasing list of probabilities, with zero padding for a finite list, and $g(y) = e^{y/2}\sqrt{p(e^y)}$. If $E_x = -\log p(x) - L(1)$ is finite, then $R(E_x^-) < x \leq R(E_x)$. The two counts in (13) give $\log x = F(E_x) + O(W^{-1})$ throughout each degenerate plateau in the counting window. Since $F'(E) = \beta(E) + O(W^{-1})$ is bounded away from zero there, its inverse $E_*(y) = F^{-1}(y)$ satisfies the energy comparison (72) on $F([\mu - 3B/2, \mu + 3B/2])$. Let χ be a smooth function equal to one on $[-1,1]$ and supported inside $(-3/2, 3/2)$, and define

$$G(y) = \chi\big((E_*(y) - \mu)/B\big)\exp\big((y - L(1) - E_*(y))/2\big)$$

on the range of F, extended by zero. Lemma 45 proves

$$\|g - G\|_2 = O(W^{-1}), \qquad \|G''\|_2 = O(W^{-1}), \qquad \|G'\|_2^2 = \frac{1}{4V} + O(W^{-2}). \tag{14}$$

The first estimate uses concentration under the original law to control the full tail. The derivative estimates follow from the logarithmic derivative $a(E) = 1 - 1/F'(E)$ of the squared uncut profile. Its Taylor coefficients about μ through degree two have orders $O(W^{-1})$, $-1/V + O(W^{-2})$, and $O(W^{-2})$, respectively. The first and third centered moments of the exact energy are 0 and $O(W)$, which controls the cross terms in the integral of a^2.

For $T_h g(y) = g(y-h)$ set $D_h = I - T_h$. The continuous rank coordinate preserves the exact error $\mathcal{E}_d = \|D_h g\|_2^2$ because $p(x/d) = p_{\lceil j/d \rceil}$ almost everywhere on $(j-1, j]$. Writing $r = g - G$, we have

$$\|D_h g\|_2^2 - \|D_h G\|_2^2 = 2\,\mathrm{Re}\langle D_h^* D_h G, r\rangle + \|D_h r\|_2^2.$$

The multiplier of $D_h^* D_h = 2I - T_h - T_{-h}$ is $2 - 2\cos(ht)$, bounded by $h^2 t^2$. Thus the last display is $O((1+h^2)W^{-2})$ by (14). The same multiplier differs from $h^2 t^2$ by $O(h^4 t^4)$, so $\|D_h G\|_2^2 = h^2\|G'\|_2^2 + O(h^4\|G''\|_2^2)$. This proves (11). □

The exact covariance supplies the counting estimates. For the open XX chain let C_N be the occupied one-particle projector and write C_N in blocks for the two halves. If s_j are the singular values of its cross block, then $s_j^2 = \lambda_j(1-\lambda_j)$ as a multiset, where λ_j are the restricted covariance eigenvalues. The Schmidt probabilities form a product of two-outcome distributions with one-mode energies $e_j = 2\,\mathrm{arcosh}(1/(2s_j))$. With zero padding, Corollary 40 compares the ordered (s_j) in ℓ^1 with error $O(1)$ to the mesh

$$\left\{\tfrac{1}{2}\,\mathrm{sech}(2\pi^2 k/L_i) : k \in \mathbb{Z},\ i = 1,2\right\}, \qquad L_i = \log N + O(1). \tag{15}$$

The two branches come from the two parity blocks of the exact sine covariance. Each is compared with a finite Hilbert matrix, then with a periodized Carleman kernel diagonal in Fourier modes.

This mesh gives $\sum_j s_j = O(\log N)$, which bounds the fixed-order derivatives of $L(z) = \sum_j \log(1 + e^{-ze_j})$ by $O(\log N)$ near $\beta = 1$. In any fixed positive energy band $[a, b]$, a mesh branch has spacing $4\pi^2/L_i$. All but $O(1)$ entries in the band match true energies in $[a/2, 2b]$, with total absolute discrepancy $O(1)$. These modes give $v_\beta \geq c \log N$. The Bernoulli characteristic factor has squared modulus $1 - 4q(1-q)\sin^2(te_j/2)$, where $q = (1 + e^{\beta e_j})^{-1}$. For small t the retained modes give a sine-square sum of order Wt^2 with $W = \log N$. For $|t| \geq \delta$ the mesh sum is bounded below by $cW - O(1 + |t|)$ by integral comparison. Replacing the mesh energies by the matched true energies costs another $O(1 + |t|)$. Choosing τ sufficiently small yields (10) through $|t| \leq \tau W$.

The entropy and information variance are sums over modes of

$$f_S(e) = \log(1 + e^{-e}) + \frac{e}{1 + e^e}, \qquad f_V(e) = \frac{e^2 e^e}{(1 + e^e)^2}.$$

As functions of $s = (2\cosh(e/2))^{-1}$ both are Lipschitz. The mesh comparison therefore changes their sums by $O(1)$. The two mesh branches give density $(L_1 + L_2)/(2\pi^2)$ in positive energy and $\int_0^\infty f_S(e)\,de = \int_0^\infty f_V(e)\,de = \pi^2/6$. This gives (2) and recovers the open-boundary entropy coefficient of Calabrese and Cardy and of Fagotti and Calabrese for XX [6, 8]. The matching leading term for V, the capacity of entanglement, agrees with the conformal Rényi calculation [9, 67]. Proposition 8 now gives (3) and (4).

For Ising, the exact Majorana decomposition identifies the mode list of the XX chain on $2L$ sites with two copies of the Ising list on L sites (Corollary 42). The Ising logarithmic partition function is therefore half that of XX, and its tilted characteristic function has modulus equal to the square root of the XX modulus. This proves Corollary 2 with $W = \log L$.

Corollary 9 (Loss of the explicit strategy)**.** *For the finite open XX and critical Ising states, the explicit measurements of Theorem 14 achieve $\mathcal{B}(p) = (2/243)\mathcal{E}_2(1 + o(1))$. The leading losses are $(\log 2)^2/(81 \log N)$ for XX and $2(\log 2)^2/(81 \log L)$ for Ising.*

Proof. The same mesh gives $b_N = -\log p_1 \sim (\log N)/12$ for XX and $b_L \sim (\log L)/24$ for Ising (Appendix F). Thus $p_1 = o(\mathcal{E}_2)$, and the achieved loss follows from the state-dependent estimate in Theorem 14. □

Corollary 43 extends the chain law uniformly to cuts between any two adjacent sites in either open chain as the shorter side grows. The logarithm of the shorter side sets the scale. The corollary also covers Ising chains of odd length.

Corollary 10 (Targets of growing dimension)**.** *Under the assumptions of Proposition 8, uniformly over integers $d \geq 2$ as $W \to \infty$,*

$$\mathcal{E}_d = 2\left(1 - \exp\left[-\frac{(\log d)^2}{8V}\right]\right) + o(1). \tag{16}$$

In particular the error tends to zero precisely when $\log d = o(\sqrt{V})$. In this regime the error is asymptotic to $(\log d)^2/(4V)$ by (11).

The optimal local-unitary root fidelity $1 - \mathcal{E}_d/2$ has the limiting Gaussian form proved for LOCC by Zhang, Liu and Chen [68, Supplement, Theorem 1]. The counting criterion establishes this form for the exact finite XX and Ising spectra.

Proof. Set $y_0 = F(\mu) = L(1) + \mu - \frac{1}{2}\log(2\pi V)$ and define the probability density $\rho_W(z) = \sqrt{V}\, g(y_0 + \sqrt{V}\, z)^2$. For $y = y_0 + \sqrt{V}\, z$ at fixed z, let $u = F^{-1}(y) - \mu$. Since F' is bounded away from zero, $u = O_z(\sqrt{W})$. The derivative bounds from Appendix F give

$$F(\mu + u) - F(\mu) = u - \frac{u^2}{2V} + O\left(\frac{|u|}{W} + \frac{|u|^3}{W^2}\right).$$

It follows that $u = \sqrt{V}\, z + O_z(1)$. By the inverse count estimate (72), the exact energy at rank e^y is $\mu + u + O(W^{-1})$. Consequently

$$\log \rho_W(z) = -\tfrac{1}{2}\log(2\pi) + y - y_0 - u + O(W^{-1}) \longrightarrow -\tfrac{1}{2}\log(2\pi) - z^2/2.$$

Both ρ_W and the standard normal density ϕ integrate to one. Dominated convergence applied to $\min(\rho_W, \phi)$ gives $\|\rho_W - \phi\|_1 \to 0$ and hence $\|\sqrt{\rho_W} - \sqrt{\phi}\|_2 \to 0$. Translation by h in y becomes translation by $h/\sqrt{V}$ in z. Since $\|I - T_a\| \le 2$ uniformly in real a, the squared translation errors of the two square-root densities differ by $o(1)$ uniformly in the shift. The Gaussian overlap is $\langle \sqrt{\phi}, T_a \sqrt{\phi} \rangle = e^{-a^2/8}$, which gives (16). □

3.1 Exact discrete towers

For an end block of a critical chain, conformal field theory describes the reduced state through an annulus Hamiltonian. Its width grows as $W = \ln(\kappa N)$ and its level degeneracies depend on the boundary operator content [46–49]. For the corresponding open-boundary conformal comparison, the free-fermion towers have odd one-body levels

$$\varepsilon_\ell = (2\ell + 1)\varepsilon, \qquad \varepsilon = \pi^2/W, \tag{17}$$

once for Ising and twice for XX [50]. The double odd tower is the comparison for the XX sizes divisible by four used in the numerical study. When $N/2$ is odd, the finite XX covariance has an exact $1/2$ mode, which contributes a flat Schmidt factor. The finite-chain theorem covers both parity sectors. Such towers also occur as exact half-chain spectra in integrable gapped models [51]. For the towers themselves, the leading embezzlement law can be proved without replacing the discrete levels by a smooth density. For a fixed integer $r \ge 1$ and $\varepsilon > 0$, consider the normalized countable spectrum

$$p_{\boldsymbol{\nu}} = Z_r(\varepsilon)^{-1} \exp\left[-\varepsilon \sum_{a=1}^{r} \sum_{\ell \ge 0} (2\ell + 1)\nu_{a,\ell}\right], \qquad Z_r(\varepsilon) = \prod_{\ell \ge 0} (1 + e^{-(2\ell+1)\varepsilon})^r. \tag{18}$$

Here every $\nu_{a,\ell}$ is 0 or 1 and only finitely many occupations are nonzero. Order the probabilities decreasingly as $(p_j)_{j \ge 1}$, put $\lambda_j = \sqrt{p_j}$, and define

$$(D_d \lambda)_j = d^{-1/2} \lambda_{\lceil j/d \rceil}, \qquad \mathcal{E}_d = \|\lambda - D_d \lambda\|_2^2, \qquad S = -\sum_j p_j \log p_j.$$

The cases $r = 1, 2$ are the single and double odd-level towers used for the Ising and XX conformal spectra, respectively.

Theorem 11 (Exact discrete tower law)**.** *Fix integers $r \ge 1$ and $d \ge 2$, and set $c_r = r\pi^2/24$. For the exact spectrum (18), as $\varepsilon \downarrow 0$,*

$$S = \frac{2c_r}{\varepsilon} + O_r(\varepsilon), \tag{19}$$

$$\mathcal{E}_d = \frac{(\log d)^2 \varepsilon}{8c_r} + O_{r,d}(\varepsilon^2), \tag{20}$$

$$\mathcal{E}_d S = \frac{(\log d)^2}{4} + O_{r,d}(\varepsilon). \tag{21}$$

All constants are independent of ε. The left-hand sides use the exact discrete spectrum with its full level multiplicities.

Appendix D proves the tower law by applying Proposition 8 to the exact spectrum with $W = \varepsilon^{-1}$. The tilted characteristic-function bound (62) for the integer tower levels gives (10) after rescaling to the physical energy spacing. This yields the $O_{r,d}(\varepsilon^2)$ remainder. Lemma 37 also compares the exact tower with the smooth Calabrese–Lefevre distribution in Hellinger distance.

3.2 Numerical comparison

For target dimension two, write $\mathcal{E}^{(r)}_{\mathrm{tower}}(b)$ for the r-copy tower curve at $b = -\ln p_1$, and $\mathcal{E}_{\mathrm{s}}(b)$ for the smooth Calabrese–Lefevre curve. The tower superscript labels level multiplicity; the subscript d in $\mathcal{E}_d$ always labels the target dimension.

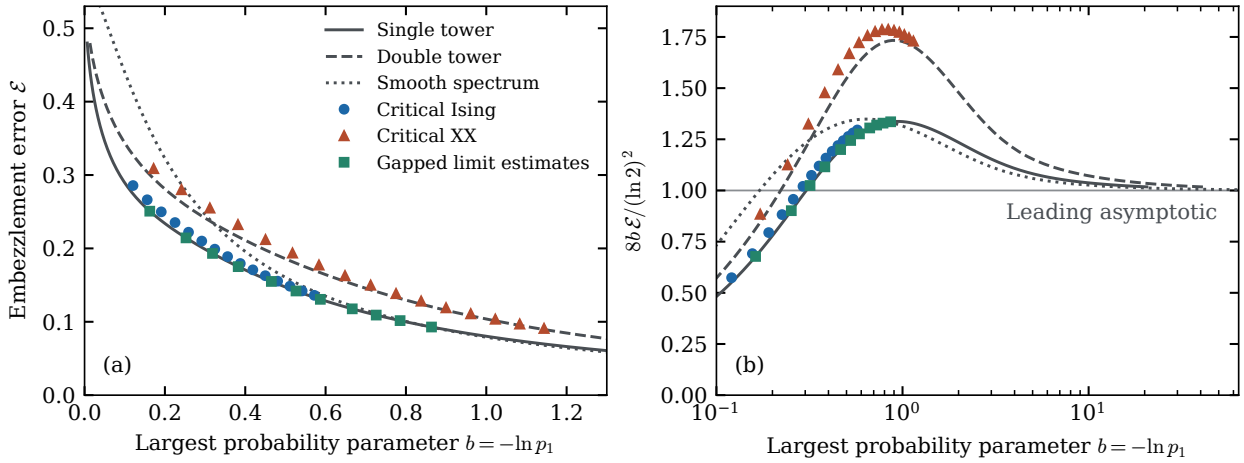


Figure 2: The finite-size law. (a) Embezzlement error against $b = -\ln p_1$ for the critical Ising chain (8 to 131,072 spins), the critical XX chain (8 to 262,144 sites) and gapped chains in the limit of long chains. Curves: prediction from equally spaced entanglement levels (17), each level occurring once or twice, and from the smooth conformal spectrum of Ref. [52]. (b) The same data and curves multiplied by $8b/(\ln 2)^2$. The proved XX, Ising and tower asymptotics correspond to the value one; it is approached slowly.

		S	b	$\mathcal{E}$	$\mathcal{E}/\mathcal{E}^{(r)}_{\text{tower}}(b)$	$\mathcal{E}/\mathcal{E}_{\text{s}}(b)$
Critical Ising	$N = 16$	0.423	0.156	0.2663	1.049	0.72
	$N = 256$	0.663	0.292	0.2100	1.043	0.84
	$N = 4{,}096$	0.895	0.419	0.1708	1.030	0.91
	$N = 65{,}536$	1.126	0.542	0.1420	1.022	0.95
	$N = 131{,}072$	1.184	0.572	0.1361	1.021	0.96
Critical XX	$N = 16$	0.714	0.241	0.2799	1.065	0.97
	$N = 256$	1.210	0.518	0.1935	1.065	1.24
	$N = 4{,}096$	1.674	0.775	0.1382	1.038	1.33
	$N = 65{,}536$	2.136	1.022	0.1034	1.020	1.34
	$N = 262{,}144$	2.367	1.144	0.0908	1.015	1.34
Staggered XX	$\xi = 20$	0.729	0.318	0.1932	1.000	0.82
	$\xi = 200$	1.114	0.526	0.1420	1.000	0.93
	$\xi = 2{,}000$	1.498	0.726	0.1092	1.000	0.98
	$\xi = 10{,}000$	1.766	0.864	0.0929	1.000	1.00

Table 1: Entanglement entropy S, $b = -\ln p_1$ and embezzlement error $\mathcal{E}$ of half-chain ground states. $\mathcal{E}^{(1)}_{\text{tower}}$ (Ising and gapped chains) and $\mathcal{E}^{(2)}_{\text{tower}}$ (XX chain) are the predictions from equally spaced levels at the same b; $\mathcal{E}_{\text{s}}$ is the prediction of the smooth conformal spectrum. For gapped chains the values estimate the long-chain limits.

Test. We computed the entanglement spectra of the ground states of three chains by the correlation-matrix method [43, 53]: the critical XX chain with open ends, the critical transverse-field Ising chain with open ends, and the XX chain with a staggered potential, which is gapped with correlation length ξ. All chains are cut in the middle. Appendix K gives the methods and checks. Figures 1(b) and 2 and Table 1 show the results.

In both critical chains $\mathcal{E}$ decreases monotonically over the sampled powers of two, spanning more than four decades in size. At the larger sizes, the increments in $1/\mathcal{E}$ grow with successive doublings. The slope of S with respect to $\ln N$ at the largest sampled doubling is 0.083335 for Ising and 0.166670 for XX, close to $c/6$ for $c = 1/2$ and $c = 1$, respectively.

The spectra have the predicted form. For the gapped chain the one-body entanglement energies satisfy $\varepsilon_\ell/\varepsilon_0 = 2\ell + 1$ to five digits, and $\mathcal{E}$ agrees with $\mathcal{E}^{(1)}_{\text{tower}}(b)$ within 2×10^{-4} for all ξ from 4 to 10^4. At criticality the levels are equally spaced only approximately. For the Ising chain, expressing ε_0 as $\pi^2/\ln(\kappa N)$ gives $\kappa = 16.9$ at 1,024 spins and 17.6 at 131,072 spins. The ratios $\varepsilon_\ell/\varepsilon_0$ for $\ell = 1, 2, 3$ are 3.13, 5.46 and 7.95 at 1,024 spins and 3.04, 5.18 and 7.39 at 131,072 spins. Accordingly $\mathcal{E}/\mathcal{E}^{(1)}_{\text{tower}}(b)$ falls from 1.05 to 1.021 for the Ising chain, and $\mathcal{E}/\mathcal{E}^{(2)}_{\text{tower}}(b)$ falls from 1.07 to 1.015 for the XX chain.

The comparison with the smooth spectrum shows which conformal data matter. At fixed b the smooth

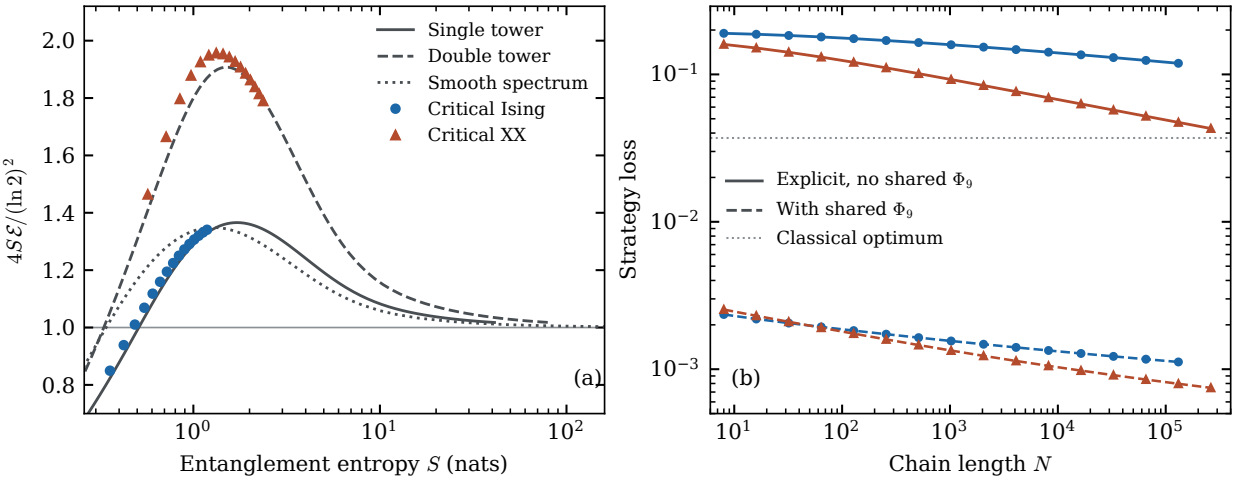


Figure 3: (a) The entropy-normalized error for the physical chains, exact towers and smooth spectrum. The exact tower and finite-chain theorems give limit one. (b) The exact loss of the explicit unassisted strategy (solid) and the loss after adding Φ_9 (dashed). The horizontal line is the classical optimum 1/27. These are achieved strategy losses, not values of the optimization over all measurements.

spectrum is a one-parameter approximation that suppresses tower degeneracies, and at the largest sizes the XX chain lies 34% above it while the Ising chain lies 4% below. What distinguishes $\mathcal{E}^{(1)}_{\text{tower}}$, $\mathcal{E}^{(2)}_{\text{tower}}$ and $\mathcal{E}_{\text{s}}$ is the operator content, that is, the degeneracies of the levels of L_0.

The sampled chains remain pre-asymptotic for $\mathcal{E}$. At $N = 4{,}096$, the XX chain gives $\mathcal{E} \simeq 0.13824$ and $V \simeq 1.58147$. The error is about 82% above the capacity expression $(\ln 2)^2/(4V) \simeq 0.07595$ and about 60% above the leading logarithmic expression $3(\ln 2)^2/(2\ln N) \simeq 0.08664$ in (3). The combination $8b\,\mathcal{E}/(\ln 2)^2$, which tends to one under Theorem 1, still equals 1.19, 1.08 and 1.016 at $b = 2$, 4 and 16 for the smooth spectrum, and 1.14 and 1.27 at $b = 3.2$ on the curves $\mathcal{E}^{(1)}_{\text{tower}}$ and $\mathcal{E}^{(2)}_{\text{tower}}$. The Ising and XX samples reach $b = 0.57$ and $b = 1.14$, respectively. On the large-S side of Figure 3(a), the entropy-normalized error comes within ten per cent of one near $S \simeq 7$ for the smooth curve and near $S \simeq 9$ and $S \simeq 15$ for the single and double towers.

Gapped chains behave like critical chains of finite length. Their $\mathcal{E}$ stops decreasing once N exceeds a few correlation lengths (Figure 1(b)). The estimated limiting value lies close to the curve $\mathcal{E}^{(1)}_{\text{tower}}(b)$ and falls slowly with ξ, from 0.193 at $\xi = 20$ to 0.142, 0.109 and 0.093 at $\xi = 200$, 2,000 and 10,000.

At the largest sizes the explicit unassisted losses are 0.11895 for Ising and 0.04295 for XX, both above the classical loss 1/27. With the additional Φ_9, the losses are 1.120×10^{-3} and 7.472×10^{-4}. These achieved losses lie below the classical bound and above the flat-spectrum floor γ. The compressed certificate of Section 5.2 establishes separation in the original game at the explicit larger length $N = 2^{84}$. Section 5.3 gives an exact four-spin separation using a tagged extension, and Section 4.2 gives a nine-qubit resource separated by the original game.

4 The game measures the embezzlement error

We first collect the properties of $\mathcal{E}$ that are used below. The bounds (24)–(26) are contained in Ref. [21, Section 6.4], and we include short proofs to keep the paper self-contained. Throughout, T denotes a random variable on $(0, \infty)$ with density $p(t) = p_{\lceil t \rceil}$, which is a continuous version of the rank of a Schmidt probability, and $X = \log_2 T$.

Proposition 12 (Properties of $\mathcal{E}$)**.** *Let ψ be a pure bipartite state with Schmidt probabilities $p_1 \geq p_2 \geq \cdots$, Schmidt coefficients λ and entanglement entropy $S = -\sum_j p_j \ln p_j \leq \infty$. The embezzlement error is positive and never exceeds its value on a product state,*

$$0 < \mathcal{E}(\psi) \leq 2 - \sqrt{2}, \tag{22}$$

with equality on the right exactly when all nonzero p_j are equal, that is, for product states and maximally entangled states. It does not change when a maximally entangled state Φ_k of any local dimension k is added,

and its square root is Lipschitz continuous in the Schmidt coefficients: if ψ' has Schmidt coefficients λ', then

$$\mathcal{E}(\psi \otimes \Phi_k) = \mathcal{E}(\psi), \qquad \left|\sqrt{\mathcal{E}(\psi)} - \sqrt{\mathcal{E}(\psi')}\right| \le 2\|\lambda - \lambda'\|. \tag{23}$$

A small error forces the Schmidt probabilities to spread over many ranks. The largest of them, and the total weight of the ranks between $r+1$ and $2r$ for every integer $r \ge 1$, satisfy

$$p_1 \le (6 + 4\sqrt{2})\,\mathcal{E}(\psi), \qquad \sum_{j=r+1}^{2r} p_j \le \sqrt{\mathcal{E}(\psi)}, \tag{24}$$

while the entanglement entropy and, when finite, the variance of X satisfy

$$\sqrt{\mathcal{E}(\psi)} \;\ge\; \frac{\ln 2}{2\,(S + 1 + 3\ln 2)} \tag{25}$$

and

$$\mathcal{E}(\psi)\,\big(4\,\mathrm{Var}(X) + 1\big) \ge 1. \tag{26}$$

Proof. We use the formula $\mathcal{E} = \|\lambda - D\lambda\|^2 = 2 - 2\langle\lambda, D\lambda\rangle$ of Lemma 7 throughout.

Range. Since λ is decreasing, $\lambda_{\lceil m/2\rceil} \ge \lambda_m$. Hence $\langle\lambda, D\lambda\rangle = \sum_m \lambda_m \lambda_{\lceil m/2\rceil}/\sqrt{2} \ge 1/\sqrt{2}$, which is the upper bound in (22). Equality requires $\lambda_m \in \{0, \lambda_{\lceil m/2\rceil}\}$ for all m, that is, a flat spectrum. Positivity follows from the first bound in (24).

Invariance and continuity. The Schmidt coefficients of $\psi \otimes \Phi_k$ are $(D_k\lambda)_m = \lambda_{\lceil m/k\rceil}/\sqrt{k}$. The map D_k is an isometry, and it commutes with $D = D_2$ because $\lceil\lceil m/k\rceil/2\rceil = \lceil m/(2k)\rceil$. Therefore $\|D_k\lambda - DD_k\lambda\| = \|D_k(\lambda - D\lambda)\| = \|\lambda - D\lambda\|$. The Lipschitz bound holds because $\sqrt{\mathcal{E}} = \|(1 - D)\lambda\|$ and $\|1 - D\| \le 2$.

Largest probability and windows. The first component of $\lambda - D\lambda$ is $\lambda_1(1 - 1/\sqrt{2})$, so $\mathcal{E} \ge p_1(1 - 1/\sqrt{2})^2 = p_1/(6 + 4\sqrt{2})$. For the second bound in (24), let $f(s) = (\ln 2)\,2^s p(2^s)$ be the density of X and $g = \sqrt{f}$. The substitution $t = 2^{s+1}$ in $\|\lambda - D\lambda\|^2 = \int_0^\infty \big(\sqrt{p(t)} - \sqrt{p(t/2)/2}\big)^2 dt$ gives

$$\mathcal{E}(\psi) = \int_{\mathbb{R}} \big(g(s+1) - g(s)\big)^2\, ds. \tag{27}$$

In other words, the embezzlement error measures how much the distribution of X changes when it is shifted by one. Put $\mu(u) = \int_u^{u+1} f$. By the Cauchy–Schwarz inequality, $\sum_{k\in\mathbb{Z}} |\mu(u+k) - \mu(u+k-1)| \le \int_{\mathbb{R}} |g(s) - g(s-1)|\,(g(s) + g(s-1))\,ds \le 2\sqrt{\mathcal{E}}$. The left-hand side is the total variation of a nonnegative summable sequence and is therefore at least twice its maximum. So $\mu(u) \le \sqrt{\mathcal{E}}$ for all u, and the choice $u = \log_2 r$ gives the claim.

Entropy. Because p is decreasing, $t\,p(t) \le \int_0^t p \le 1$, hence $-\ln p(T) \ge \ln T$ and $S \ge (\ln 2)\,\mathbb{E}[X]$. The negative part of X contributes $\int_0^1 p_1 \log_2(1/t)\,dt = p_1/\ln 2 \le 1/\ln 2$. Put $h = \sqrt{\mathcal{E}}$. For $x \ge 0$ the interval $(1, 2^x]$ is covered by at most $x+1$ windows of the form $(r, 2r]$, and $p_1 = 2\mu(-1) \le 2h$, so $\Pr[X \le x] \le (x+3)h$ and $\mathbb{E}[X_+] \ge \int_0^\infty (1 - (x+3)h)_+\,dx = (1 - 3h)_+^2/(2h)$. If $h < 1/3$ this gives $S \ge (\ln 2)\big(\frac{1}{2h} - 3\big) - 1$, which is (25). If $h \ge 1/3$ then (25) holds because its right-hand side is below 0.12.

Variance. Assume X has finite second moment; otherwise the variance bound is interpreted as vacuous. The laws of X and $X - 1$ have means that differ by one. With $\mu_X = \mathbb{E}[X]$ and $c = \mu_X - \frac{1}{2}$ we obtain $1 = \int (s - c)\big(f(s) - f(s+1)\big)ds \le \sqrt{\mathcal{E}}\,\big(\|(s-c)g(s)\|_2 + \|(s-c)g(s+1)\|_2\big) = 2\sqrt{\mathcal{E}}\,\sqrt{\mathrm{Var}(X) + \frac{1}{4}}$, which is (26). □

Remark 13. The power of S in (25) cannot be improved for general states, since the sine profiles in Section 4.2 have $\mathcal{E} = O(K^{-2})$ and $S = \Theta(K)$. Their shell distribution keeps a fixed fraction of its mass at ranks $2^{\Theta(K)}$, giving the entropy estimate. For exact conformal tower spectra $\mathcal{E}$ is of order $1/S$ (Section 3), and (26) is then the relevant bound. The second bound in (24) controls the weight of every factor-of-two window of ranks. In particular the quantity η_{emb} of Ref. [25], which decides universal LOCC embezzlement [23], satisfies $\eta_{\mathrm{emb}} \le 2\sqrt{\mathcal{E}}$. Explicit Gaussian embezzlement protocols for free-fermion ground states are given in Ref. [24], which also notes that their convergence with the system size is very slow.

The numerical lower comparison. The proof in Appendix B extracts the magic-square sector with a $210\sqrt{N}$ error, where N is its total forbidden mass. A polar decomposition then aligns the nine label sectors, at a cost bounded by twice the square root of their total mismatch. These estimates give

$$\mathcal{E}(\psi) \le C\,\varepsilon, \qquad C = \frac{243(210^2 + 448)}{8} = \frac{2{,}706{,}291}{2},$$

for every strategy of loss ε. Local projective dilations preserve the nonzero Schmidt spectrum, so the bound holds for arbitrary POVMs and arbitrary Schmidt rank. Taking the infimum of the loss proves the lower bound in Theorem 3.

4.1 An explicit finite-resource strategy

Write $\lambda_i = \sqrt{p_i}$ in decreasing order and pad the sequence with zeroes. Use labels $j \in \mathbb{Z}/9\mathbb{Z}$ and set $\lambda_{j,n} = \lambda_{9(n-1)+j+1}$ for $j = 0, \dots, 8$. For each j and $m \geq 1$, define

$$a_{jm} = \lambda_{j,m}, \qquad b_{jm,k} = \lambda_{j+1+2k+r,\lceil m/2 \rceil}, \qquad r = (m-1) \bmod 2, \quad k = 0,1,2,3,$$

where the first index of λ is read modulo nine. For $b = (b_0, b_1, b_2, b_3)$ define $\bar{b} = (b_0 + b_1 + b_2 + b_3)/4$ and

$$q_1 = \tfrac{1}{2}(b_0 - b_1 + b_2 - b_3), \quad q_2 = \tfrac{1}{2}(b_0 + b_1 - b_2 - b_3), \quad q_3 = \tfrac{1}{2}(b_0 - b_1 - b_2 + b_3).$$

Set

$$\mathcal{B}(p) = \frac{1}{243} \sum_{j,m} \left[(\sqrt{2} a_{jm} - \bar{b}_{jm})^2 + 8(q_{jm,1}^2 + q_{jm,3}^2) + \frac{25}{4} q_{jm,2}^2 \right]. \tag{28}$$

Theorem 14. *There is a strategy for G on ψ with losing probability $\mathcal{B}(p)$ using no additional shared state. In particular,*

$$1 - \omega(\psi) \leq \min\{1/27, \mathcal{B}(p)\}, \qquad \mathcal{B}(p) \leq \frac{7}{3} \mathcal{E}(\psi).$$

If ψ has Schmidt rank d, the sum in (28) can be restricted to $1 \leq m \leq 2\lceil d/9 \rceil$.

Proof. Use the real Pauli square with $Z = \mathrm{diag}(1,-1)$, $X = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, $Y = \left(\begin{smallmatrix} 0 & -i \\ i & 0 \end{smallmatrix}\right)$, and with tensor products written without the $\otimes$ symbol,

$$\begin{pmatrix} ZI & IZ & ZZ \\ IX & XI & XX \\ ZX & XZ & YY \end{pmatrix}.$$

The entries in a row or column commute and their products give the required parity. On $\mathbb{C}^5 = \mathbb{C}f \oplus (\mathbb{C}^2 \otimes \mathbb{C}^2)$ use the joint spectral projections of a row or column, and the projection onto f for the flag. For a sign triple (s_1, s_2, s_3) of the required parity the projection on the four-dimensional summand is $\prod_i (I + s_i M_i)/2$, where M_1, M_2, M_3 are the relevant cells in order. The four row-2 basis vectors are

$$(e_0, e_1, e_2, e_3) = (|++\rangle_X, |-+\rangle_X, |+-\rangle_X, |--\rangle_X).$$

The selected row-1 and column-1 vectors are $r_y = |0y\rangle_Z$ and $c_x = |0\rangle_Z \otimes |(-1)^x\rangle_X$. For T_x the sign projections are onto $(f \pm c_x)/\sqrt{2}$, and the three triple projections are those of column 1 other than c_x. For U_y use $(f \pm r_y)/\sqrt{2}$ and the other row-1 triple projections. These are complete orthogonal measurements.

For a question in copy j, both players use the same identification of Schmidt coordinates with five-level blocks. The coordinate (j, m) is (f, m) and the coordinate $(j + 1 + 2k + r, \lceil m/2 \rceil)$ is (e_k, m). This is a bijection between the padded countable coordinate space and $\{f, e_0, e_1, e_2, e_3\} \times \mathbb{N}$. Use the direct sum of the five-level measurements over m. For finite Schmidt rank, the identification on the Schmidt support is an isometry into $\mathbb{C}^5 \otimes \mathbb{C}^{2\lceil d/9 \rceil}$ and the compressed projections are a POVM. One can extend the POVM off the Schmidt support by any fixed outcome. Bob measures the original label for Lab. Alice's row-1 flag selects label j and her row-2 $+++$ selects e_0, the pair of labels $j+1, j+2$. Both label tests pass exactly. If p consists of ninefold equal blocks, the same construction has loss $2\mathcal{E}(p)/243$, proving the assisted assertion of Theorem 3.

In one block the state is the unnormalised vector $a|ff\rangle + \sum_k b_k |e_k e_k\rangle$. The block contributions add, since the copy's projections are diagonal in the block index. For each row-column test the flag component has zero loss, and the nonflag loss is $\frac{1}{4} \sum_e (b_e - b_{\pi(e)})^2$, where the cell operator permutes the $X \otimes X$ basis by π up to signs. Each of the three nonidentity flips of the X bits occurs twice and the other cells are diagonal. The sum over nine tests is therefore

$$\sum_{e<e'} (b_e - b_{e'})^2 = 4 \sum_e (b_e - \bar{b})^2.$$

In (T_x, C_1) the sign sector groups the flag with c_x. On this block-diagonal state there are no flag-to-nonflag cross terms in the loss. Thus a loss is precisely a mismatch in the column-1 basis on the nonflag sector.

Changing the first bit from X to Z gives total loss $2(q_1^2+q_3^2)$ over the two values of x. Likewise (R_1,U_y) compares outcomes in the $Z\otimes Z$ basis, where the four diagonal entries of the nonflag coefficient matrix are $\bar b$. This gives $2\sum_e (b_e-\bar b)^2$ over the two values of y.

When both answers are signs in (T_x,U_y), the restricted state is

$$a|ff\rangle + (-1)^{xy}\frac{b_{2x}+b_{2x+1}}{2\sqrt{2}}|c_x r_y\rangle.$$

Measuring the indicated sign projections gives loss $\frac12(a-(b_{2x}+b_{2x+1})/(2\sqrt{2}))^2$. The sum over (x,y) is $(\sqrt{2}a-\bar b)^2+q_2^2/4$. The four contributions add to (28) because $\sum_e (b_e-\bar b)^2 = q_1^2+q_2^2+q_3^2$ and the remaining question pairs are free.

For the uniform bound let P be orthogonal projection onto sequences constant on each consecutive block of nine coordinates. Put $v=\|(1-P)\lambda\|^2$ and $h=\|(1-D)\lambda\|=\sqrt{\mathcal{E}(\psi)}$. The quadratic form in (28) is invariant under a simultaneous cyclic shift of all label blocks, so its bilinear form vanishes between $P\lambda$ and $(1-P)\lambda$. This follows by averaging the latter argument over the nine shifts. On $P\lambda$ its value is $2\|(1-D)P\lambda\|^2/243$, since the four b entries are equal. For each copy the quadratic form inside the sum is at most $8(a^2+\sum_k b_k^2)$. Indeed, in the orthonormal coordinates $a, 2\bar b, q_1, q_2, q_3$ the first square has coefficient norm squared $9/4$, and the other coefficients are at most 8. Each copy uses every coordinate exactly once. Hence

$$\mathcal{B}(p) \le \frac{2}{243}\|(1-D)P\lambda\|^2 + \frac{8}{27}v \le \frac{2}{243}(h+\sqrt{8v/9})^2 + \frac{8}{27}v. \tag{29}$$

For the second inequality use $(1-D)P\lambda = P(1-D)\lambda + PD(1-P)\lambda$. For one zero-mean block $x\in\mathbb{R}^9$, applying D and averaging the resulting two blocks gives two constant vectors with entries $\pm(x_1+x_2+x_3+x_4-x_6-x_7-x_8-x_9)/(9\sqrt{2})$. Thus $\|PD(1-P)\|\le\sqrt{8/9}$.

The variance of nine real numbers in a range of length R is at most $20R^2/9$, where the variance here means the sum of squared deviations from the mean. To see this, maximise the convex sum of squared deviations over the nine-dimensional box by choosing each coordinate at an endpoint. If r coordinates are at one endpoint, the sum is $r(9-r)R^2/9$. The ranges of disjoint consecutive blocks of the decreasing sequence λ sum to at most λ_1. Their squared ranges after the first two blocks sum to at most λ_{19}^2. In particular $v\le(20/9)p_1$, and a better estimate follows from the first 19 coordinates. Let T act on $x\in\mathbb{R}^{19}$ by $(Tx)_i = x_i - x_{\lceil i/2\rceil}/\sqrt{2}$, and set

$$Q = \operatorname{diag}(I_9 - J_9/9, I_9 - J_9/9, 20/9),$$

where J_9 has all entries one. Then $v\le x^{\mathsf T}Qx$ at $x=(\lambda_1,\dots,\lambda_{19})$. The 19×19 matrix $(15/2)T^{\mathsf T}T - Q$ is positive definite. An exact certificate is given below. It follows that $v\le(15/2)\|Tx\|^2\le(15/2)h^2$. Equation (29) gives

$$\mathcal{B}(p) \le \left[\frac{20}{9} + \frac{2}{243}(1+\sqrt{20/3})^2\right]\mathcal{E}(\psi) < \frac{7}{3}\mathcal{E}(\psi).$$

Finally, in each flagged magic-square copy a deterministic strategy loses at least one pair. If all row-column tests pass, the six flag statuses agree. Playing only flags forces the four sign tests to implement the contradictory classical CHSH constraints, while playing no flags forces the contradictory row and column parities of the magic square. Conversely, Bob can answer label 0 and the players use all flags and all signs $+$ in copy 0. For $j\neq0$ use

$$R_1 = C_1 = (-,+,-), \qquad R_2 = R_3 = C_2 = (+,t,t), \qquad C_3 = (-,t,t),$$

and $T_x = C_1$, $U_y = R_1$, where $t=+$ exactly when $0\in\{j+1,j+2\}$. The copy loses just at (R_3,C_1) and both label tests pass. Copy 0 loses just at (T_1,U_1). Thus the classical loss is exactly $9/243 = 1/27$. □

Exact matrix certificate. Start with $H=(15/2)T^{\mathsf T}T-Q$. For $k=1,\dots,19$, record $d_k=H_{kk}$ and replace every remaining entry H_{ij}, $i,j>k$, by $H_{ij}-H_{ik}H_{kj}/d_k$. With arithmetic in $\mathbb{Q}(\sqrt{2})$ the integers $\lfloor 1000 d_k\rfloor$ are

$$(3504, 6418, 10169, 8726, 11381, 10778, 11296, 10649, 11831, 7230,$$
$$3906, 2185, 3944, 2287, 3512, 972, 4004, 2448, 145).$$

These positive pivots certify positive definiteness by successive Schur complements. The verification uses only rational operations and comparison of a rational number with a rational multiple of $\sqrt{2}$ by squaring.

Leading coefficient for spread-out spectra. In the proof above, $|\|(1-D)P\lambda\| - h| \le 2\sqrt{v}$. Using the orthogonal splitting of (28) gives

$$\left|\mathcal{B}(p) - (2/243)\mathcal{E}(\psi)\right| \le \frac{8}{243}\sqrt{v\,\mathcal{E}(\psi)} + \frac{80}{243}v, \qquad v \le \frac{20}{9}p_1.$$

Thus the explicit strategy uses no shared ancilla and has loss $(2/243)\mathcal{E}(\psi)(1+o(1))$ along any sequence for which $p_1 = o(\mathcal{E}(\psi))$.

Implementation in a known Schmidt basis. For copy j, reversibly decompose a zero-based Schmidt index i into its quotient and remainder upon division by nine. If the remainder is j, keep the quotient as block index and set the five-level register to f. Otherwise subtract j modulo nine, subtract one, and write the result as $2k + r$ with $0 \le k \le 3$ and $r = 0, 1$. Set the register to e_k and double the old quotient and add r to obtain the block index. The inverse recovers the original quotient and remainder from these registers. With binary index registers, reversible arithmetic for this isometry and its inverse has a gate count and workspace polynomial in $\log d$. It is followed by a fixed five-level measurement independent of d. The change to the Schmidt basis is a separate state-dependent operation.

Corollary 15. *Let $p^{(d)}$ be the first d Schmidt probabilities renormalised by $1-\tau$, where $\tau = \sum_{i>d} p_i < 1$. There is a strategy on ψ with loss at most*

$$(1-\tau)\min\{1/27, \mathcal{B}(p^{(d)})\} + \tau/27.$$

For any state ρ at trace distance at most η from $|\psi\rangle\langle\psi|$, the same strategy on ρ has loss at most this bound plus η.

Proof. Use the better of the explicit and classical strategies on the Schmidt prefix and the classical strategy on its orthogonal complement. The local measurements are block diagonal and both sides' Schmidt projections agree on the state, so the loss is the probability-weighted sum of the two losses. For the noisy state use $0 \le L \le I$ and $|\operatorname{Tr} L(\rho-\sigma)| \le \frac{1}{2}\|\rho-\sigma\|_1$ for states ρ, σ. □

4.2 An explicit lower bound for maximally entangled states

Mančinska proves a dimension-independent gap for maximally entangled states on the single flagged magic square and constructs a rank-five perfect resource [28, Corollary 4 and Proposition 1]. The existence of a positive gap for G already follows by restriction to its nine copies. The next theorem gives an explicit constant and quantitative stability bounds.

Theorem 16. *For every finite local dimension and every strategy on Φ_d with arbitrary POVMs, the losing probability in G is at least*

$$\gamma = \frac{3-2\sqrt{2}}{243} > 7.06 \times 10^{-4}. \tag{30}$$

The bound holds for mixtures of maximally entangled states on arbitrary Schmidt supports. At trace distance η from such a mixture the losing probability is at least $\gamma - \eta$. For a pure state with decreasing Schmidt amplitudes λ_i and a flat rank-d state in the same local spaces,

$$1 - \omega(\psi) \ge \left(\sqrt{\gamma} - \sqrt{2 - 2d^{-1/2}\sum_{i=1}^{d}\lambda_i}\right)_+^2.$$

Proof. Compress all effects to the Schmidt supports and absorb the transpose into Bob's effects. The resulting probabilities are $\tau(AB)$, with $\tau = \operatorname{Tr}/d$. Write $\|A\|_2^2 = \tau(A^*A)$. For effects P, Q, the binary mismatch $\mu = \tau(P + Q - 2PQ)$ bounds both $|\tau(P) - \tau(Q)|$ and $\|\sqrt{P} - \sqrt{Q}\|_2^2$. The first assertion follows from the two nonnegative mismatches $\tau(P(I-Q))$ and $\tau((I-P)Q)$. The second follows from $\tau(\sqrt{P}\sqrt{Q}) \ge \tau(P\sqrt{Q}) \ge \tau(PQ)$ by positivity and the trace property.

In a fixed copy, write P_i, Q_j for the row and column flag effects, and m_{ij} for the conditional loss at (R_i, C_j). Let $M = \sum_{i,j} m_{ij}$, let N be the sum of the four conditional losses at (T_x, C_1) and (R_1, U_y), and

let L be the sum of the four conditional losses at (T_x, U_y). Write α for the loss at (R_1, Lab) and B for the matching label effect. Use the positive contraction

$$W = \left(\frac{\sqrt{P_1} + \sqrt{Q_1}}{2} \right)^2, \qquad w = \tau(W).$$

Each m_{ij} bounds the binary flag mismatch. The coefficient matrix

$$\frac{1}{9} \begin{pmatrix} 5 & 2 & 2 \\ 2 & -1 & -1 \\ 2 & -1 & -1 \end{pmatrix}$$

has row sums and column sums $(1, 0, 0)$ and squared Frobenius norm $5/9$. Thus $\sqrt{P_1} - \sqrt{Q_1}$ is the corresponding linear combination of $\sqrt{P_i} - \sqrt{Q_j}$, giving $\|\sqrt{P_1} - \sqrt{Q_1}\|_2^2 \le 5M/9$. In addition, the flag comparison graph has three edge-disjoint paths from R_1 to C_1, one direct and the other two through (C_i, R_i) for $i = 2, 3$. Summing the triangle inequalities for the trace differences along these paths gives $|\tau(P_1) - \tau(Q_1)| \le M/3$. Consequently,

$$w = \tfrac{1}{2}\big(\tau(P_1) + \tau(Q_1)\big) - \tfrac{1}{4}\|\sqrt{P_1} - \sqrt{Q_1}\|_2^2 \ge \tau(B) - \alpha - \tfrac{11}{36}M. \tag{31}$$

Let E_x, F_y denote the sums of the sign effects of T_x, U_y, and let S_x, H_y denote their signed differences. Put

$$d_x = \tau(W(I - E_x)), \qquad e_y = \tau(W(I - F_y)), \qquad D = \sum_x d_x + \sum_y e_y.$$

The calibration loss at (T_x, C_1) bounds $\tau(Q_1(I - E_x))$, since a flag on C_1 requires a sign on T_x. The loss at (R_1, U_y) likewise bounds $\tau(P_1(I - F_y))$. In each of the four norms defining the square roots of d_x, e_y, replace $\sqrt{W}$ by the corresponding flag square root. The change is bounded by $\|\sqrt{P_1} - \sqrt{Q_1}\|_2/2$ since the remaining factor is a contraction. Taking the Euclidean norm of the four bounds gives

$$\sqrt{D} \le \sqrt{N} + \sqrt{5M/9}. \tag{32}$$

Write $t_{xy} = (-1)^{xy}$. For each losing pair of sign effects (A, D') at (T_x, U_y), use the two Hilbert–Schmidt vectors $\sqrt{A}\sqrt{D'}$ and $\sqrt{A}W\sqrt{D'}$. The sum of the squared norms of the first vectors is L. For the second vectors, the sum is

$$q = \frac{1}{2}\left(\sum_{x,y} \tau(WE_xWF_y) - C \right), \qquad C = \sum_{x,y} t_{xy}\tau(WS_xWH_y).$$

This follows by expanding the two losing sign combinations per pair. Their inner products sum, in real part, to

$$R = \frac{1}{2}\operatorname{Re}\sum_{x,y} \tau(WE_xF_y - t_{xy}WS_xH_y).$$

Therefore $R \le \sqrt{Lq}$. Since $0 \le W, E_x, F_y \le I$, the sum of $\tau(WE_xWF_y)$ is at most $2\sum_x \tau(WE_x)$ and also at most $2\sum_y \tau(WF_y)$. Averaging gives

$$R \le \sqrt{Lq}, \qquad 0 \le q \le 2w - \tfrac{1}{2}D - \tfrac{1}{2}C. \tag{33}$$

For Hermitian S_x, H_y, split the Hilbert–Schmidt inner products of $S_x\sqrt{W}, H_y\sqrt{W}$ with the weights W and $I - W$. The contribution from W is C, the CHSH sum of the vectors $\sqrt{W}S_x\sqrt{W}$ and $\sqrt{W}H_y\sqrt{W}$. Let U be the sum of their four squared norms. The CHSH coefficient matrix has operator norm $\sqrt{2}$, so $|C| \le U/\sqrt{2}$. The contribution from $I - W$ is at most the sum of the four squared norms of $(I-W)^{1/2}S_x\sqrt{W}, (I-W)^{1/2}H_y\sqrt{W}$ divided by $\sqrt{2}$. Each signed difference satisfies $S_x^2 \le E_x$ or $H_y^2 \le F_y$. Indeed, the row operator $[\sqrt{A_+}\ \sqrt{A_-}]$ is a contraction and the signed difference factors through it, with right factor the column of $\sqrt{A_+}, -\sqrt{A_-}$. Thus the sum of the four squared norms in the $I - W$ contribution is at most $4w - D - U$. We obtain

$$\operatorname{Re}\sum_{x,y} t_{xy}\tau(WS_xH_y) \le \frac{4w - D}{\sqrt{2}} + C - |C|.$$

Expanding E_xF_y around the identity gives $\operatorname{Re}\sum_{x,y}\tau(WE_xF_y)\geq 4w-3D$. In fact the linear terms sum to $-2D$ and the four quadratic terms have total absolute value at most $\sum_{x,y}\sqrt{d_xe_y}\leq D$ by Cauchy–Schwarz and the contraction bounds $(I-E_x)^2\leq I-E_x$ and $(I-F_y)^2\leq I-F_y$. Combining these estimates gives

$$R\geq(2-\sqrt{2})w-\left(\tfrac{3}{2}-\tfrac{1}{2\sqrt{2}}\right)D+\tfrac{1}{2}(|C|-C).$$

Set $u=(2-\sqrt{2})^{-1}$, $c=(2-\sqrt{2})/2$, and $k=(10-\sqrt{2})/8$. Use $\sqrt{Lq}\leq uL+q/(4u)$ and (33). The term $-C/(8u)-(|C|-C)/2$ is nonpositive for every real C, so the preceding lower bound for R implies $cw\leq uL+kD$. Substituting (31) and (32) gives

$$c\tau(B)\leq uL+c\alpha+\begin{pmatrix}\sqrt{M} & \sqrt{N}\end{pmatrix}\begin{pmatrix}1-2\sqrt{2}/9 & k\sqrt{5}/3\\ k\sqrt{5}/3 & k\end{pmatrix}\begin{pmatrix}\sqrt{M}\\ \sqrt{N}\end{pmatrix}.$$

The displayed matrix is bounded above by uI: the difference has first diagonal entry $13\sqrt{2}/18$ and determinant $(5-2\sqrt{2})/288$, both positive. Since $c\leq u$, we conclude $c\tau(B)\leq u(L+\alpha+M+N)$. Sum over the nine copies and use that the label effects sum to the identity. All counted losses are disjoint, so the total conditional loss over the 243 pairs is at least $c/u=3-2\sqrt{2}$.

Linearity for fixed effects and trace-distance continuity give the mixed-state extensions. For pure states, $\big|\|K^{1/2}\psi\|-\|K^{1/2}\phi\|\big|\leq\|\psi-\phi\|$ for any loss operator $0\leq K\leq I$. Take ϕ flat on the first d Schmidt basis vectors, with overlap $d^{-1/2}\sum_{i=1}^{d}\lambda_i$. □

A weighted variant. Assign raw weight 2 to each (R_1,C_1), weight 1 to each of the four sign pairs and the four calibration pairs in a copy, and weight $1/5$ to each (R_1,Lab). Any additional pairs can receive arbitrary nonnegative weights. The weighted sum of losses on every Φ_d is at least $3-2\sqrt{2}$. Indeed, using only the loss β at (R_1,C_1), the preceding argument has $w\geq\tau(B)-\alpha-3\beta/4$ and $\sqrt{D}\leq\sqrt{N}+\sqrt{\beta}$. The last matrix in the proof is replaced by $\left(\begin{smallmatrix}k+3c/4 & k\\ k & k\end{smallmatrix}\right)$, which is bounded above by $u\operatorname{diag}(2,1)$. The difference has first diagonal entry $3\sqrt{2}/2$ and determinant $9/32-\sqrt{2}/16$, both positive. Also $c/u<1/5$. Summing the resulting bound over copies gives the raw floor, to be divided by the total raw weight for a game value.

An exact finite separation. Set $(x_0,\dots,x_5)=(15,11,10,8,6,3)$. Let χ have decreasing Schmidt amplitudes $x_0/\sqrt{555}$ at rank one and $x_k/\sqrt{555\cdot 2^{k-1}}$ at ranks $2^{k-1}<i\leq 2^k$ for $1\leq k\leq 5$. These amplitudes have squared norm one, and with $x_6=0$ their duplication error is

$$\mathcal{E}(\chi)=\frac{(1-1/\sqrt{2})^2x_0^2+(x_1-x_0/\sqrt{2})^2+\sum_{k=2}^{6}(x_k-x_{k-1})^2}{555}=\frac{598-390\sqrt{2}}{555}.$$

Prepare $\chi\otimes\Phi_9$, which has Schmidt rank 288 and fits in nine qubits per party. The strategy of Theorem 14 has loss $2\mathcal{E}(\chi)/243<\gamma$, since $330\sqrt{2}<469$. A circuit can prepare the six shell weights, form the uniform superposition within each shell controlled by its index, and copy the computational index into the second register before distributing the registers. The leading nonzero bit gives the shell index for reversible erasure.

A finite state separated by the two bounds. For an integer $K\geq 5$ set $\theta=\pi/(K+3)$, $x_k=\sin((k+2)\theta)$ for $1\leq k\leq K$, $x_0=\sqrt{2}x_1$, and $x_{K+1}=0$. A rank-2^K state χ has normalised Schmidt amplitudes proportional to x_0 at rank 1 and to $x_k/\sqrt{2^{k-1}}$ on the ranks $2^{k-1}<i\leq 2^k$. These amplitudes are decreasing because $x_1\leq x_0$ and $x_{k+1}/x_k\leq(k+3)/(k+2)\leq 4/3$ for $k\geq 1$, by monotonicity of $\sin u/u$ on $(0,\pi)$. Putting $A=\sum_{k=0}^{K}x_k^2$, the duplication formula yields

$$A\mathcal{E}(\chi)=(\sqrt{2}-1)^2x_1^2+\sum_{k=2}^{K+1}(x_k-x_{k-1})^2=2(1-\cos\theta)\sum_{k=1}^{K}x_k^2+\big[\sin(2\theta)/\sin(3\theta)-(2\sqrt{2}-2)\big]x_1^2.$$

The first equality follows by summing the square of $(1-D)\lambda$ over the dyadic rank intervals. The second uses the sine recurrence with fictitious left value $\sin(2\theta)$. Since $\cos\theta>9/10$, one has $\sin(2\theta)/\sin(3\theta)=2\cos\theta/(4\cos^2\theta-1)<45/56<2\sqrt{2}-2$. Hence $\mathcal{E}(\chi)\leq\theta^2$. Prepare $\psi=\chi\otimes\Phi_9$ so that the ninefold amplitudes are part of the target spectrum. For $K=9$, this shared state fits in 13 qubits per party, and the measurements above give

$$\mathcal{B}(p)=\frac{2}{243}\mathcal{E}(\chi)\leq\frac{2\pi^2}{243\cdot 12^2}<5.65\times10^{-4}<\gamma.$$

The Schmidt basis can be the computational basis. A circuit can prepare the shared shell state by preparing the $K+1$ shell weights, forming a uniform superposition within each shell controlled by the shell index, and copying the computational index into the second register before distributing the registers. The shell index can be erased reversibly because the leading nonzero bit gives the shell. This circuit and the local measurements use a number of elementary gates polynomial in K, at fixed precision. The inequality for $L^{1/2}$ also implies that a phase-aligned preparation error of at most 2×10^{-3} in vector norm still gives loss below γ with these measurements.

5 Finite-chain separations

Three constructions distinguish finite critical-chain resources from every maximally entangled state. The original game has an assisted certificate at $N = 2^{19}$, using the chain together with one shared nine-dimensional maximally entangled register, and a chain-only certificate at $N = 2^{84}$. A four-spin construction uses a tagged tilted CHSH test and the chain ground state alone.

5.1 The original game with a shared nine-dimensional register

Theorem 17 (Assisted finite XX separation)**.** *Let ψ_N be the ground state of the half-filled open XX chain on N sites, cut in the middle. In the original uniform game G, $\psi_{2^{19}} \otimes \Phi_9$ admits a strategy with loss below* 0.000703*, and $\psi_{2^{20}} \otimes \Phi_9$ admits one with loss below* 0.000663*. Both losses are strictly below $\gamma = (3-2\sqrt{2})/243$, the floor of Theorem 16 for every maximally entangled state of arbitrary finite rank and arbitrary local POVMs.*

Proof. The strategy of Theorem 14 on $\psi_N \otimes \Phi_9$ has loss exactly $2\mathcal{E}(\psi_N)/243$. Appendix C.1 bounds this expression using the exact finite XX covariance, certified quadrature and spectral errors, and directed integer rank calculations. The resulting upper bounds are 0.000702783850 and 0.000662385080, respectively. □

The auxiliary register is part of the shared input. Providing the same register to a maximally entangled competitor gives $\Phi_D \otimes \Phi_9 \simeq \Phi_{9D}$. The prescribed assisted strategy beats γ exactly when

$$\mathcal{E}(\psi_N) < \frac{3-2\sqrt{2}}{2} = 0.0857864376\ldots.$$

The numerical estimates for $\mathcal{E}(\psi_N)$ at $N = 2^{18}, 2^{19}, 2^{20}$ round to 0.0908, 0.0854, and 0.0805, respectively. The certified intervals in Table 6 lie above the threshold at $N = 2^{18}$ and below it at the other two sizes.

5.2 An explicit finite chain for the original game

Theorem 18. *The ground state of the open XX chain of length $N = 2^{84}$, cut in the middle, has a strategy in the uniform game G with loss below* 0.000461*. The chain state is the strategy's sole shared resource. This loss is strictly below the floor for every maximally entangled state in Theorem 16.*

The chain length is far beyond the sizes of the numerical sweep. The certificate uses a 1418×1418 approximation to one parity block of the exact cross covariance and integer rank counts for a finite model of its Schmidt spectrum. The dimension of the covariance calculation grows with the number of dyadic intervals in the half-chain. The model strategy has loss below 0.000410720762. Propagating the spectral approximation and arithmetic bounds gives the certified loss bound in the theorem for the physical ground state. Appendix C gives the complete calculation.

5.3 A four-spin separation with tilted CHSH

The four-spin construction realizes the known tilted CHSH separation on an XX-chain ground state [74, 75]. For $0 < \alpha \leq 2$, define T_α with binary answers $a, b \in \{-1, 1\}$. Each of the four pairs $(x, y) \in \{0, 1\}^2$ is asked with probability $1/(4+\alpha)$ and wins when $ab = (-1)^{xy}$. The pair $(0, *)$ is asked with probability $\alpha/(4+\alpha)$ and wins when $a = 1$. The question 0 is the same in both rules. If A_x, B_y denote the differences of the two answer effects, the loss is $(4 + \alpha - S_\alpha)/(2(4+\alpha))$ where

$$S_\alpha = \langle \alpha A_0 + A_0(B_0 + B_1) + A_1(B_0 - B_1) \rangle \,. \tag{34}$$

Let H_α choose G with probability $1/10$ and T_α with probability $9/10$. Each player receives the branch tag as part of the question.

Theorem 19. *The ground state of the four-site open XX chain, cut after site two, has a strategy on H_1 with loss less than 0.177. Every maximally entangled state, of any finite rank and with arbitrary local POVMs, has loss at least 0.180. The ground state is the sole shared resource, and the strategy plays G classically. The same loss floor holds for mixtures of maximally entangled states.*

Proof. On Φ_d, absorb Bob's transpose into his effects and write $\tau = \mathrm{Tr}\,/d$. For fixed B_0, B_1, maximizing over the selfadjoint contractions A_0, A_1 allows them to be reflections. Indeed, $\max_{-I\le A\le I} \tau(AK) = \tau|K|$ for every selfadjoint K, and the maximum is attained by the spectral sign of K with either sign on its kernel. Optimizing over Bob's two contractions now yields

$$\alpha\tau(A_0) + \tau|A_0 + A_1| + \tau|A_0 - A_1|.$$

The selfadjoint operator $K = (A_0A_1 + A_1A_0)/2$ commutes with the reflections. On its k eigenspace, $|A_0 \pm A_1| = \sqrt{2 \pm 2k}\, I$. For $-1 < k < 1$, the reflection $(A_1 - kA_0)/\sqrt{1-k^2}$ anticommutes with A_0, so $\mathrm{Tr}\, A_0 = 0$ on this eigenspace. Its contribution per dimension is at most $2\sqrt{2}$. At $k = \pm 1$ the contribution is at most $2 + \alpha$. Thus

$$S_\alpha(\Phi_d) \le \max\{2\sqrt{2}, 2 + \alpha\}. \tag{35}$$

Local effects can always be compressed to the Schmidt supports, so this argument covers arbitrary local POVMs. Linearity for fixed measurements extends the corresponding loss bound to mixtures of flat states.

Use the hopping convention $-\sum_{j=1}^{3}(a_j^* a_{j+1} + a_{j+1}^* a_j)$ for the open XX chain after Jordan–Wigner transformation. Reversing the hopping sign is an on-site unitary change of basis. The occupied standing waves are $v_1 = \sqrt{2/5}\,(s, t, t, s)$ and $v_2 = \sqrt{2/5}\,(t, s, -s, -t)$, where $s = \sin(\pi/5)$ and $t = \sin(2\pi/5)$. In the left half their covariance is

$$\frac{2}{5}\begin{pmatrix} s^2 + t^2 & 2st \\ 2st & s^2 + t^2 \end{pmatrix} = \begin{pmatrix} 1/2 & 1/\sqrt{5} \\ 1/\sqrt{5} & 1/2 \end{pmatrix}.$$

In particular $\nu = 1/2 + 1/\sqrt{5}$ is its larger eigenvalue. Define the left orbitals $\ell_1 = (1, 1)/\sqrt{2}$, $\ell_2 = (-1, 1)/\sqrt{2}$ and the right orbitals $r_1 = (1, -1)/\sqrt{2}$, $r_2 = (1, 1)/\sqrt{2}$. The occupied vectors $(v_1 + v_2)/\sqrt{2}$ and $(v_1 - v_2)/\sqrt{2}$ are respectively $\sqrt{\nu}\,\ell_1 + \sqrt{1-\nu}\,r_1$ and $\sqrt{1-\nu}\,\ell_2 + \sqrt{\nu}\,r_2$.

Use local occupation bases in these orbitals with left creation operators ordered before right ones. If (n_1, n_2) are the left occupations in the two occupied orbital pairs, the right occupations are $(1 - n_1, 1 - n_2)$ and the Slater determinant has phase $(-1)^{(1-n_1)n_2}$ in that component, up to an overall phase. The local basis changes

$$|n_1 n_2\rangle_A \longmapsto (-1)^{(1-n_1)n_2}|1 - n_1, n_2\rangle_A, \qquad |m_1 m_2\rangle_B \longmapsto |m_1, 1 - m_2\rangle_B$$

therefore give exactly two Schmidt qubit pairs, both in $|\xi\rangle = \sqrt{\nu}|00\rangle + \sqrt{1-\nu}|11\rangle$. On the first pair measure $A_0 = Z$, $A_1 = X$ and $B_0 = (\sqrt{5}Z + X)/\sqrt{6}$, $B_1 = (\sqrt{5}Z - X)/\sqrt{6}$. Its three expectations $\langle Z \otimes I\rangle$, $\langle Z \otimes Z\rangle$, and $\langle X \otimes X\rangle$ are $2/\sqrt{5}$, 1, and $1/\sqrt{5}$. The score in (34) is consequently

$$S_\alpha(\psi_4) \ge \frac{2\sqrt{6} + 2\alpha}{\sqrt{5}}. \tag{36}$$

The other pair is ignored. In the G branch use the deterministic strategy of Theorem 14. With $\alpha = 1$ the total loss on ψ_4 is at most

$$\frac{1}{270} + \frac{9}{100}\left(5 - \frac{2\sqrt{6} + 2}{\sqrt{5}}\right) < \frac{177}{1000}.$$

For example, $\sqrt{6} > 2449/1000$ and $\sqrt{5} < 2237/1000$ show $(2\sqrt{6} + 2)/\sqrt{5} > 37/12$. Equation (35) gives loss at least $(9/100)(5 - 3) = 9/50$ on every Φ_d. □

Question weights. The separation also holds with the rational weight $\alpha = 5/6$. One can draw the four CHSH pairs with six units of weight each and the marginal pair with five units. The tracial score bound is $17/6$ and the four-site strategy gives score at least $(2\sqrt{6} + 5/3)/\sqrt{5}$. The total loss is less than 0.181 on the ground state and at least $27/145 > 0.186$ on all flat states. For the fixed four-site observables in the proof, the unnormalized score gap over (35) is largest at $\alpha = 2\sqrt{2} - 2$.

Stability under preparation error. For either question distribution, write L_{flat} for the floor from (35) and L_4 for the achieved four-site loss. The exact formulas give $L_{\mathrm{flat}} - L_4 > 0.0039$ at $\alpha = 1$ and $L_{\mathrm{flat}} - L_4 > 0.0058$ at $\alpha = 5/6$. Both inequalities follow by using $\sqrt{6} > 2449489/10^6$ and $\sqrt{5} < 2236068/10^6$ in the score. Trace distance η from the four-site state raises the loss of its fixed strategy by at most η. Thus the separation survives preparation error in trace distance up to 0.0039 or 0.0058, respectively.

The optimal benchmark and the type obstruction. Acín, Massar and Pironio give the sharp tilted CHSH score $\sqrt{8+2\alpha^2}$ [74], and Bamps and Pironio give sum-of-squares decompositions [75]. For commuting local reflections A_0, A_1, B_0, B_1, set $C = (B_0 + B_1)/2$, $D = (B_0 - B_1)/2$, $U = (A_0A_1 + A_1A_0)/2$ and $V = (A_0A_1 - A_1A_0)/(2i)$. Then $C^2 + D^2 = I$, $CD + DC = 0$, and $U^2 + V^2 = I$. If W is the operator in (34),

$$W^2 = (4+\alpha^2)I + R, \qquad R = 4\alpha C + 4\alpha UD + 8V(iCD).$$

The three terms of R anticommute pairwise. For the commuting positive contractions $u = U^2$ and $c = C^2$ this gives

$$\frac{R^2}{16} = \alpha^2 u + (1-u)\big[\alpha^2 c + 4c(1-c)\big] \le \frac{(4+\alpha^2)^2}{16} I.$$

The scalar bound follows by maximizing the quadratic on $0 \le c \le 1$ and using $\alpha^2 \le (4+\alpha^2)^2/16$. Hence $\|W\|^2 \le 8 + 2\alpha^2$. Every selfadjoint contraction is an average of its spectral threshold reflections over thresholds in $[-1, 1]$. Since W is affine in each observable, the same score bound holds for contractions in commuting local von Neumann algebras. Equality for $0 < \alpha < 2$ is achieved on $\cos\theta|00\rangle + \sin\theta|11\rangle$ with $\sin(2\theta) = \sqrt{(4-\alpha^2)/(4+\alpha^2)}$, using $A_0 = Z$, $A_1 = X$ and $B_{0,1} = (Z \pm \sin(2\theta)X)/\sqrt{1+\sin^2(2\theta)}$. The endpoint $\alpha = 2$ is attained on a product state.

Consequently the loss of H_α is at least

$$L^*_\alpha = \frac{9}{10}\,\frac{4 + \alpha - \sqrt{8+2\alpha^2}}{2(4+\alpha)}.$$

Equality forces a perfect strategy in the G branch. In particular, the original obstruction to exact play in the presence of a semifinite local trace applies to attaining this benchmark. For $0 < \alpha < 2$, the ground states of the finite open XX chains approach L^*_α as $N \to \infty$. The G branch approaches perfect play by Theorem 4. For T_α, the exact finite-chain one-body entanglement energies contain a mode tending to the strictly positive energy $\log\cot^2\theta$. Indeed, in any fixed neighbourhood of this energy there are order $\log N$ mesh entries from Corollary 40. The bounded sum of singular-value errors forces at least one true mode in the neighbourhood for all sufficiently large N. Measuring that mode approaches the optimal two-qubit score. The two branch strategies can be selected independently by the question tags.

6 Critical chains

The model. The XX chain is $H = \frac{1}{2}\sum_j(\sigma^x_j\sigma^x_{j+1} + \sigma^y_j\sigma^y_{j+1})$, which the Jordan–Wigner transformation [33] turns into free fermions hopping on the chain, $H = \sum_j(a^*_j a_{j+1} + a^*_{j+1}a_j)$, the model of Ref. [11, Example 23]. The fermionic ground state ω on the infinite chain is gauge-invariant and quasi-free, with two-point function

$$\omega(a^*_x a_y) = C_{xy} = \frac{1}{2\pi}\int_{\pi/2}^{3\pi/2} e^{ik(x-y)}\,dk, \tag{37}$$

the filled Fermi sea of the modes with negative energy $2\cos k$. On the two-sided infinite chain we use the Jordan–Wigner transformation of Refs. [34, 35]. The spin ground state ω_{spin} is even and agrees with ω on even elements. Alice's half is $\{1, 2, \dots\}$ and Bob's is $\{\dots, -1, 0\}$. Conjugating every other spin by σ^z and rescaling by a positive constant gives the zero-field isotropic XY Hamiltonian in the convention of Refs. [5, 72]. Matsui proved that its ground-state half-chain factors are type III_1 [5, Theorem 4.3]. Keyl, Matsui, Schlingemann and Werner record this result and establish Haag duality for the physical half-chain pair [72, Corollary 6.8 and Proposition 6.9].

More generally, van Luijk, Stottmeister and Wilming consider translation-invariant free-fermion Hamiltonians with continuous finite-matrix symbols for which the positive-energy spectral projection is piecewise continuous and has at least one jump [11, Assumption 20 and Definition 21]. Their theorem gives a pure, gauge-invariant quasi-free ground state with type-III_1 half-chain factors [11, Theorem 25]. The physical

bipartition is into complementary left and right half-chains. The fermionic parties use the parity-even observable algebras on the even GNS sector. In the Jordan–Wigner spin dual they use the full spin algebras of the two halves. The physical algebras are each other's commutants in both representations [11, Proposition 8 and Remark 10].

Proof of Theorem 4 for the infinite chains. In a bipartite system given by a type-III_1 factor with separable predual and its commutant, every unit vector is bipartite embezzling [13, Theorems 11 and 63]. Finite truncations of a van Dam–Hayden state give finite-dimensional strategies with loss approaching zero by Theorem 14. Embezzle the state of one of these strategies into two initially pure local ancillas to within vector distance ε. Conjugating the target measurements by the local embezzling unitaries and compressing to the initial ancilla vectors gives POVMs in the physical algebras. The induced loss differs from the target loss by at most 2ε, since the loss operator is a contraction. This gives value one at every unit vector in each of the ground-state sectors above, with the stated half-chain bipartition. □

Proposition 20 (A modular obstruction)**.** *Let the marginal φ of a unit vector Ω on M_A be faithful. If 2 is absent from the point spectrum of Δ_φ on $\overline{M_A\Omega}$, every POVM strategy for G in the commuting algebras M_A, M_B has positive loss.*

Proof. By Lemma 6, perfect play would give $\Delta_\varphi^{it} w_j\Omega = 2^{it} w_j\Omega$ for every real t, with $\|w_j\Omega\|^2 = \varphi(F_j) = 1/9$. This would give a nonzero modular eigenvector of eigenvalue two. Van Luijk, Stottmeister, Werner and Wilming established the point-spectrum obstruction for exact embezzlement of ancillary states [13, Proposition 35 and Corollary 36]. □

Proof of Theorem 5. With Jordan–Wigner strings starting at site 1, the right half-chain algebra $\mathfrak{A}_R$ is isomorphic to the algebra of canonical anticommutation relations over $\ell^2(\mathbb{N})$. Under this isomorphism the restriction of ω_{spin} is the gauge-invariant quasi-free state ω_C with $C = (C_{xy})_{x,y\geq 1}$ from (37). This C is the Toeplitz operator on $\ell^2(\mathbb{N})$ whose symbol is the indicator function of the arc $(\pi/2, 3\pi/2)$. A Toeplitz operator with a real, bounded, non-constant symbol has no eigenvalues [36]. In particular C and $1 - C$ are injective. This makes the Araki–Wyss vacuum separating in the cyclic representation of ω_C. The compression of the physical factor M_A to $\overline{M_A\Omega}$ is a nonzero normal representation of a factor and hence faithful. Thus the physical marginal φ is faithful. In the Araki–Wyss representation the cyclic space is the antisymmetric Fock space over $\ell^2(\mathbb{N}) \oplus \overline{\ell^2(\mathbb{N})}$. The modular unitary group is the second quantization of the one-particle group with generator $K \oplus (-\overline{K})$, up to an overall sign convention, where K is the Borel function $s \mapsto \ln(s/(1-s))$ of C [37–40]. The endpoint spectral projections of C vanish, and the scalar function is injective on $(0, 1)$. It follows that the one-particle generator has no eigenvalues.

In fact the only modular eigenvectors are multiples of Ω, with eigenvalue one. On the full n-fold tensor product, $n \geq 1$, the generator is a sum of commuting copies of the one-particle generator. In a product vector its spectral measure is a convolution of atomless measures, hence atomless. Product vectors are total, so every singleton spectral projection vanishes. The antisymmetric subspace reduces the group and therefore has no eigenvalues either.

Proposition 20 now excludes perfect play. □

Lemma 21 (Local approximation of measurements)**.** *Let $\mathfrak{A}_1 \subset \mathfrak{A}_2 \subset \cdots$ be finite-dimensional unital $*$-subalgebras of a von Neumann algebra M on a separable Hilbert space whose union is strongly dense in M. For every POVM $(A_a)_{a=1}^n$ in M, every vector Ω and every $\delta > 0$ there are k and a POVM $(A'_a)_{a=1}^n$ in $\mathfrak{A}_k$ with $\|(A_a - A'_a)\Omega\| < \delta$ for all a.*

Proof. By the Kaplansky density theorem, and because positive contractions of the norm closure of $\bigcup_k \mathfrak{A}_k$ are norm limits of positive contractions in the $\mathfrak{A}_k$, there are positive contractions $T_a \in \mathfrak{A}_k$ converging strongly to A_a as $k \to \infty$. Then $S = \sum_a T_a$ converges strongly to 1 and $0 \leq S \leq n$. Let $\gamma(s) = \max(s, \frac{1}{2})^{-1/2}$. Continuous functions act strongly continuously on bounded sets of self-adjoint operators [41], so $\gamma(S) \to 1$ and $\gamma(S)T_a\gamma(S) \to A_a$ strongly. These operators lie in $\mathfrak{A}_k$, are positive, and sum to $S\gamma(S)^2 \leq 1$. Adding the deficit $1 - S\gamma(S)^2 = (1 - 2S)_+$, which tends strongly to zero, to the first of them gives the POVM (A'_a). □

Proof of Theorem 4 for finite chains. Label the sites of the open chain by $x = -N/2 + 1, \ldots, N/2$, with the cut between 0 and 1 and N even, so that the ground state ψ_N is unique. Its two-point function is $C^{(N)}_{xy} = \frac{2}{N+1} \sum_{k>N/2} \sin(\theta_k i) \sin(\theta_k j)$ with $\theta_k = \pi k/(N+1)$, $i = x + N/2$ and $j = y + N/2$. Writing the product of sines as a difference of cosines, the term with $i - j = x - y$ is a Riemann sum for (37), and the term with $i + j = N + x + y$ is $\frac{1}{N+1} \sum_k (-1)^k \cos(\theta_k(x + y - 1))$, an alternating sum of a smooth function,

which is $O(1/N)$. Hence $C_{xy}^{(N)} \to C_{xy}$ for fixed x, y. Both states are gauge-invariant and quasi-free, so by Wick's theorem all correlation functions in a fixed window $\{-k+1, \ldots, k\}$ converge. In the spin picture both states are even, and even operators of the window are even operators of the fermions in the window. Therefore the reduced density matrices of ψ_N on the window converge in norm to that of ω_{spin}.

Fix $\delta > 0$. At the infinite ground state choose embezzling unitaries $U_A \in M_A \overline{\otimes} M_2$ and $U_B \in M_B \overline{\otimes} M_2$ with squared error below δ. Each unitary is the exponential of a bounded self-adjoint operator. Kaplansky density, followed by exponentiation, approximates it strongly by unitaries supported on finitely many sites and the local ancilla. Their joint error can therefore be made less than 2δ. Expanding this squared norm expresses it as the expectation of a fixed bounded operator on the finite window and the two ancillas. Local convergence of ψ_N then gives the same error below 3δ for all sufficiently large even N. Hence $\mathcal{E}(\psi_N) \to 0$ along the full sequence. The upper bound in Theorem 3 gives $\omega(\psi_N) \to 1$. □

This local approximation argument is the mechanism behind the finite-family results in Ref. [11]. It uses convergence of all fixed-window states and does not require the robust lower constant C. The same argument applies to any locally convergent finite sequence whose limiting physical pair is bipartite embezzling.

7 Gapped chains and singlet products

Theorem 22 (An entropy bound)**.** *Every pure bipartite state of finite entanglement entropy S satisfies*

$$\mathcal{E} \geq \frac{(\ln 2)^2}{4(S + 1 + 3\ln 2)^2}.$$

Consequently, for a uniformly gapped family of one-dimensional Hamiltonians with fixed local dimension, uniformly bounded interaction range and strength, and unique ground states, the game loss is bounded below uniformly in the chain length.

Proof. The inequality is (25). The area law bounds S by a constant depending on the uniform gap, interaction strength and local dimension [3, 42]. Apply Theorem 3. □

Uniformly bounded finite-interval entropy also implies a type I half-chain in the pure infinite state [4]. In the corresponding tensor-product representation the state has a normalizable Schmidt sequence, so $\mathcal{E} > 0$ and the game has a strictly positive loss floor there as well.

Theorem 23 (The same entropy law, a different type)**.** *For every $c > 0$ there is a pure spin-chain state whose centered restrictions have $S_N = (c/6)\ln N + O_c(1)$ and $\mathcal{E} = 2 - \sqrt{2}$, and whose infinite half-chains have type II_∞. Its finite and infinite game losses are at least γ. The state can be the unique ground state of a gapless, inhomogeneous interaction that decays exponentially with the distance between the interacting sites.*

Proof. Put $P = \{\lceil 2^{6k/c} \rceil : k = 0, 1, \ldots\}$ and place a singlet on every pair $(1-x, x)$ with $x \in P$. Fix every unused spin in a pure one-site state. For even N, the centered interval contains n_N singlet pairs, all crossing the cut. Counting the indices k with $\lceil 2^{6k/c} \rceil \leq N/2$ gives $\lfloor c \ln(N/2)/(6\ln 2) \rfloor + 1$. Counting distinct sites subtracts the repetitions from rounding, whose total is finite for fixed c because successive unrounded values eventually differ by more than one. Thus $n_N = c\ln N/(6\ln 2) + O_c(1)$. The nonzero Schmidt probabilities are all 2^{-n_N}, so $S_N = n_N \ln 2 = (c/6)\ln N + O_c(1)$ and $\mathcal{E} = 2 - \sqrt{2}$. Theorem 16 gives the finite loss floor.

In the infinite right half, the infinitely many singlet endpoints generate the hyperfinite II_1 factor in its tracial representation. The unused spins generate a type I_∞ factor in their pure representation. Their tensor product is type II_∞, and the tensor-product commutation theorem gives Haag duality. The singlet algebras act by left and right multiplication on $L^2(R, \tau)$ for the hyperfinite II_1 factor R. Compress both unused type I factors to their pure vectors. The remaining POVMs have correlations $\tau(AB)$, so the normalized-trace proof of Theorem 16 gives the same γ.

For a parent interaction, put $e^{-\alpha(2x-1)}(1 - |s\rangle\langle s|)$ on each pair $(1-x, x)$, where $|s\rangle$ is its singlet and $\alpha > 0$, and a one-site projector fixing each unused spin. These disjoint positive terms have exactly the stated product ground state. On any finite union of pairs and unused sites, local ground-state stability forces the unique zero-energy product state; these restrictions determine the infinite state. The pair couplings tend to zero, giving gaplessness, while their norms decay exponentially in their diameters. □

State or family	Infinite half-chain type	Finite entropy	Game behavior
Sparse crossing singlets	II_∞	$(c/6)\ln N + O_c(1)$	Loss at least γ
Uniformly gapped chain	I	Bounded	Uniform positive loss
Critical XX chain	III_1	$S_N \sim (\ln N)/6$	Finite loss tends to zero; infinite vacuum value one is not attained
XX with the quartic impurity of Section 8	III_1	Not used in the proof	Finite loss tends to zero; every sector vector has value one
Free massive field, complementary wedges	III_1	No finite reduced-density entropy	Vacuum value one is not attained

Table 2: The type column refers to the infinite system. Every finite spin chain has type I local algebras. A score for one finite state does not by itself determine an infinite-system type.

Random-singlet fixed-point states provide a familiar physical version of the entropy contrast [25, 44, 45]. Liu and Yi already obtain several half-chain types with the same $\Theta(\ln \ell)$ interval-entropy order in translation-invariant pure states [63]. Their examples have superpolynomially decaying parent interactions. Their remaining locality question concerns the interacting exponentially local class, rather than translation invariance alone.

8 Interacting bulk chains and local perturbations

The same spectral law holds for the XXZ corner-transfer-matrix spectrum near the isotropic point. At fixed critical coupling, explicit finite-depth circuit Hamiltonians retain the type-III_1 half-chain structure, and a cut-preserving family retains the entire finite Schmidt spectrum. A quartic impurity provides a further nonquasifree extension.

8.1 The XXZ chain approaching the isotropic point

The spin-$\frac{1}{2}$ antiferromagnetic XXZ chain has Hamiltonian

$$H_\Delta = \sum_{j\in\mathbb{Z}} \big(X_j X_{j+1} + Y_j Y_{j+1} + \Delta Z_j Z_{j+1}\big).$$

The $Z_j Z_{j+1}$ term gives a nearest-neighbour density interaction under the Jordan–Wigner map. It acts throughout the chain. For $\Delta > 1$, there are two boundary-selected thermodynamic ground states with Néel order [73, Proposition 1.3 and Theorem 1.4]. Peschel, Kaulke and Legeza describe the half-chain spectrum using corner transfer matrices (CTMs) [51, Sec. 3]. In the symmetry-broken Néel convention, the CTM Schmidt probabilities at a single cut are [49, Eqs. (30)–(37)]

$$p_{\boldsymbol{\nu}} = \mathcal{Z}(t)^{-1} \exp\left[-t\sum_{j=1}^{\infty} j\nu_j\right], \qquad \mathcal{Z}(t) = \prod_{j=1}^{\infty}(1+e^{-tj}), \qquad t = 2\eta, \quad \eta = \operatorname{arcosh}\Delta, \tag{38}$$

Here $\boldsymbol{\nu} = (\nu_j)_{j\geq 1}$ ranges over binary occupation sequences of finite support. Expanding the convergent product $\mathcal{Z}(t)$ sums the unnormalized weights. We take this spectrum as input and write $\psi_t = \sum_{\boldsymbol{\nu}} \sqrt{p_{\boldsymbol{\nu}}}\, |\boldsymbol{\nu}\rangle \otimes |\boldsymbol{\nu}\rangle$ for the corresponding bipartite state in the occupation bases. Alba, Calabrese and Tonni give the exact count in terms of partitions into distinct parts, its large-level asymptotic, and its near-critical Bessel approximation [49, Eqs. (39)–(42) and (47)].

Theorem 24 (Near-critical law for the XXZ CTM spectrum)**.** *Let S and $\mathcal{E}_d$ be the entropy and squared*

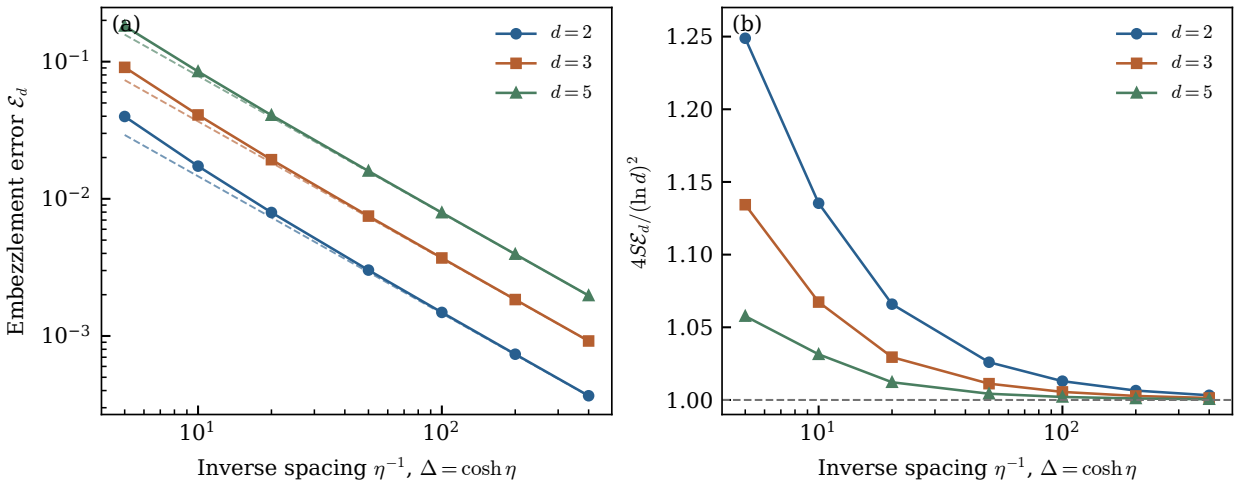


Figure 4: The XXZ corner-transfer-matrix spectrum in the symmetry-broken convention. (a) The error for targets $d = 2, 3, 5$, computed with exact integer level multiplicities. Dashed curves show $3\eta(\ln d)^2/\pi^2$. (b) The product $4S\mathcal{E}_d/(\ln d)^2$ approaches one. Moving right sends $\eta = \operatorname{arcosh}\Delta$ to zero. The omitted rank tail contributes less than 10^{-15} at each point. The plotted values use floating-point evaluation.

local-unitary embezzlement error of ψ_t with spectrum (38). For each fixed integer $d \geq 2$, as $\eta = t/2 \downarrow 0$,

$$S = \frac{\pi^2}{12\eta} - \frac{\log 2}{2} + O\Big(\eta^{-1} e^{-\pi^2/\eta}\Big), \tag{39}$$

$$\mathcal{E}_d = \frac{3\eta(\log d)^2}{\pi^2} + O_d(\eta^2), \tag{40}$$

$$\mathcal{E}_d S = \frac{(\log d)^2}{4} + O_d(\eta). \tag{41}$$

The symmetric CTM spectrum adds the binary mode $j = 0$. Its error is identical at every η and its entropy is larger by $\log 2$.

Calabrese and Cardy obtained the leading entropy divergence in the symmetric convention [6, Eq. (82)]. The constant in the symmetry-broken convention follows from the Rényi expansion of Alba, Calabrese and Tonni [49, Eq. (45)]. The exponential correction follows from the resummed entropy formula of Calabrese, Cardy and Peschel [7, Eq. (19)]. Appendix H derives (39) from the modular identity and verifies the counting criterion for the exact spectrum, with all level multiplicities included. The correlation length of the XXZ chain satisfies $\log \xi = \pi^2/(2\eta) + O(1)$ [49, Eq. (44)]. The CTM error also has the form

$$\mathcal{E}_d = \frac{3(\log d)^2}{2\log \xi} + O_d\big((\log \xi)^{-2}\big).$$

Corollary 25 (Game loss for the XXZ CTM spectrum)**.** *On ψ_t, the measurements of Theorem 14 have losing probability*

$$\mathcal{B}(p) = \frac{2\eta(\log 2)^2}{81\pi^2} + O(\eta^2).$$

The optimal game loss on ψ_t is $\Theta(\eta)$, and the game value tends to one as $\eta \downarrow 0$.

Proof. The largest weight in (38) is $p_1 = \exp[-\pi^2/(24\eta) + O(1)]$. The estimate following Theorem 14 gives $\mathcal{B}(p) = (2/243)\mathcal{E}_2 + O(\sqrt{p_1\mathcal{E}_2} + p_1)$. Substitute (40). The lower bound in Theorem 3 gives the order of the optimal loss. □

Proposition 26 (Wick violation in the XXZ Néel phases)**.** *For $1 < \Delta \leq 21/20$, the Jordan–Wigner fermion state of each boundary-selected Néel phase violates the gauge-invariant Wick identity on two neighboring sites.*

Proof. The two boundary limits of Ref. [73, Proposition 1.3] are $U(1)$-invariant because the finite ground vectors have definite total Z spin. The staggered Z rotation in their Eq. (1.20) leaves Z_j unchanged and conjugates translation to translation followed by a uniform Z rotation. Thus one-site translation still

exchanges the two phases in the convention of H_Δ. Global spin flip exchanges the boundary fields and hence the limiting states. It leaves the bond Hamiltonian invariant and sends the even Jordan–Wigner hopping operator $a_j^* a_{j+1}$ to its adjoint. Consequently the local Z expectations alternate between m and $-m$, the bond energy e_Δ is independent of the bond, and $|\omega(a_j^* a_{j+1})| = \alpha$ is also independent of the bond.

Put $\eta = \operatorname{arcosh} \Delta$. In the finite-volume loop representation of these same states [73, Theorem 4.2 and Eqs. (4.7)–(4.10)], condition on the unoriented loops. The two orientations of the loop through a site have probabilities $e^{\pm\eta}/(e^\eta + e^{-\eta})$, with the signs exchanged for the opposite boundary condition. Reversing that orientation flips the site's Pauli Z value. Thus the conditional mean has absolute value $\tanh \eta$. Averaging and taking the boundary limits give $m^2 \le \tanh^2 \eta = 1 - \Delta^{-2}$.

On an interval of N sites, the fermionic covariance is a positive contraction. A diagonal unitary makes all nearest off-diagonal entries equal to α. Its trace against the open-chain adjacency matrix is then $2(N-1)\alpha$, at most the sum of the positive adjacency eigenvalues. These are the positive members of $2\cos(k\pi/(N+1))$, $1 \le k \le N$. Dividing by $2(N-1)$ and taking the limit gives $\alpha \le 1/\pi$.

The bond energy equals the thermodynamic ground energy per site. To see this, let H_I be the open Hamiltonian of an interval I, let B_I be the two bonds crossing its boundary, and reset I to an open-chain ground vector. Summing the ground-state condition $\omega(A^*[H_\Delta, A]) \ge 0$ over the reset channel's Kraus operators gives $\omega(H_I) \le E_0(H_I) + 2\|B_I\|$. The reverse bound is variational. At $\Delta = 1$ the exact Pauli-normalized energy is $e_1 = 1 - 4\log 2$ [76]. An $SU(2)$-averaged finite isotropic ground state gives one third of its isotropic energy to each spin component. Using it as a trial state for H_Δ and taking the thermodynamic limit gives $e_\Delta \le e_1(\Delta + 2)/3$.

The number relation $Z_j = 2a_j^* a_j - 1$ gives the Wick prediction $\omega(Z_j Z_{j+1}) = -m^2 - 4\alpha^2$. The hopping term has expectation at least -4α. The actual defect in the Wick identity therefore satisfies

$$\begin{aligned}\Delta\big(\omega(Z_j Z_{j+1}) + m^2 + 4\alpha^2\big) &\le e_1(\Delta+2)/3 + 4/\pi + 4\Delta/\pi^2 + \Delta - \Delta^{-1} \\ &\le \frac{61}{60}(1 - 4\log 2) + \frac{4}{\pi} + \frac{21}{20}\left(\frac{4}{\pi^2} + \frac{41}{441}\right) < 0.\end{aligned}$$

The first bound increases with Δ because $e_1 > -3$. The last strict inequality follows, for example, from $\log 2 > 0.6931$ and $\pi > 3.1415$. The $U(1)$ symmetry sets the anomalous two-point expectations to zero, so the defect excludes quasifreeness. □

8.2 Critical bulk Hamiltonians obtained by local circuits

There is also an exact critical bulk construction at fixed coupling. Write $h_j = (X_j X_{j+1} + Y_j Y_{j+1})/2$. Controlled-phase gates turn the XX interaction into density-dependent hopping throughout the chain. Two choices distinguish preservation of the algebraic type from preservation of the entire finite-chain Schmidt spectrum.

Theorem 27 (Critical interacting bulk circuits)**.** *The one-spin translation-invariant, finite-range interaction*

$$H_{\mathrm{bulk}} = \sum_{j \in \mathbb{Z}} Z_{j-1} h_j Z_{j+2}$$

has a pure gapless ground state whose physical half-chain factors are type III_1 *and satisfy Haag duality. The value of* G *is one at every vector in its ground-state sector. For its open finite-volume circuit ground states, the Bell-pair embezzlement error tends to zero and the game value tends to one.*

The period-two bulk interaction

$$H_{\mathrm{pair}} = \sum_{j \in \mathbb{Z}} \big(h_{2j+1} + Z_{2j+1} h_{2j+2} Z_{2j+4}\big)$$

has the same type and gaplessness properties. At the cut between 0 *and* 1*, it has exactly the XX ground state's game value and nonattainment property. On centred open intervals of* $4m$ *spins its Schmidt spectrum equals that of the XX chain. Consequently every finite-chain embezzlement law of Theorem 1 holds unchanged for these interacting ground states.*

Proof. Let $C_{j,j+1} = \mathrm{diag}(1, 1, 1, -1)$ on neighbouring spins. Conjugation by this gate sends X_j and Y_j to $X_j Z_{j+1}$ and $Y_j Z_{j+1}$, respectively, and fixes both Z operators. The gate commutes with h_j. The product

over all bonds defines an involutive automorphism α: the gates commute and can be applied in two layers of disjoint bonds. It commutes with one-site translation and satisfies

$$\alpha(h_j) = Z_{j-1} h_j Z_{j+2}.$$

The state $\omega_{\mathrm{XX}} \circ \alpha$ is a pure ground state of the conjugate dynamics. Its GNS Hamiltonian is unitarily equivalent to the XX GNS Hamiltonian and is therefore gapless.

At the cut, all gates except $C_{0,1}$ act separately in the two halves. Thus $\alpha = \mathrm{Ad}(C_{0,1}) \circ (\alpha_L \otimes \alpha_R)$. Proposition 47 shows that the represented half-chain pair is simultaneously unitarily conjugate to the XX pair. Its type, Haag duality and sector-wide value one follow. In a finite interval, conjugate the open XX Hamiltonian by the product of its internal bond gates. The resulting Hamiltonian agrees with H_{bulk} except for its boundary terms. Its ground state converges locally to $\omega_{\mathrm{XX}} \circ \alpha$. Finite-support approximations to the latter state's embezzling unitaries and game measurements, as in the proof of Theorem 4, transfer both limiting assertions to the finite intervals.

For H_{pair}, apply the gates only on the pairs $\{2j+1, 2j+2\}$. They fix the hopping within each pair and dress the interpair hopping by the two displayed Z factors. No gate crosses the indicated cut. The automorphism preserves both half-chain algebras, and each finite-volume circuit is a tensor product of a left and a right unitary. This preserves every Schmidt coefficient, every state-specific game value, and whether its supremum is attained. □

These interactions contain genuine higher-order terms in the original Jordan–Wigner fermions: the dressed hopping is multiplied by $(1-2n_{j-1})(1-2n_{j+2})$, producing quartic and sextic terms. Their exact circuit equivalence to XX explains why their critical spectral laws can be transferred without approximation. The XXZ CTM calculation supplies a different family of spectra to which the same counting criterion applies.

8.3 A quartic impurity in the XX representation

Write $a(f)$ for the fermionic annihilator smeared with $f \in \ell^2(\mathbb{Z})$. Let h be the single-particle nearest-neighbour hopping operator $(hf)_j = f_{j-1} + f_{j+1}$, and let δ_j denote a site basis vector. Put

$$f = h\delta_0/\sqrt{2}, \qquad g = h\delta_1/\sqrt{2}, \qquad V = a^*(f)a(f)a^*(g)a(g).$$

The two vectors are orthonormal. Thus V is a self-adjoint projection. It is even and its Jordan–Wigner image V_{s} is a spin operator supported on $[-1, 2]$. Let H_λ denote the spin XX interaction perturbed by λV_{s}, where $\lambda \in \mathbb{R}$.

Theorem 28 (A pure ground state in the XX sector)**.** *For $0 < |\lambda| < 1/\sqrt{8}$, H_λ has a pure ground state in the unperturbed XX GNS representation, distinct from the original ground state. Its two physical half-chains are type III_1 factors in Haag duality. The ground-state GNS Hamiltonian has positive spectrum accumulating at zero. In particular the value of the game G is one in this ground state and at every vector in its sector, by Theorem 4 and its sector-wide proof.*

Corollary 29 (A ground state that violates the quasifree description)**.** *For $0 < |\lambda| < \left(\frac{1}{2} - \frac{4}{3\pi}\right)^2/(4\sqrt{8})$, the ground state in Theorem 28 can be chosen gauge invariant with a pure, nonquasifree Jordan–Wigner fermionic state.*

Corollary 30 (Finite impurity chains)**.** *Under the coupling hypothesis of Theorem 28, every local subsequential limit of the centred even-length open-chain ground states is a pure ground state in the XX sector. Their finite-chain game value tends to one for the game G. Their Bell-pair embezzlement error also tends to zero.*

The choice $f = h\delta_0/\sqrt{2}$, $g = h\delta_1/\sqrt{2}$ suppresses the interaction at the Fermi points. A pull-through estimate bounds the expected number of particle–hole excitations by $8\lambda^2$, uniformly in the volume. This retains a nonzero overlap with the free vacuum and places an infinite ground state in the same irreducible representation. Compactness of the excitation vectors gives the finite-chain conclusion. The four-excitation component of the quartic interaction then excludes a quasifree description at small nonzero coupling. Full proofs are in Appendix I.

9 Quantum fields

Corollary 31 (Vacuum of a free field)**.** *Let Ω be the vacuum of the free scalar field of mass $m > 0$ in $d+1$ space-time dimensions, with $d \geq 1$. If Alice and Bob measure in complementary wedges, then $\omega(\Omega) = 1$ and every strategy loses with positive probability. For any split pair $M_A \subseteq \mathcal{N} \subseteq M_B'$ with $\mathcal{N}$ a type I factor, the loss of every strategy is at least $\mathcal{E}(\rho_{\mathcal{N}})/C > 0$, where $\mathcal{E}(\rho_{\mathcal{N}})$ is the Bell-pair error (9) evaluated on the eigenvalues of the vacuum density matrix on $\mathcal{N}$.*

Proof. The vacuum is cyclic and separating for the wedge algebra by the Reeh–Schlieder property and locality [10]. The Bisognano–Wichmann theorem identifies complementary wedge algebras as commutants and the vacuum modular group with the Lorentz boosts [54]. The wedge algebra is a factor of type III_1 [10, 55]. Van Luijk, Stottmeister, Werner and Wilming proved bipartite embezzlement for the wedge vacuum [13, Theorem 97] and reported it in Ref. [14]. The ancilla argument in the proof of Theorem 4 gives value one for G.

For the free massive scalar field, write a point of the positive mass shell as

$$p^0 = m_\perp \cosh\theta, \qquad p^1 = m_\perp \sinh\theta, \qquad m_\perp = \sqrt{m^2 + |p_\perp|^2}.$$

Then $d^d p/(2p^0) = \frac{1}{2}\, d\theta\, d^{d-1}p_\perp$, and a boost in the wedge direction translates the rapidity θ. Fourier transformation in θ identifies the one-particle boost generator with multiplication by a real coordinate on a space with Lebesgue measure, so its vector spectral measures have no atoms. On each positive-particle sector of symmetric Fock space the generator is the restriction of a sum of one-particle generators. Product vectors have convolution spectral measures, which are atomless, and are total in the unsymmetrized tensor power. The spectral projections onto singletons therefore vanish on each positive-particle sector. Only the vacuum line contributes a modular eigenvector. The Bisognano–Wichmann identification of the vacuum modular group with the boosts thus excludes 2 from the point spectrum of the modular operator. Proposition 20 rules out perfect play.

For a split pair, $\mathcal{H} = \mathcal{H}_1 \otimes \mathcal{H}_2$ with $\mathcal{N} = \mathcal{B}(\mathcal{H}_1) \otimes 1$. Every strategy is a tensor-product strategy on the pure state Ω, whose Schmidt probabilities are the eigenvalues of $\rho_{\mathcal{N}}$. Theorem 3 together with (22) gives the bound. For example, for double cones $\mathcal{O}_1, \mathcal{O}_2$ with $\overline{\mathcal{O}_1} \subset \mathcal{O}_2$, the split property supplies the required factor for $\mathcal{M}(\mathcal{O}_1)$ and $\mathcal{M}(\mathcal{O}_2')$ [57, 58]. □

Summers and Werner showed that the vacuum violates the Bell–CHSH inequality maximally for complementary wedges [56, 62]. The first statement of Corollary 31 is the corresponding result for a game that no system of type I or II can win.

10 What the game reads

The game tests approximate invariance under adding a Bell pair, hence the ratio two in the spectrum. This is a state-specific test. For a factor and its full commutant, so that Haag duality holds, the classification uses the stronger statement that the value is one at *every faithful normal state*. Under the flow criterion of Ref. [21, Section 4], that statement is equivalent to $1/2 \in S(M_A)$: it holds for III_1 and also for III_λ with $\lambda = 2^{-1/k}$ for a positive integer k. A single fixed state with a very high score does not establish this all-states property.

Using the ratio-three game and its corresponding factor criterion from Ref. [21], we can combine that game with G by sending both players a common tag telling them which test is being played. With equal tag probabilities the resulting value is exactly $(\omega_2 + \omega_3)/2$, because strategies on the two tagged question sets can be chosen independently. Value one at every faithful normal state then requires both $1/2$ and $1/3$ in the factor invariant and singles out III_1. The incommensurate-ratio mechanism is already part of the flow-of-weights account of embezzlement [13]; the tagged game is an operational way of combining the tests.

Finite tensor powers show the same selectivity. For pairs with Schmidt probabilities $(2/3, 1/3)$, the Bell-pair error decreases, whereas ratios three and four retain a positive defect (Table 3). Extensive entropy is compatible with either behavior.

The positive floors follow directly from the multiplicative spacing. If the nonzero probabilities lie in $c\, r^{\mathbb{Z}}$ with $r > 2$, put $a = 1/\sqrt{2}$, $b = \sqrt{r/2}$ and $t_m = (D\lambda)_m/\lambda_m$ wherever $\lambda_m > 0$. Each t_m lies outside (a, b), so

$(a+b)t_m \le t_m^2 + ab$. Multiply by λ_m^2 and sum; since $\sum_{\lambda_m>0} \lambda_m^2 t_m^2 \le 1$, this gives

$$\mathcal{E} \ge 2 - 2\frac{1+ab}{a+b} > 0.$$

For ratios three and four the bounds are respectively $2-(1+\sqrt{3})/\sqrt{2} = 0.0681483\ldots$ and $2-4\sqrt{2}/3 = 0.1143819\ldots$, uniformly in the number of copies.

Schmidt probabilities	Ratio	$n=1$	16	64	256	1000
$(2/3, 1/3)$	2	0.3905	0.1392	0.0417	0.0179	0.0091
$(3/4, 1/4)$	3	0.3270	0.1079	0.0718	0.0682	0.0681
$(4/5, 1/5)$	4	0.3029	0.1286	0.1146	0.1144	0.1144

Table 3: Bell-pair embezzlement error for n copies of the indicated two-qubit state, computed with exact binomial multiplicities.

The explicit strategy also clarifies the experimental requirements. In a known Schmidt basis, the measurements reduce to reversible quotient and remainder arithmetic followed by one fixed five-level measurement. Their dependence on Schmidt rank is polynomial in its logarithm. Learning or implementing that basis for a many-body state is a separate task; entanglement-Hamiltonian spectroscopy and tomography address aspects of it [59, 60]. The finite sine-profile resource avoids this step by choosing the computational basis from the outset. Its separation from all flat-spectrum resources is quantitative and survives the preparation error stated in Section 4.2.

The spectral counting criterion determines the sharp extraction cost and its growing-target crossover. The original game has an assisted certificate at $N = 2^{19}$ with one shared Φ_9, and a chain-only certificate at $N = 2^{84}$. The four-spin witness comes from tilted CHSH, with classical play in the original game branch. The half-chain type guarantees value one throughout the critical XX sector, and the modular spectrum obstructs attainment at the ground state. The XXZ law concerns the corner-transfer-matrix spectrum as the anisotropy approaches one from above, with the physical spectral identification taken as input. The cut-preserving bulk circuit retains the finite critical XX Schmidt spectrum by exact conjugacy.

How this work was done

The authors chose the questions and directed the work, which used AI models from two developers: Claude Fable 5.2 (Anthropic) and GPT-6 Astra and GPT-6.1 Sol (OpenAI). Before release the manuscript was checked refutation-first by fresh model instances that had not worked on it: an earlier revision in three independent passes by Claude Opus 5.5, two on the proofs and one on priority and credit, and the following revision by GPT-6.1 Sol, which also reran the chain-only certificate. This version applies the wording fixes from that check, made with Claude Opus 5.5. The game constants, loss bounds and separations rest on exact or outward-rounded certificates that the accompanying archive re-checks; the other numerical values are floating-point computations with the error control of Appendix K. The theorems are proved in the text; none is formalized in a proof assistant. The authors take responsibility for every claim.

Reproducibility

The accompanying source archive contains the numerical data, exact game certificates, and programs used to reproduce the calculations and figures. The game quadratic form is evaluated from all 243 question pairs, with its matrix identity checked in $\mathbb{Q}(\sqrt{2})$. Numerical methods and truncation estimates are given in Appendix K.

A The game

We reproduce the definition of Ref. [21, Section 2.1]; the complete rule table is in the ancillary files of that paper.

The flagged magic-square game G_{FMS} [28]. Let $\mathsf{R} = \{+++, +--, -+-, --+\}$ be the sign triples of product $+1$, and C_j the sign triples of product η_j, where $\eta = (+,+,-)$. Fix $r_0 = c_0 = +++$ and $r_1 = c_1 = +--$. Alice's questions are R_1, R_2, R_3, T_0, T_1 and Bob's are C_1, C_2, C_3, U_0, U_1. To a row question R_i Alice answers a triple in R or the flag f; to a column question C_j Bob answers a triple in C_j or f. To T_x Alice answers a triple in $\mathsf{C}_1 \setminus \{c_x\}$ or a sign; to U_y Bob answers a triple in $\mathsf{R} \setminus \{r_y\}$ or a sign.

The rules are as follows. At a pair (R_i, C_j), two flags win and a single flag loses; if neither player flags, Alice's sign for column j must equal Bob's sign for row i, as in the magic-square game. At (T_x, C_1), Alice must answer a sign exactly when Bob's answer is f or c_x, and if she answers a triple, Bob must answer the same triple. The pair (R_1, U_y) has the same rule with the roles exchanged and r_y in place of c_x. At (T_x, U_y), two signs s and s' must satisfy $ss' = (-1)^{xy}$, and every other pair of answers wins. All remaining question pairs are won with any answers. The 25 question pairs are asked with equal probability. The ideal strategy uses the state $\big(|ff\rangle + \sqrt{2}\sum_{i=1}^{4}|ii\rangle\big)/3$ on $\mathbb{C}^5 \otimes \mathbb{C}^5$, a flag level and two qubits per player, with the magic-square measurements on the qubits; the factor $\sqrt{2}$ is the ratio tested by the questions T_x and U_y.

The game G. Take nine copies of G_{FMS}, indexed by $j \in \mathbb{Z}/9\mathbb{Z}$, and one further question Lab for Bob with answers $\ell \in \mathbb{Z}/9\mathbb{Z}$. Besides the 9×25 pairs inside the copies there are 18 pairs

$$\begin{aligned}(R_1^j, \mathrm{Lab}): &\quad \text{win iff } \big[\text{Alice answers } f\big] \Leftrightarrow \big[\ell = j\big],\\ (R_2^j, \mathrm{Lab}): &\quad \text{win iff } \big[\text{Alice answers } +++\big] \Leftrightarrow \big[\ell \in \{j+1, j+2\}\big].\end{aligned}$$

The 243 pairs are asked with equal probability. The first family makes the nine flags a partition of unity, each matched with a label of Bob. The second identifies one of the four two-qubit levels of copy j with the flags of copies $j+1$ and $j+2$. Together with the ratio $\sqrt{2}$ this yields (7).

The explicit measurements and their loss for general Schmidt spectra are given in Section 4.1.

B The explicit robustness constant

The lower comparison of Ref. [21] uses a flagged-sector estimate from Ref. [32]. We obtain explicit constants for the sector estimate and the nine-copy comparison. Write $\mathrm{Sch}(v)$ for the decreasing Schmidt-amplitude list of an arbitrary vector v, padded by zeroes. Write $k \times s$ for k copies of a list s, arranged in decreasing order. We use

$$\|\,\mathrm{Sch}(v) - \mathrm{Sch}(w)\|_2 \le \|v - w\|. \tag{42}$$

This follows by applying von Neumann's trace inequality to the Hilbert–Schmidt coefficient operators of the two vectors and expanding both squares. Finite-rank approximation gives the same inequality in arbitrary Hilbert spaces.

Lemma 32 (Quantitative flagged-sector estimate)**.** *Let ψ be a vector and the measurements of one flagged magic-square copy be projective. Let F be Alice's R_1 flag projection and let N be the sum of the forbidden masses on the 25 internal pairs, evaluated on ψ. Then*

$$\|\,\mathrm{Sch}((I-F)\psi) - 4 \times (\sqrt{2}\;\mathrm{Sch}(F\psi))\|_2 \le 210\sqrt{N}. \tag{43}$$

Proof. We omit tensor identities for local operators. Denote Alice's row flag projections by F_i, and Bob's column flag projections by G_j. Put $F = F_1$, $G_\star = G_1$, and $\mu = (I-F)(I-G_\star)\psi$. For the row–column pair (i,j) let u_{ij}^2 be its forbidden mass and let $M = \sum_{ij} u_{ij}^2$. Define

$$d_A = \|F(I-G_\star)\psi\|, \qquad d_B = \|(I-F)G_\star\psi\|, \qquad d^2 = d_A^2 + d_B^2 \le u_{11}^2.$$

The last inequality and $\|(F_i - G_j)\psi\| \le u_{ij}$ use the flag-mismatch rule.

Extracting the magic-square state. For each cell let A_{ij}, B_{ij} be the two local sign observables, giving every flag the sign $+1$. They are self-adjoint unitaries and satisfy $\|(A_{ij} - B_{ij})\psi\| \le \beta_{ij}$, where $\beta_{ij} = 2u_{ij}$. Alice's rows and Bob's columns commute within each context. Alice's row products and Bob's first two column products are I, and

$$B_{33} = -B_{13}B_{23} + 2G_3.$$

Denote the generators at cells $(11, 12, 21, 22)$ by (a, b, c, d_0) on Alice and by the same primed symbols on Bob. A word of k local observables on ψ differs from the reversed word on the other party by at most the

sum of the k cell bounds β_{ij}. To see this, replace the rightmost factor on ψ and commute it to the left of the remaining factors. In particular, a difference L of two signed local words, followed by a one-letter suffix s, has $\|Ls\psi\| \le \|L\psi\| + 2\beta_s$.

Write $W = \sum_{ij} \beta_{ij}$, and let $[q,r] = qr - rq$ and $\{q,r\} = qr + rq$. On either party the three expressions

$$d_0\{b,c\}a, \qquad b\{a,d_0\}c, \qquad c\{a,d_0\}b$$

applied to ψ differ from $2G_3\psi$ by at most W. In each expression the two terms are a product of column pairs and a product of row pairs. The column pairs are the cells $31, 32$ on Bob, and their product transfers to Alice's A_{33}. The row pairs are the cells $13, 23$ on Alice, and their product transfers to Bob's $B_{13}B_{23}$. The remaining transfer is the cell 33. Expanding each product on the other party by the reversed-word rule costs in total each of the nine bounds β_{ij} once. Reversing the order of the two row or column pairs is allowed at the transfer, by commutation in Bob's column 3 or Alice's row 3. Also, if two generators q, r commute on the other party, then $\|[q,r]\psi\| \le 2(\beta_q + \beta_r)$ by two word transfers.

We use the real two-qubit Pauli group $\mathcal{P}$ generated by $a = ZI$, $b = IZ$, $c = IX$, $d_0 = XI$ and $J = -I$, with the usual Pauli matrices Z, X. Its 32 elements have unique signed normal words in any fixed ordering of a, b, c, d_0, with each generator appearing at most once. The anticommuting pairs are (a, d_0) and (b, c), and the other pairs commute. For $g \in \mathcal{P}$, let A_g evaluate its signed normal word in the physical observables, using the order (c, d_0, a, b). Define B_g on Bob using (b, d_0, a, c). The local isometries

$$V_A v = 32^{-1/2} \sum_{g \in \mathcal{P}} |g\rangle A_g v, \qquad V_B v = 32^{-1/2} \sum_{g \in \mathcal{P}} |g\rangle B_g v$$

have their registers in the -1 eigenspace of $\rho(J)$, where $\rho(h)|g\rangle = |gh^{-1}\rangle$ is the right regular action. The exact commutation of the last two generators in each order gives $V_A a = \rho(a) V_A$, $V_A b = \rho(b) V_A$, and $V_B a' = \rho(a) V_B$, $V_B c' = \rho(c) V_B$ on the full input spaces.

For a generator h write $\alpha_{Ah} = \|(V_A A_h - \rho(h) V_A)\mu\|$, and define α_{Bh} in the same way. These norms are the root-mean-square errors of rewriting the physical word for g followed by h into the word for gh. Put

$$L_A = W + 2u_{13} + 2d_A, \qquad L_B = W + 2u_{13}.$$

The table gives the rewriting bounds on $(I - G_\star)\psi$ for Alice and $(I - F)\psi$ for Bob. The three columns cover presence of only the first, only the second, and both of the last two letters in the word for g. With neither letter present the rewriting is exact.

	first	second	both
A, c	$2(\beta_a + \beta_c)$	$L_A + 2\beta_a$	$L_A + 2\beta_a + 2(\beta_a + \beta_c + \beta_b)$
A, d_0	$L_A + 2\beta_c$	$2(\beta_b + \beta_{d_0})$	$L_A + 2(\beta_b + \beta_{d_0})$
B, b	$2(\beta_a + \beta_b)$	$L_B + 2\beta_a$	$L_B + 2\beta_a + 2(\beta_a + \beta_b + \beta_c)$
B, d_0	$L_B + 2\beta_b$	$2(\beta_c + \beta_{d_0})$	$L_B + 2(\beta_c + \beta_{d_0})$

For the anticommutator swaps the flag correction costs at most $2\|(I - G_\star)G_3\psi\| \le 2u_{13} + 2d_A$ on Alice and $2\|(I - F)G_3\psi\| \le 2u_{13}$ on Bob. When the swap has no suffix, transfer the trailing letter of the displayed anticommutator identity to the other party before projecting. Use a for the pair (b, c), c on Alice for (a, d_0), and b on Bob for (a, d_0). The transferred letter commutes with the opposite flag $G_\star$ or F. When the (a, d_0) swap already has a suffix, use the identity with that suffix and project directly. In the remaining swaps only the commutator bound and, if present, one suffix transfer are needed.

On μ add $2d_A$ or $2d_B$ to each nonzero case in the table, because the difference of the two signed words has norm at most two. Each case has register weight $1/4$. Write $v = (u_{11}, u_{12}, \ldots, u_{33}, d_A, d_B)^{\mathsf{T}}$ in row-major order, and let L_{Ah} and L_{Bh} be the coefficient matrices of the three bounds for each row of the table, including the additions $2d_A$ or $2d_B$. Thus $\alpha_{Ah} \le \|L_{Ah} v\|_2 / 2$ and likewise for Bob.

The consistency errors $\delta_h = \|(A_h - B_h)\mu\|$ satisfy

$$\delta_a \le 2u_{11}, \quad \delta_b \le 2u_{12} + 2d_B, \quad \delta_c \le 2u_{21} + 2d_A, \quad \delta_{d_0} \le 2u_{22} + u_{12} + u_{21} + d_A + d_B.$$

For a, b, c, use commutation with the corresponding anchor flags. For d_0, the cell difference commutes with G_2 and with F_2. Compare μ to $(I - G_2)\psi$ or $(I - F_2)\psi$, at distances at most $d_B + u_{12}$ and $d_A + u_{21}$, respectively, and average the two bounds. Write $\ell_h(v)$ for the corresponding linear upper bound on δ_h.

The -1 register sector carries ZI, IZ, IX, XI on two qubits and an identity on a multiplicity space. Indeed the four joint eigenspaces of $\rho(a), \rho(b)$ are identified with the joint $+1$ eigenspace by $\rho(d_0), \rho(c)$,

which flip their respective signs and commute. Let $w = (V_A \otimes V_B)\mu$. The four simultaneous involutions $K_h = \rho_A(h)\rho_B(h)$ commute and their joint $+1$ projection Π has range $\Phi_4 \otimes \mathcal{K}_A \otimes \mathcal{K}_B$. The generators ZI, IZ first enforce equal bits on both registers and IX, XI then enforce equal coefficients on the four equal-bit basis vectors. Put $w_0 = \Pi w = \Phi_4 \otimes j$ and $\eta = \|w - w_0\|$. Since $\|(I - K_h)w\| \le \alpha_{Ah} + \delta_h + \alpha_{Bh}$,

$$4\eta^2 \le \sum_h (\alpha_{Ah} + \delta_h + \alpha_{Bh})^2 \le v^{\mathsf{T}} H v.$$

The first inequality follows from $I - \Pi \le \sum_h (I - K_h)/2$. For the second, use weighted Cauchy–Schwarz with weights $(1, 6)$ for b, $(1, 5)$ for c, and $(2, 6, 5)$ for d_0. In terms of the linear forms above the resulting matrix is given by

$$\begin{aligned} v^{\mathsf{T}} H v = {} & \ell_a(v)^2 + 7\ell_b(v)^2 + 6\ell_c(v)^2 + \tfrac{13}{2}\ell_{d_0}(v)^2 \\ & + \tfrac{7}{24}\|L_{Bb}v\|_2^2 + \tfrac{3}{10}\|L_{Ac}v\|_2^2 + \tfrac{13}{24}\|L_{Ad_0}v\|_2^2 + \tfrac{13}{20}\|L_{Bd_0}v\|_2^2. \end{aligned}$$

All its entries are nonnegative. For

$$D_* = \operatorname{diag}(.561, 1, 1, 1, 1, 1, 1, 1, 1, .439, .439), \qquad q = (1000, 623, 452, 634, 517, 226, 226, 226, 226, 833, 553)^{\mathsf{T}},$$

the integers $\lceil 1000(Hq)_i/(D_*q)_i \rceil$ in order are

$$(521543, 521579, 521308, 521552, 521430, 521308, 521308, 521308, 521308, 521546, 521180).$$

Thus $Hq \le 521.6 D_* q$. Using $2v_iv_j \le v_i^2 q_j/q_i + v_j^2 q_i/q_j$ entrywise gives $v^{\mathsf{T}} H v \le 521.6 v^{\mathsf{T}} D_* v \le 521.6M$, the last inequality by $d_A^2 + d_B^2 \le u_{11}^2$. This gives $\eta \le 11.42\sqrt{M}$.

Comparing the flag and magic-square sectors. Embed each local space into a direct sum of a flag space and an extracted magic-square space, by $v \mapsto Fv \oplus V_A(I - F)v$ for Alice and $v \mapsto G_\star v \oplus V_B(I - G_\star)v$ for Bob. Let P, Q be the two flag-space projections and let Ψ be the embedded state. Write $\varphi = PQ\Psi$, $\zeta = \varphi + w_0$, and $e = \Psi - \zeta$. The components of e in $P(I - Q)$, $(I - P)Q$ and $(I - P)(I - Q)$ have norms d_A, d_B, η, respectively. The measurements transport to the images of the isometries and can be extended as projective measurements by assigning the orthogonal complement to one outcome.

On either magic register let R_y be the projection onto $r_y = |0\rangle|y\rangle$ and C_x the projection onto $c_x = |0\rangle(|0\rangle + (-1)^x|1\rangle)/\sqrt{2}$, tensored with the identity on its multiplicity and original local space. Extend these projections by zero on the flag spaces and use superscripts A, B for the party. The exact intertwining for Alice's a, b and Bob's a', c' means that $P_y = P + R_y^A$ and $Q_x = Q + C_x^B$ act on Ψ as the calibrated physical coarse-grainings f or r_y and f or c_x. Put $E_x^0 = P + C_x^A$ and $H_y^0 = Q + R_y^B$. These satisfy $E_x^0 \zeta = Q_x \zeta$ and $H_y^0 \zeta = P_y \zeta$.

Let E_x, H_y be the actual sign projections of T_x, U_y, and let t_x, u'_y be the forbidden masses at (T_x, C_1) and (R_1, U_y). The mismatch rule gives $\|(E_x - Q_x)\Psi\| \le \sqrt{t_x}$ and $\|(H_y - P_y)\Psi\| \le \sqrt{u'_y}$. Set $T = \sum_x t_x + \sum_y u'_y$. Define $\xi_{xy} = E_x^0 H_y^0 \zeta$. Using the two exact relations on ζ gives

$$\begin{aligned} E_x H_y \Psi - \xi_{xy} = {} & H_y(E_x - Q_x)\Psi + E_x^0(H_y - P_y)\Psi \\ & + H_y(Q_x - E_x^0)e + E_x^0 P_y e. \end{aligned}$$

The first two terms together have norm at most $2\sqrt{T}$ in the direct sum over x, y. For the third, each difference of projections has operator norm at most one, so the direct-sum norm is at most $2\sqrt{d^2 + \eta^2}$. For the fourth, $E_x^0 P_y$ is the identity on Alice's flag and equals $C_x^A R_y^A$ on her magic space, with $\sum_{xy} R_y^A C_x^A R_y^A \le I$. Hence

$$\left(\sum_{xy} \|E_x H_y \Psi - \xi_{xy}\|^2 \right)^{1/2} \le 2\sqrt{T} + 2\sqrt{d^2 + \eta^2} + \sqrt{\eta^2 + 4d_A^2 + d_B^2}. \tag{44}$$

Let U_x, V_y be the local sign unitaries, acting by the measured sign on the sign subspace and by $+1$ on its complement. Put $t_{xy} = (-1)^{xy}$ and let S sum the forbidden masses on the four sign pairs. Then

$$\sum_{xy} \|(U_x - t_{xy} V_y) E_x H_y \Psi\|^2 = 4S.$$

The ideal support vector is $\xi_{xy} = \varphi + \chi_{xy}$ with $\chi_{xy} = t_{xy}|c_x r_y\rangle h$ and $h = j/(2\sqrt{2})$, by the overlap of the two selected magic-register vectors. Define $b_{xy} = (U_x - t_{xy}V_y)\xi_{xy}$ and $B^2 = \sum_{xy} \|b_{xy}\|^2$. Equation (44) gives

$$B \le 2\sqrt{S} + 4\sqrt{T} + 4\sqrt{d^2 + \eta^2} + 2\sqrt{\eta^2 + 4d_A^2 + d_B^2}.$$

To compare φ and h, project the four b_{xy} onto PQ. The resulting differences are $PU_x\varphi - t_{xy}QV_y\varphi$. For four vectors v_x, w_y in a common space, $2w_0 = (w_0 - v_0) + (v_0 - w_1) + (w_1 + v_1) + (w_0 - v_1)$. Cauchy–Schwarz therefore bounds $\|w_0\|^2$ by the sum of the four squared differences $\|v_x - t_{xy}w_y\|^2$, and the same holds for v_0. In particular, $\|QV_0\varphi\|^2 \le \sum_{xy} \|PQb_{xy}\|^2$. Next project onto $C_x^A R_y^B$ and remove the explicit qubit factors and the scalar sign t_{xy}. The four differences now involve the compressions of U_x to c_x and of V_y to r_y, acting on the common vector h. This gives $\|C_0^A U_0 \chi_{00}\|^2 \le \sum_{xy} \|C_x^A R_y^B b_{xy}\|^2$.

The components of $U_0\chi_{00} - V_0\varphi$ in PQ and C_0^A have exactly these two bounded norms. Its remaining components, in $P(I-Q)$ and $I - P - C_0^A$, are $P(I-Q)b_{00}$ and $(I - P - C_0^A)R_0^B b_{00}$, respectively. These are orthogonal to one another and to the two projections of b_{00} used in the cycle bounds. Adding the squared norms yields $\|U_0\chi_{00} - V_0\varphi\| \le B$. By (42) and local unitarity, $\|\operatorname{Sch}(h) - \operatorname{Sch}(\varphi)\|_2 \le B$.

The list of w_0 is $4 \times (\sqrt{2}\operatorname{Sch}(h))$. The extracted $F\psi$ is within d_A of φ, and the extracted $(I-F)\psi$ is within $\sqrt{d_B^2 + \eta^2}$ of w_0. The left side of (43) is therefore bounded by

$$\sqrt{d_B^2 + \eta^2} + 2\sqrt{2}\, d_A + 2\sqrt{2}\, B \le 209.6\sqrt{M} + 8\sqrt{2}\sqrt{T} + 4\sqrt{2}\sqrt{S}.$$

For the numerical bound use $\sqrt{1 + 11.42^2} < 11.464$, $\sqrt{4 + 11.42^2} < 11.594$ and $2\sqrt{2} < 2.828428$. Since $209.6^2 + 160 < 210^2$ and $M + T + S = N$, Cauchy–Schwarz proves (43). □

Lower bound in Theorem 3. First make the measurements projective by local isometries. For a finite POVM $(M_a)_a$ on $\mathcal{H}$, its isometry $Wv = \sum_a \sqrt{M_a} v \otimes |a\rangle$ maps into an outcome register space $\mathcal{H} \otimes \mathbb{C}^m$. A common enlarged space is $\mathcal{H} \oplus (\mathcal{H} \otimes \mathbb{C}^m)$, with the input in the first summand and m at least the largest number of outcomes. For each question the self-adjoint unitary $(v, w) \mapsto (W^*w, Wv + (I - WW^*)w)$ exchanges the input with the range of W. Conjugating the outcome-coordinate projections, with the first summand and any unused coordinates assigned to a fixed outcome, gives the required projective measurement. This dilation preserves correlations and the nonzero Schmidt amplitudes.

Write $\Lambda = \operatorname{Sch}(\psi)$, Q_j for Bob's label projections, and F_j for the nine R_1 flags on Alice. Let $d_j = \|(F_j - Q_j)\psi\|$, $N_L = \sum_j d_j^2$, and $N_F = \sum_j N_j$, where N_j is the internal forbidden mass of copy j. For the strategy's losing probability ε the disjoint budgets give $N_F + N_L \le 243\varepsilon$.

To separate the label sectors, use the local column operator $Tv = \sum_j F_j v \otimes |j\rangle$ on Alice, with $A = T^*T = \sum_j F_j$. Add an unused register entry and extend the partial isometry in the polar decomposition $T = VA^{1/2}$ to an isometry V by mapping $\ker A$ into that entry. Since $A \ge F_j$, operator monotonicity of the square root gives $A^{1/2} \ge F_j$. On the orthogonal Bob labels this yields $A^{1/2} \ge \sum_j F_j Q_j$, and

$$\|(V - T)\psi\|^2 = \|(I - A^{1/2})\psi\|^2 \le \langle \psi, (A + I - 2\textstyle\sum_j F_j Q_j)\psi\rangle = N_L.$$

Apply the exact label isometry on Bob. For any label subset I_0, the raw map T on $\sum_{j\in I_0} Q_j\psi$ then differs from $\sum_{j\in I_0} Q_j\psi \otimes |j,j\rangle$ by a vector of squared norm $\sum_{j\in I_0,k} \|Q_j(F_k - Q_k)\psi\|^2 \le N_L$. The same input restricted to I_0 can only decrease the $(V - T)$ error. Thus the two local isometries give a vector within $2\sqrt{N_L}$ of the target. Let $S_j = \operatorname{Sch}(Q_j\psi)$ and let $\bigsqcup$ denote sorted union. The target is orthogonal on both sides, so

$$\left\| \operatorname{Sch}\left(\sum_{j\in I_0} Q_j\psi \right) - \bigsqcup_{j\in I_0} S_j \right\|_2 \le 2\sqrt{N_L}.$$

Use this estimate on each complementary label set and replace F_j by Q_j in (43) using (42). Then

$$\left\| \bigsqcup_{k\ne j} S_k - 4 \times (\sqrt{2}S_j) \right\|_2 \le 210\sqrt{N_j} + (1 + 2\sqrt{2})d_j + 2\sqrt{N_L}.$$

Write $U = \bigsqcup_j S_j$ and $\Delta(s) = \|2 \times s - \sqrt{2}\, s\|_2$. Concatenate the nine list comparisons and sort each side. Sorting cannot increase the ℓ^2 distance, by the rearrangement inequality. Each S_k appears eight times on the left and four times on the right, giving

$$2\Delta(U) \le 210\sqrt{N_F} + (7 + 2\sqrt{2})\sqrt{N_L}.$$

The all-label estimate gives $\|\Lambda - U\|_2 \le 2\sqrt{N_L}$. Since Δ is $2\sqrt{2}$-Lipschitz,

$$\Delta(\Lambda) \le 105\sqrt{N_F} + c_L\sqrt{N_L}, \qquad c_L = \frac{7}{2} + 5\sqrt{2}, \qquad c_L^2 < 112.$$

By Lemma 7, $\mathcal{E}(\psi) = \Delta(\Lambda)^2/2$. Cauchy–Schwarz and the disjoint budgets now give

$$\mathcal{E}(\psi) \le \frac{243(210^2 + 448)}{8}\,\varepsilon = 1{,}353{,}145.5\,\varepsilon.$$

This holds for every strategy and hence also for $\varepsilon = 1 - \omega(\psi)$. □

C Certified finite-chain separations

This appendix proves Theorems 18 and 17. We first give the chain-only certificate and then the assisted certificate at smaller sizes.

Exact cross covariance and dyadic quadrature. Put $M = N + 1$ and $m = N/4$. The exact covariance in the proof of Corollary 40 has two cross-cut parity blocks. On the even-even block, multiplying rows and columns by signs gives the symmetric matrix with entries

$$B_{ij} = \frac{1}{2M}\left[\csc\left(\frac{\pi(2i + 2j - 1)}{2M}\right) + \sec\left(\frac{\pi(2i - 2j)}{2M}\right)\right], \qquad 1 \le i, j \le m. \tag{45}$$

The two parity blocks have identical singular-value lists. In fact the left covariance C_A satisfies $\Gamma C_A \Gamma = I - C_A$, where Γ alternates signs between the sites. This follows from the sign reversal of the hopping Hamiltonian under Γ and the absence of a zero energy at even length. Thus in parity order $C_A - I/2$ has off-diagonal block D and its adjoint, with D square. The product of the full covariance cross block and its adjoint equals $C_A - C_A^2$, whose parity blocks are $I/4 - DD^*$ and $I/4 - D^*D$.

Partition the index interval into the integer bins $[a, \min(2a - 1, m)]$ for $a = 1, 2, 4, \dots, m$. On a bin of L points use Gaussian quadrature of order $k = \min(18, L)$ for counting measure. For a bin with midpoint b the nodes are $b + L\mu_i$, where μ_i are eigenvalues of the $k \times k$ Jacobi matrix with diagonal zero and off-diagonal entries

$$\beta_j = \frac{j}{2L}\sqrt{\frac{L^2 - j^2}{4j^2 - 1}}, \qquad 1 \le j < k.$$

The weights are $L|v_i(1)|^2$ for unit eigenvectors v_i. These recurrence coefficients follow by taking the ratios of the squared norms of monic orthogonal polynomials on $0, \dots, L - 1$. For degree j the monic polynomial is the jth forward difference of $x^{\underline{j}}(x - L)^{\underline{j}}$ divided by $(2j)!/j!$. Summation by parts gives orthogonality to lower degrees and squared norm $(j!)^4\binom{L+j}{2j+1}/(2j)!$ by Vandermonde's identity. The symmetric grid gives the zero diagonal after centering. The quadrature moments agree through degree $2k - 1$ because a path from the first basis vector back to itself in that many Jacobi steps cannot use any index beyond k. The nodes lie in the bin since the Jacobi matrix compresses multiplication by the centered grid coordinate. At $k = L$ the nodes and weights are the original integers and unit weights.

Form the exact quadrature matrix K by evaluating (45) at the nodes and multiplying each entry by the geometric mean of the associated weights. Write $s(A)$ for the decreasing singular values of a matrix A, padding by zeros when comparing lists. We show

$$\|s(B) - s(K)\|_{\ell^2} < 6 \times 10^{-6}. \tag{46}$$

For a pair of bins, decompose the kernel into $1/(\pi w) + M^{-1}[f(w/M) + g(z/M)]$ with $w = 2i + 2j - 1$, $z = 2i - 2j$, $f(t) = (2\sin(\pi t/2))^{-1} - (\pi t)^{-1}$, and $g(t) = (2\cos(\pi t/2))^{-1}$. Approximate each of the three terms by its degree-17 Taylor polynomial about the two bin midpoints. These polynomials in the bin variables have separate degrees at most 17. The corresponding weighted quadrature matrix therefore has the same nonzero singular values as the polynomial matrix evaluated on the full integer grid. This follows from the isometric evaluation of polynomials on each bin by exact quadrature for their pairwise inner products.

For the first term, $|w - w_0|/w_0 \le 1/3$ on a dyadic bin pair, so the geometric series error is at most $3^{-18}/(\pi w)$. On the original grid, the Frobenius norm of $1/(\pi w)$ is at most $\sqrt{1 + \ln(2m)}/(2\pi) < 4/3$ by summing over $i + j - 1$. The corresponding quadrature Frobenius norm is at most twice as large, since the

maximum of w on a bin pair is less than twice its minimum and the quadrature weights sum to the number of grid points in each bin. On $|t| = 3/2$, $|f(t)| < 1$. Indeed, either $|\Im t| \geq 1/2$ or $\sqrt{2} \leq |\Re t| \leq 3/2$, and each case gives $|\sin(\pi t/2)| \geq 1/\sqrt{2}$. On $|t| = 3/4$ we have $|g(t)| \leq 4/3$ since $|\cos(\pi t/2)| \geq \cos(3\pi/8) > 3/8$. These bounds hold in the enclosed disks by analyticity. Write t_0 for a bin-pair center and δ for the argument's maximum deviation on the bin rectangle. The bin lengths give $\delta \leq 1/4$. The argument interval is symmetric about t_0, so $\delta \leq R_0 - |t_0|$, with $R_0 = 1$ for f and $R_0 = 1/2$ for g. Consequently $\delta/(R - |t_0|) \leq (1/4)/(R - R_0 + 1/4)$, where $R = 3/2$ and $3/4$, respectively. These ratios are at most $1/3$ and $1/2$. The Cauchy estimates bound the two Taylor remainders by $(3/2)3^{-18}$ and $(8/3)2^{-18}$. Their contribution to the sum of the two Frobenius errors is at most their sum times $2m/M < 1/2$. The full sum is less than 6×10^{-6}. The singular-value perturbation inequality in Frobenius norm gives (46).

Verification of the compressed covariance. The code `certify_xx84_modes.py` uses binary64 eigensolvers to propose eigenpairs. It checks them by the following enclosures. Write $\widehat{K}$ for the symmetric binary64 matrix produced by the code. For round-to-nearest arithmetic put $u_0 = 2^{-53}$ and $\gamma_j = ju_0/(1 - ju_0)$. A product of j relative rounding factors or their inverses differs from one by at most γ_j.

1. *Nodes and weights.* Rational square comparisons verify each Jacobi entry to within 2×10^{-15}, so the operator-norm perturbation is at most 4×10^{-15}. For each proposed pair (μ, v), exact rational checks give $|\|v\|^2 - 1| < 2 \times 10^{-14}$ and rounded-matrix residual norm below 2×10^{-14}. The normalized residual on the exact matrix is bounded by

$$\frac{2 \times 10^{-14}}{\sqrt{1 - 2 \times 10^{-14}}} + 4 \times 10^{-15} < r_J, \qquad r_J = 4 \times 10^{-14}.$$

   The computed gaps exceed 10^{-3}, so the disjoint intervals of radius r_J each contain a distinct exact eigenvalue and exhaust the spectrum. Expanding $v/\|v\|$ in the exact eigenvectors bounds the norm of its component perpendicular to the matching eigenvector by $r_J/(10^{-3} - r_J)$. After sign alignment and restoring the original norm, the vector error is bounded by $\sqrt{2}\, r_J/(10^{-3} - r_J) + 2 \times 10^{-14} < b_v = 5.66 \times 10^{-11}$. Since the checked first-coordinate square of v exceeds 10^{-3}, the relative weight error after the two multiplications is at most

$$\frac{2b_v + b_v^2 + \gamma_2(1 + 2 \times 10^{-14})}{10^{-3} - 2b_v - b_v^2} < \epsilon_w, \qquad \epsilon_w = 2 \times 10^{-7}.$$

   The bin start a and length L are powers of two, and $L \leq a$. In forming $a + (L - 1)/2 + L\mu$, the multiplication by L is exact and $|\mu| \leq 1/2 + r_J$. Thus the node's absolute error is bounded by

$$Lr_J + a\gamma_2 + \tfrac{L}{2}\gamma_3 + L(\tfrac{1}{2} + r_J)u_0 < a\epsilon_x, \qquad \epsilon_x = 5 \times 10^{-14}.$$

   The exact node is at least a, giving a relative error below ϵ_x. The integer nodes and unit weights in the short bins are represented exactly.
2. *Kernel entries.* The Machin enclosure verifies an absolute error below 10^{-15} for the binary64 value of π, hence relative error below $\epsilon_\pi = 10^{-15}/3$. Rounding $H = 2M$ to $2N$ gives relative error $1/M < u_0$. For nodes x, y let $t_+ = \pi(2x + 2y - 1)/H$ and $t_- = \pi|2x - 2y|/H$. Substituting the rounded nodes into $2x + 2y - 1$ contributes relative error at most $4\epsilon_x/3$ since $2x + 2y - 1 \geq 3$. The rounded nodes are also at least one and their doubling is exact. The addition and subtraction in the rounded numerator give relative error at most $(4/3)u_0 + u_0(1 + (4/3)u_0) < 3u_0$. For the difference argument, substituting the nodes contributes absolute error at most $(\pi/2)\epsilon_x$, since $x + y \leq 2m$. With the remaining rounding factors and the enclosure of π, these estimates give

$$\left|\frac{\widehat{t}_+}{t_+} - 1\right| \leq (1 + 4\epsilon_x/3)(1 + 3u_0)(1 + \epsilon_\pi)(1 + \gamma_3) - 1 < 3 \times 10^{-13},$$
$$|\widehat{t}_- - t_-| \leq \tfrac{\pi}{2}\epsilon_x + (\tfrac{\pi}{4} + \tfrac{\pi}{2}\epsilon_x)\big[(1 + \epsilon_\pi)(1 + \gamma_4) - 1\big] < 3 \times 10^{-13}.$$

   The rounded arguments satisfy $|\widehat{t}_+| \leq 1.58$ and $|\widehat{t}_-| \leq 0.8$. Their nonzero magnitudes exceed 2^{-137}, since doubled rounded nodes are multiples of 2^{-51} and the rounded denominator is 2^{85}. Thus the argument squares are normal when nonzero. For the degree-20 even Taylor polynomials the first omitted terms are below 10^{-17} and the sums of absolute terms are below 3 on these intervals. The ratios of adjacent term magnitudes are below 0.42, so each Horner sum retains at least half the magnitude of its coefficient. The smallest coefficient has magnitude above 2^{-66}. This also keeps nonzero Horner products in the normal

range. Each term passes through at most 31 rounding factors, counting coefficient rounding, the powers of the rounded square, and the Horner multiplications and additions. The polynomial evaluation and truncation error is therefore bounded by $3\gamma_{31} + 10^{-17} < 10^{-13}$ for each function.
At the exact arguments, $\operatorname{sinc}(t_+) \geq 2/\pi$ and $\cos(t_-) \geq 1/\sqrt{2}$. The function $\operatorname{sinc}(t) = \int_0^1 \cos(zt)\,dz$ has Lipschitz constant $1/2$, and cosine has Lipschitz constant one. With $b = 3 \times 10^{-13}$, the relative errors of the evaluated sine and cosine are bounded by

$$(1+b)\big[1 + (\pi/2)(10^{-13} + (\pi/4)b)\big](1+u_0) - 1 < 10^{-12},$$
$$\sqrt{2}(10^{-13} + b) < 10^{-12},$$

respectively. The rounded reciprocal of either factor then has relative error at most $(10^{-12} + u_0)/(1 - 10^{-12}) < 2 \times 10^{-12}$.
Let p be the product of the two exact weights, $\widehat{p}$ their rounded counterpart, and h the proposed square root of $\widehat{p}$. The rounded weights lie between 2^{-10} and 2^{83}. The rounded square and division in the root check produce a quotient within 10^{-14} of one. The final subtraction of one is exact by Sterbenz's lemma. The check forces the rounded square and quotient into the normal range, so allowing for those two roundings gives

$$\left|\frac{h^2}{\widehat{p}} - 1\right| \leq (1 + 10^{-14})(1 + \gamma_2) - 1 < 1.1 \times 10^{-14}.$$

The weight enclosures then put h^2/p between $(1-\epsilon_w)^2(1-u_0)(1 - 1.1 \times 10^{-14})$ and $(1+\epsilon_w)^2(1 + u_0)(1 + 1.1 \times 10^{-14})$. These endpoints lie strictly between $(1 - 3 \times 10^{-7})^2$ and $(1 + 3 \times 10^{-7})^2$. The two reciprocal terms are positive, so their sum has the same relative error bound before rounding. Including the addition, denominator rounding, division and final multiplication gives

$$\left|\frac{\widehat{K}_{ij}}{K_{ij}} - 1\right| \leq (1 + 2 \times 10^{-12})(1 + 3 \times 10^{-7})(1 + \gamma_4) - 1 < 10^{-6}.$$

3. *Frobenius norm and eigenpairs.* The entry calculation uses normal numbers or zero throughout. In particular its positive entries exceed 2^{-100} by the weight bounds and the upper bounds on the evaluated sine and cosine. The rounded sum of squared entries is below four. With $d < 1600$ the matrix dimension, squaring and the two sums use at most $2d + 1$ rounding factors per term. The absolute error is below $4\gamma_{2d+1}/(1 - \gamma_{2d+1}) < 3 \times 10^{-12}$. Hence

$$\|K - \widehat{K}\|_F \leq \frac{10^{-6}}{1 - 10^{-6}}\|\widehat{K}\|_F < \frac{10^{-6}\sqrt{4 + 3 \times 10^{-12}}}{1 - 10^{-6}} < 2.1 \times 10^{-6}.$$

For each proposed leading eigenpair (μ, v) the code verifies $|\mu| < 1$, a computed squared norm within 10^{-11} of one, and each computed residual component below 10^{-14} in absolute value. The positive-sum error bound gives $\|v\| > 0.99$. The arithmetic error in the matrix-vector product is at most $\gamma_{d+1}\|\widehat{K}\|_F\|v\|$ apart from underflow. Allowing 2^{-1075} per scalar operation for gradual underflow, its total contribution after summation, taking norms and normalizing is below 2^{-1000}. The scalar multiplication contributes at most $u_0\|v\|$ apart from underflow, and the final subtraction costs a factor $1/(1-u_0)$ in recovering the norm before rounding. Thus

$$\frac{\|\widehat{K}v - \mu v\|}{\|v\|} < \frac{40 \times 10^{-14}}{0.99(1-u_0)} + 2.01\gamma_{d+1} + u_0 + 2^{-1000} < 3 \times 10^{-12}.$$

The selected signed intervals with this radius are disjoint and each contains an exact eigenvalue. Subtracting the sum of the squared lower bounds on their absolute values from the upper bound on the squared Frobenius norm controls the remaining squared singular mass.

For the finite model take amplitudes with single-mode ratios $r_i = (32767/32768)^{k_i}$, where the 81 integers k_i are listed in Table 4. Each mode occurs twice. The model singular values are $s_i^* = r_i/(1 + r_i^2)$. The code encloses each r_i by binary exponentiation with directed decimal rounding and compares s_i^* to the interval for the assigned exact singular value. The reference run bounds the omitted squared mass and the squared error of the retained singular values by 3.46×10^{-11} and 4.73×10^{-11}. Match each selected exact singular value to its assigned s_i^* and every other singular value to zero. Sorting both lists in decreasing order cannot increase their ℓ^2 distance, by the rearrangement inequality. The verifier checks that the two squared errors sum to less than $(4 \times 10^{-5})^2$. Combining this bound with the two matrix approximation errors gives one-parity singular-list distance at most 4.81×10^{-5}. The verifier also checks that the largest model singular value plus this distance is below 0.499.

2676	8030	13389	18758	24138	29530	34936	40355	45787
51231	56686	62153	67629	73114	78607	84109	89617	95132
100653	106180	111713	117250	122793	128340	133892	139448	145007
150571	156139	161710	167284	172862	178443	184027	189614	195204
200797	206393	211991	217592	223195	228801	234410	240021	245634
251249	256867	262486	268108	273732	279358	284986	290616	296248
301882	307517	313155	318794	324435	330078	335722	341368	347016
352665	358316	363968	369622	375278	380935	386594	392254	397915
403578	409242	414908	420575	426243	431913	437584	443256	448929

Table 4: The 81 exponents of the finite geometric model, read row by row. Each exponent occurs twice in the Schmidt-mode product. The common amplitude ratio is 32767/32768.

State and game loss. The Schmidt angle of a mode with singular value s is $\theta(s) = \frac{1}{2}\arcsin(2s)$. The occupation and its complement are the squares of the two Schmidt amplitudes. The derivative of θ is less than 16 on $0 \le s \le 0.499$. Choose the finite model state $\widehat{\psi}$ in the exact Schmidt mode bases of the chain and freeze the omitted modes. The two parity lists agree and each mode overlap is the cosine of the angle difference. Hence

$$\|\psi_N - \widehat{\psi}\|^2 = 2\left(1 - \prod_i \cos^2(\theta(s_i) - \theta(s_i^*))\right) \le 2\sum_i (\theta(s_i) - \theta(s_i^*))^2,$$

which gives vector distance less than $16\sqrt{2}\,(4.81 \times 10^{-5}) < 0.0011063 < 3/2500$.

Scoring the geometric model. Let $q = 32767/32768$, and let k_i be the integer exponents in Table 4, one for each of the 81 retained mode pairs. The model uses two modes with Schmidt ratio q^{k_i} for every i. Put $p_0 = \prod_i (1 + q^{2k_i})^{-2}$ and let n_e be the coefficient of z^e in $\prod_i (1 + z^{k_i})^2$. The Schmidt amplitudes are $\sqrt{p_0}q^e$, with multiplicity n_e in increasing order of e. Retain the prefix $e \le 380000$. For its unnormalised amplitudes λ, put $\tau = 1 - \|\lambda\|_2^2$, $E = \|(I - D)\lambda\|_2^2$, and $v = \|(I - P)\lambda\|_2^2$, where P averages consecutive groups of nine amplitudes and the prefix is padded with zeros. The players project locally onto the corresponding Schmidt-prefix subspaces, which gives a common branch on $\widehat{\psi}$. Apply (29) to the prefix by homogeneity and use the classical strategy on the complement, with loss contribution $\tau/27$. This gives

$$L_{\text{model}} \le \frac{2(\sqrt{E} + \sqrt{8v/9})^2 + 72v + 9\tau}{243}.$$

The integer recurrence for n_e adds $2n_{e-k_i} + n_{e-2k_i}$ for each mode pair, processing e in decreasing order and using zero for a negative index. The prefix has rank 185,926,736. The score verifier uses this run representation throughout. For E, it intersects intervals of equal amplitudes with the intervals obtained by doubling their endpoints. The overlap on each intersection is the product of the corresponding two amplitudes, its integer length, and $1/\sqrt{2}$. To compute v it counts only the nine-element groups that cross an interval endpoint or the end of the prefix. Their contribution is the sum of the nine squared amplitudes minus the square of their sum divided by nine.

Every interval endpoint in the verifier is an integer multiple of 10^{-45}. Addition is exact on this grid. Products and quotients use integer floor and ceiling, and square roots use integer square roots of the endpoints multiplied by 10^{45}. This gives $\tau < 0.000082218$, $E < 0.013735797$, $v < 0.000800859$, and $L_{\text{model}} < 0.000410720762$. Use the same local POVMs on ψ_N. Their average losing effect F satisfies $0 \le F \le I$, so $\sqrt{L_{\text{XX}}} = \|F^{1/2}\psi_N\| \le \|F^{1/2}\widehat{\psi}\| + \|\psi_N - \widehat{\psi}\|$. With the aligned vector-distance bound 3/2500, the physical loss obeys

$$L_{\text{XX}} \le (\sqrt{L_{\text{model}}} + 3/2500)^2.$$

Using the directed upper endpoint for L_{model}, the integer score verifier `model_score.py` gives the upper endpoint

$$0.000460799754443163771129846265247324951980458 < 0.000461 < \frac{3 - 2\sqrt{2}}{243}.$$

C.1 The finite XX certificate with a shared nine-dimensional register

Let L_N denote the loss of the explicit strategy on $\psi_{\text{XX},N} \otimes \Phi_9$. Repeating each Schmidt amplitude nine times and dividing it by 3 commutes with duplication by two and preserves the norm. The ninefold blocks in Theorem 14 therefore give $L_N = 2\mathcal{E}(\psi_{\text{XX},N})/243$. We bound this loss at $N = 2^{18}, 2^{19}, 2^{20}$.

Covariance compression. Use the exact parity kernel (45), whose singular list occurs twice, and the dyadic quadrature above with order $k = \min(24, L)$ on a bin of length L. The polynomial comparison leading to (46) now uses degree 23. For $m = N/4 \le 2^{18}$, the reciprocal kernel $1/(\pi w)$ has Frobenius norm at most $\sqrt{1 + \ln(2m)}/(2\pi) < 1$ on the integer grid and less than 2 on the quadrature grid. The same geometric and Cauchy remainder estimates with this order give

$$\|s(B) - s(K)\|_2 \le 3\,3^{-24} + \tfrac{1}{2}\left[(3/2)3^{-24} + (8/3)2^{-24}\right] < 7.95 \times 10^{-8}.$$

Here K is the exact weighted quadrature kernel and singular lists are padded by zeroes. The dimensions of K are $296, 320, 344$, respectively.

Arithmetic and spectral errors. The verifier `assisted_xx_certificate.py` uses floating-point eigensolvers to propose roots and eigenpairs. On each bin with $L > 24$ it evaluates the rational Jacobi determinant polynomials by $p_0(x) = 1,\ p_1(x) = x,\ p_{j+1}(x) = xp_j(x) - \beta_j^2 p_{j-1}(x)$. The 24 disjoint sign-changing brackets for p_{24} are bisected on the grid $10^{-28}\mathbb{Z}$. The resulting brackets enclose every Jacobi eigenvalue. For a root μ, the Jacobi eigenvector with first entry one has entries $p_j(\mu)/(\beta_1 \cdots \beta_j)$. The quadrature weight is L divided by the sum of their squares. This gives intervals for the exact nodes and weights.

With directed arithmetic on $10^{-45}\mathbb{Z}$, the code encloses π by Machin's formula and evaluates the degree-24 cosine and sinc polynomials on the argument intervals. For $|t| \le 2$ the alternating remainders are bounded by $2^{26}/26! < 10^{-18}$. The entry enclosures give a symmetric midpoint matrix $\widehat{K}$ and $\|K - \widehat{K}\|_{\mathrm{F}} < 3.32 \times 10^{-18}$ in all three runs.

For each of 24 proposed eigenvectors v, the verifier takes μ to be the nearest grid point to its exact Rayleigh quotient. Integer arithmetic gives an upper bound on $\|\widehat{K}v - \mu v\|_2/\|v\|_2$. The disjoint residual intervals contain distinct eigenvalues. Subtracting the squares of the lower bounds on their absolute values from the exact $\|\widehat{K}\|_{\mathrm{F}}^2$ bounds the omitted squared singular mass by 3×10^{-15}. For a model ratio $r_i = q^{k_i}$ with $q = 32767/32768$, the model singular value is $s_i^* = r_i/(1 + r_i^2)$. Table 5 gives the three lists of 24 exponents. The verifier encloses their ratios by directed binary exponentiation. The sum of squared differences between the matched model entries and the retained singular values is less than 1.21×10^{-11}. Sorting the matched lists, with omitted entries paired with zero, can only decrease their ℓ^2 distance. Combining the three errors gives the one-parity distance $\|s(B) - s^*\|_2 < 3.55 \times 10^{-6}$ in all three runs. The verifier uses the individual enclosures for each size.

The physical and model singular values lie below $1/2$. A singular value s gives the mode Schmidt angle $\theta(s) = \frac{1}{2}\arcsin(2s)$. On $[0, s_{\max}]$ the derivative is at most $(1 - 4s_{\max}^2)^{-1/2}$, with $s_{\max}$ bounded by the largest model value plus the one-parity error. Placing the model in the physical Schmidt mode bases gives squared vector distance $2(1 - \prod_i \cos^2(\theta(s_i) - \theta(s_i^*)))$, which is at most $2\sum_i (\theta(s_i) - \theta(s_i^*))^2$. This uses both copies of the one-parity list. Rearrangement again decreases the distance. The verifier checks that the resulting bound on the ordered amplitude distance fits the fixed rational budget d in Table 6.

N	Exponents k_i in increasing order
2^{18}	11037, 33596, 57136, 81569, 106688, 132338, 158423, 184879, 211661, 238735, 266074, 293658, 321469, 349490, 377711, 406118, 434702, 463456, 492369, 521437, 550651, 580008, 609500, 639124
2^{19}	10537, 32018, 54344, 77460, 101191, 125398, 149994, 174920, 200135, 225608, 251316, 277239, 303362, 329670, 356152, 382798, 409599, 436548, 463637, 490860, 518212, 545687, 573281, 600990
2^{20}	10080, 30585, 51822, 73761, 96253, 119175, 142448, 166016, 189843, 213900, 238165, 262622, 287256, 312054, 337007, 362104, 387338, 412703, 438192, 463798, 489518, 515347, 541280, 567314

Table 5: The three finite geometric models for the assisted certificates. Each exponent occurs twice; the common amplitude ratio is 32767/32768.

Exact ranks and score transfer. Each ratio q^{k_i} occurs twice in the normalized geometric model. The amplitudes are $\sqrt{p_0}q^e$, where $p_0 = \prod_i (1 + q^{2k_i})^{-2}$, in increasing order of e, with multiplicities from $\prod_i (1 + z^{k_i})^2$. Keep the unnormalized prefix u at $e \le 400000$. The integer recurrence and rank interval intersections used for the 2^{84} model give its mass and duplication overlap in directed arithmetic. Put $E_0 = \|(I - D)u\|_2^2$ and $\tau = 1 - \|u\|_2^2$. The full model's overlap with its duplication is at least the prefix overlap by positivity, so its duplication defect is at most $E_0 + 2\tau$. Since D is an isometry, the two norm comparisons with the full model give

$$\max\{0, \sqrt{E_0} - 2(d + \sqrt{\tau})\} \le \sqrt{\mathcal{E}(\psi_{\mathrm{XX},N})} \le \sqrt{E_0 + 2\tau} + 2d.$$

The upper bound thus pays the omitted geometric mass linearly before transferring the square root of the defect. The verifier checks that each omitted prefix mass satisfies $\tau < 1.30 \times 10^{-8}$ and uses the fixed d budget in each row to obtain the loss intervals.

N	Prefix rank	d upper bound	Certified interval for $\mathcal{E}(\psi_N)$	Certified interval for L_N
2^{18}	4375	1.363×10^{-5}	[0.0906646, 0.0908053]	[0.000746210786, 0.000747368515]
2^{19}	5642	9.187×10^{-6}	[0.0852473, 0.0853883]	[0.000701624233, 0.000702783850]
2^{20}	7295	1.684×10^{-5}	[0.0803123, 0.0804798]	[0.000661007046, 0.000662385080]

Table 6: Certified embezzlement errors of the exact XX ground states ψ_N and losses of the prescribed strategy on $\psi_N \otimes \Phi_9$. The intervals round the verifier's endpoints outwards. The amplitude-distance bound d includes covariance compression, spectral truncation and arithmetic errors.

The floor $\gamma = (3 - 2\sqrt{2})/243$ lies between 0.0007060612150362 and 0.0007060612150363. The certified assisted losses are below 0.000703 and 0.000663 at $N = 2^{19}$ and $N = 2^{20}$, respectively.

D Proof of the exact tower law

D.1 Logarithmic rank and a quantitative transfer inequality

For any probability density p on $(0, \infty)$, set

$$g_p(y) = e^{y/2}\sqrt{p(e^y)}, \qquad y \in \mathbb{R}, \qquad (T_h g)(y) = g(y - h).$$

Then $\|g_p\|_2 = 1$ and, for $h = \log d$,

$$\mathcal{E}_d(p) := \int_0^\infty \left(\sqrt{p(x)} - d^{-1/2}\sqrt{p(x/d)}\right)^2 dx = \|g_p - T_h g_p\|_2^2. \tag{47}$$

When $p(x) = p_j$ on $(j-1, j]$ and d is an integer, the integral is exactly the discrete Schmidt-coefficient error. In particular, the passage to a continuous rank variable introduces no approximation.

Lemma 33 (Hellinger transfer)**.** *For two probability densities p, q on $(0, \infty)$ and any $d > 0$,*

$$\left|\sqrt{\mathcal{E}_d(p)} - \sqrt{\mathcal{E}_d(q)}\right| \le 2\|\sqrt{p} - \sqrt{q}\|_2. \tag{48}$$

Consequently, if $b_n \to \infty$, $\mathcal{E}_d(q_n) = a/b_n + O(b_n^{-3/2})$, and $\|\sqrt{p_n} - \sqrt{q_n}\|_2 = O(b_n^{-1})$, then $\mathcal{E}_d(p_n) = a/b_n + O(b_n^{-3/2})$. If the latter distance is only $o(b_n^{-1/2})$, the conclusion is $\mathcal{E}_d(p_n) = a/b_n + o(b_n^{-1})$.

Proof. The change of variables $x = e^y$ is an isometry on square roots of densities. The operator $I - T_h$ has norm at most two. Apply the reverse triangle inequality to $(I - T_h)g_p$ and $(I - T_h)g_q$ and then square. □

D.2 The smooth Calabrese–Lefevre distribution

For $b > 0$, define q_b on $(0, \infty)$ by

$$q_b(x) = \begin{cases} e^{-b}, & 0 < x \le 1, \\ e^{-b - z(x)^2/(4b)}, & x > 1, \end{cases} \qquad I_0(z(x)) = x, \quad z(x) > 0. \tag{49}$$

The constant piece $(0, 1]$ represents the largest eigenvalue. It is essential for exact normalization; it has exponentially small weight in the estimates below.

Proposition 34 (Quantitative smooth law)**.** *The function q_b is a probability density. For every fixed $d > 0$, as $b \to \infty$,*

$$-\int_0^\infty q_b(x) \log q_b(x)\, dx = 2b, \qquad \mathcal{E}_d(q_b) = \frac{(\log d)^2}{8b} + O_d(b^{-3/2}). \tag{50}$$

The logarithmic-rank square root $g_b = g_{q_b}$ obeys

$$\|g_b'\|_2^2 = \frac{1}{8b} + O(b^{-3/2}), \qquad \|(g_b'')_{\mathrm{ac}}\|_2 = O(b^{-1}). \tag{51}$$

The only singular part of g_b'' is the atom $-e^{-b/2}(2b)^{-1}\delta_0$.

Proof. Write $t = z^2/(4b)$. The measure induced by $q_b(x)\,dx$ on $t \geq 0$ has an atom e^{-b} at zero and continuous part

$$e^{-b-t}\,dI_0(2\sqrt{bt}).$$

Using $I_0(2\sqrt{bt}) = \sum_{k\geq 0}(bt)^k/(k!)^2$, Tonelli's theorem gives, for $s > -1$,

$$\mathbb{E}e^{-sT} = e^{-b}\left(1 + \int_0^\infty e^{-(1+s)t}\,dI_0(2\sqrt{bt})\right) = \exp\left(\frac{b}{1+s} - b\right). \tag{52}$$

At $s = 0$ this proves normalization. Equivalently, T is the sum of a Poisson(b) number of independent exponential random variables of mean one. Its cumulants are $\kappa_k(T) = b\,k!$, so

$$\mathbb{E}T = b, \qquad \mathbb{E}(T-b)^2 = 2b, \qquad \mathbb{E}(T-b)^4 = 12b^2 + 24b. \tag{53}$$

Since $-\log q_b = b + T$, its entropy is exactly $2b$.

For $y > 0$, put $z = z(e^y)$ and

$$R(z) = I_1(z)/I_0(z), \qquad B(z) = z/R(z), \qquad a_b(y) = \frac{d}{dy}\log\big(e^y q_b(e^y)\big) = 1 - \frac{B(z)}{2b}.$$

For $y < 0$ the score a_b is one. The following elementary bounds are used:

$$z \leq B(z) \leq z + 2 \quad (z > 0), \qquad \sup_{z>0}\left|\frac{B'(z)}{R(z)}\right| < \infty. \tag{54}$$

For completeness, $0 < R < 1$ follows from the integral representation of I_0 and I_1. The Riccati equation $R' = 1 - R^2 - R/z$, together with the expansions at zero, shows that $R(z) > z/(z+2)$: at any prospective crossing of that lower comparison function, the derivative of their difference is $3z/(z+2)^2 > 0$. This proves the first pair of bounds. At zero, $R = z/2 + O(z^3)$ and $B = 2 + z^2/4 + O(z^4)$, whence $B'/R \to 1$. At infinity the fixed-order Bessel expansions give $R = 1 - 1/(2z) + O(z^{-2})$ and $B' = 1 + O(z^{-2})$. Thus $B'/R \to 1$ there as well, and continuity on compact subintervals proves the last bound. The recurrence and differentiated expansions used here are also recorded in NIST DLMF, Sections 10.29 and 10.40.

Under the probability measure in (52), let $Z = 2\sqrt{bT}$ and $U = T - b$. The exact identity

$$Z - 2b = U - \frac{U^2}{(\sqrt{T} + \sqrt{b})^2}$$

and (53) imply

$$\|Z - 2b - U\|_{L^2} = O(1), \qquad \mathbb{E}(Z-2b)^2 = 2b + O(\sqrt{b}), \qquad \mathbb{E}|Z-2b|^4 = O(b^2) \quad (b \geq 1). \tag{55}$$

For the last estimate use $|Z - 2b| \leq 2|U|$. Extend B continuously by $B(0) = 2$. Except at the atom $T = 0$, the score is $1 - B(Z)/(2b)$. At that atom its actual value is one; substituting $1 - B(0)/(2b)$ changes its second and fourth moments by exponentially small quantities. Therefore (54) and (55) give

$$\int a_b^2 g_b^2\,dy = \frac{1}{2b} + O(b^{-3/2}), \qquad \int a_b^4 g_b^2\,dy = O(b^{-2}). \tag{56}$$

Because $g_b' = a_b g_b/2$ almost everywhere, the first estimate proves the first part of (51). For $y > 0$,

$$|a_b'(y)| = \frac{1}{2b}\left|\frac{B'(z)}{R(z)}\right| = O(b^{-1}), \qquad g_b'' = \left(\frac{a_b^2}{4} + \frac{a_b'}{2}\right) g_b.$$

For $y < 0$, $g_b'' = g_b/4$ and its squared norm on that half-line is $e^{-b}/16$. These facts and (56) prove the second part of (51). The function g_b is continuous at zero. Its derivative jumps by $J_b = -e^{-b/2}/(2b)$, giving the asserted singular part of its second distributional derivative. In particular $g_b \in H^1(\mathbb{R})$; we do not claim $g_b \in H^2(\mathbb{R})$.

Let $v_b = (g_b'')_{\mathrm{ac}}$. Integrating the distributional derivative of g_b' over an interval gives, for each real s,

$$\|T_s g_b' - g_b'\|_2 \leq |s|\|v_b\|_2 + |J_b|\sqrt{|s|}.$$

For fixed $h = \log d$ and the usual orientation when $h < 0$,

$$g_b - T_h g_b = \int_0^h T_s g_b'\,ds, \quad \|g_b - T_h g_b - h g_b'\|_2 \leq \frac{h^2}{2}\|v_b\|_2 + \frac{2}{3}|h|^{3/2}|J_b| = O_h(b^{-1}).$$

Since $\|g_b'\|_2 = O(b^{-1/2})$, squaring the last approximation and applying (47) proves (50). □

D.3 Counting the exact discrete levels

Let

$$F_r(q) = \prod_{\ell \geq 0} (1 + q^{2\ell+1})^r = \sum_{n \geq 0} a_r(n) q^n, \qquad A_r(n) = \sum_{k=0}^{n} a_r(k), \qquad c = c_r.$$

The coefficient $a_r(n)$ is the exact degeneracy of the level εn. We take $A_r(-1) = 0$.

Lemma 35 (Counting with the required remainder)**.** *For each fixed positive integer r, as $n \to \infty$ through integers,*

$$a_r(n) = \frac{c^{1/4}}{2\sqrt{\pi}}\, n^{-3/4} e^{2\sqrt{cn}} \big(1 + O_r(n^{-1/2})\big), \tag{57}$$

$$A_r(n) = \frac{c^{-1/4}}{2\sqrt{\pi}}\, n^{-1/4} e^{2\sqrt{cn}} \big(1 + O_r(n^{-1/2})\big) = I_0(2\sqrt{cn}) \big(1 + O_r(n^{-1/2})\big). \tag{58}$$

Moreover, if $L_r(t) = \log F_r(e^{-t})$, then

$$L_r(t) = c/t + O_r(t), \qquad L_r'(t) = -c/t^2 + O_r(1) \quad (t \downarrow 0). \tag{59}$$

Proof. Here is a direct saddle-point proof, including the minor-arc estimate needed to exclude an unaccounted lattice periodicity. Set $f(x) = \log(1 + e^{-x})$. Midpoint Euler–Maclaurin, applied to f and to $x^k f^{(k)}(x)$, gives for each fixed $k \geq 0$

$$L_r^{(k)}(t) = (-1)^k k! c\, t^{-k-1} + O_{r,k}(t^{1-k}). \tag{60}$$

Indeed all the functions in question are smooth at zero, have integrable second derivatives, and decay exponentially with their derivatives at infinity; the midpoint-sum error is $O(t)$ before multiplication by t^{-k}. Integration by parts gives $\int_0^\infty x^k f^{(k)}(x)\, dx = (-1)^k k! \int_0^\infty f(x)\, dx$, and the last integral is $\pi^2/12$. In a complex disk $|w - t| \leq t/2$ we also have $|L_r^{(k)}(w)| \leq C_{r,k} t^{-k-1}$. This follows either from the same sum or from Cauchy's estimate after noting $|L_r(w)| \leq C_r/t$ for $\operatorname{Re} w \geq t/2$ and $|\operatorname{Im} w| \leq t/2$; the factors are nonzero there.

Under the Boltzmann distribution with parameter t, the integer energy N_t is the sum of independent Bernoulli occupations times their odd level numbers. Its characteristic function is

$$\phi_t(\theta) = F_r(e^{-t+i\theta})/F_r(e^{-t}).$$

Retaining only odd levels $2j + 1 \leq 1/t$ in the modulus of this product gives

$$|\phi_t(\theta)| \leq \exp\left[-c_0 \sum_{j=0}^{M-1} \sin^2((2j+1)\theta/2)\right], \qquad M \asymp t^{-1},$$

with a positive c_0 independent of small t. For $|\theta| \leq \pi$,

$$\sum_{j=0}^{M-1} \sin^2((2j+1)\theta/2) \geq c_1 M \min\{1, M^2\theta^2\}. \tag{61}$$

To check this, use $\sin u \geq 2u/\pi$ when $|\theta| \leq \pi/(2M)$. In the middle region use the identity $M/2 - \sin(2M\theta)/(4\sin\theta)$ and $|\sin\theta| \geq 1/M$. In the remaining region near $\pm\pi$, write $\delta = \pi - |\theta| \leq \pi/(2M)$; the same expression is $M/2 + \sin(2M\delta)/(4\sin\delta) \geq M/2$. Consequently

$$|\phi_t(\theta)| \leq \exp[-c_2 \min\{t^{-1}, \theta^2 t^{-3}\}]. \tag{62}$$

Take $t = \sqrt{c/n}$. Fourier inversion gives

$$a_r(n) = \frac{e^{tn + L_r(t)}}{2\pi} \int_{-\pi}^{\pi} \phi_t(\theta) e^{-in\theta}\, d\theta.$$

On the central arc set $\theta = t^{3/2}u$ with $|u| \leq t^{-1/12}$. Taylor's theorem and (60) give

$$L_r(t - i\theta) - L_r(t) - in\theta = -cu^2 - ic\sqrt{t}\, u^3 + O_r\Big(tu^4 + t^2u^2 + t^{3/2}|u| + t^{3/2}|u|^5\Big).$$

The error estimate is uniform on that arc. The term of order $\sqrt{t}$ is purely imaginary and odd. Expanding the exponential, integrating on the symmetric interval and using Gaussian domination therefore gives

$$\int_{|\theta|\le t^{3/2-1/12}} \phi_t(\theta) e^{-in\theta}\, d\theta = t^{3/2}\sqrt{\pi/c}\,(1+O_r(t)).$$

For clarity, the integrated error is bounded by a constant times $t\int_{\mathbb{R}} e^{-cu^2/2}(1+|u|^{10})\,du$; odd imaginary terms integrate to zero because the values at u and $-u$ are complex conjugates. The complement of the central arc is $O(\exp[-c_3 t^{-1/6}])$ by (62). Finally $tn + L_r(t) = 2\sqrt{cn} + O_r(t)$, which proves (57) with the asserted relative $O(t)$ remainder, rather than merely an ordinary central limit estimate.

To sum the coefficients, put $H(x) = I_0(2\sqrt{cx})$ for $x \ge 0$. The fixed-order Bessel expansion implies

$$H'(x) = \sqrt{c/x}\, I_1(2\sqrt{cx}) = \frac{c^{1/4}}{2\sqrt{\pi}}\, x^{-3/4} e^{2\sqrt{cx}}(1+O_r(x^{-1/2})).$$

Thus $a_r(k) = H'(k)(1+O_r(k^{-1/2}))$ for all sufficiently large k. The function H' is eventually increasing. Integral comparison then gives $\sum_{k\le n} H'(k) = H(n) + O_r(H'(n)+1)$. The total of the relative-error terms is $O_r(n^{-1/2}H(n))$: split at $n/2$, use the exponential separation below $n/2$, and bound $k^{-1/2} \le \sqrt{2/n}$ above it. Since $H'(n)/H(n) = O_r(n^{-1/2})$, this proves (58). □

Remark 36 (Independent number-theoretic check). For $r = 1, 2$ the coefficient estimate also follows from Chern's eta-quotient theorem [64, Theorem 1.1]. Indeed

$$F_r(q) = \frac{(q^2;q^2)_\infty^{2r}}{(q;q)_\infty^r (q^4;q^4)_\infty^r}.$$

In his notation $m = (1, 2, 4)$, $\delta = (-r, 2r, -r)$, $\Delta_1 = 0$, $\Delta_2 = -r$, and $\Delta_3(k)$ equals $r/4, -2r, r$ when k is odd, $k \equiv 2 \pmod 4$, and $k \equiv 0 \pmod 4$, respectively. His hypothesis (1.8) holds for $1 \le r \le 24$. The $k = 1$ term is $\sqrt{c/(n-r/24)}I_1(2\sqrt{c(n-r/24)})$; every other exponentially growing term has at most half its exponential rate. This checks the normalization of (57). The direct proof above does not need the restriction $r \le 24$.

D.4 Uniform inversion and completion of the proof

Lemma 37 (Uniform microscopic comparison)**.** *Let $p_{r,\varepsilon}(x) = p_j$ on $(j-1, j]$ for the ordered spectrum (18), and put $b = c_r/\varepsilon$. There is a constant $C_r < \infty$ such that, for all sufficiently small $\varepsilon > 0$,*

$$\operatorname*{ess\,sup}_{x>0} \left|\log \frac{p_{r,\varepsilon}(x)}{q_b(x)}\right| \le C_r\varepsilon, \qquad \|\sqrt{p_{r,\varepsilon}} - \sqrt{q_b}\|_2 = O_r(\varepsilon). \tag{63}$$

Proof. Fix r and write $c = c_r$ and $H(u) = I_0(2\sqrt{cu})$. For $x > 1$ define $u(x) = H^{-1}(x)$, and put $u(x) = 0$ for $0 < x \le 1$. If $A_r(n-1) < x \le A_r(n)$, the exact level at rank x is n. We claim

$$|u(x) - n| \le K_r \tag{64}$$

with a constant independent of both n and x. By (58), $\log A_r(n) - \log H(n) = O_r(n^{-1/2})$, whereas

$$\frac{d}{du}\log H(u) = \sqrt{c/u}\,\frac{I_1(2\sqrt{cu})}{I_0(2\sqrt{cu})} = \sqrt{c/u} + O_r(u^{-1}).$$

Choose a fixed, sufficiently large K_r. For all sufficiently large n, these two estimates imply $H(n-K_r) \le A_r(n-1)$ and $A_r(n) \le H(n+K_r)$. Monotonicity of H proves (64) there. There are only finitely many smaller levels, and their rank intervals are bounded, so increasing K_r proves it everywhere. Empty intervals caused by zero degeneracies require no treatment; for example $a_1(2) = 0$, while $a_1(n) > 0$ for every $n \ge 3$.

At the rank under consideration,

$$p_{r,\varepsilon}(x) = e^{-L_r(\varepsilon)-\varepsilon n}, \qquad q_b(x) = e^{-b-\varepsilon u(x)}.$$

Since $L_r(\varepsilon) = b + O_r(\varepsilon)$, the logarithm of their ratio is uniformly $O_r(\varepsilon)$, proving the first claim. Both functions are normalized, and pointwise $|\sqrt{p} - \sqrt{q}| \le (e^{C_r\varepsilon/2} - 1)\sqrt{q}$ after enlarging the constant, so integration proves the second claim. □

Proof of Theorem 11. The entropy of the odd tower is $S = L_r(\varepsilon) - \varepsilon L_r'(\varepsilon)$. Equation (60) gives

$$S = 2c_r/\varepsilon + O_r(\varepsilon), \qquad V = \varepsilon^2 L_r''(\varepsilon) = 2c_r/\varepsilon + O_r(\varepsilon).$$

Apply Proposition 8 with $W = \varepsilon^{-1}$ and $L(z) = L_r(\varepsilon z)$. The real and complex derivative bounds in Lemma 35 give $v_\beta \asymp_r W$ and the required analytic bounds. The characteristic function of the physical energy under tilt β is $\phi_{\varepsilon\beta}(\varepsilon t)$ in the notation of that lemma. For $|t| \le \pi W$, the odd-lattice estimate (62) gives

$$|\phi_{\varepsilon\beta}(\varepsilon t)| \le \exp\bigl(-cW \min\{1, t^2\}\bigr)$$

uniformly in β near one. This proves the hypotheses of the counting criterion. Its conclusion is

$$\mathcal{E}_d = \frac{(\log d)^2}{4V} + O_{r,d}(\varepsilon^2) = \frac{(\log d)^2 \varepsilon}{8c_r} + O_{r,d}(\varepsilon^2), \qquad S\mathcal{E}_d = \frac{(\log d)^2}{4} + O_{r,d}(\varepsilon).$$

□

D.5 A transfer condition for other spectra

The same leading law transfers to other spectra under a quantitative Hellinger estimate.

Corollary 38 (A sufficient transfer condition)**.** *Let $(p_{N,j})$ be any sequence of decreasing probability spectra, let $p_N(x) = p_{N,j}$ on $(j-1, j]$, and let $b_N \to \infty$. Suppose*

$$S(p_N)/(2b_N) \longrightarrow 1, \qquad \|\sqrt{p_N} - \sqrt{q_{b_N}}\|_2 = o(b_N^{-1/2}). \tag{65}$$

Then, for every fixed integer $d \ge 2$, $\mathcal{E}_d(p_N)S(p_N) \to (\log d)^2/4$. In particular the second hypothesis holds if $\operatorname{ess\,sup}_{x>0} |\log(p_N(x)/q_{b_N}(x))| = o(b_N^{-1/2})$.

Proof. The first assertion follows directly from Lemma 33 and Proposition 34. The final sufficient criterion follows from the same pointwise square-root estimate used in Lemma 37. □

For finite-dimensional physical spectra the final, global log-ratio criterion cannot hold because the spectrum has a finite-rank tail; the Hellinger condition in (65) is the usable criterion and includes the tail mass. It may also be replaced by the strictly weaker, direct condition $\|(I - T_{\log d})(g_{p_N} - g_{q_{b_N}})\|_2 = o(b_N^{-1/2})$. Neither leading Rényi entropy asymptotics nor weak Gaussian convergence of logarithmic rank proves these microscopic conditions. Such a transfer therefore requires more than the leading entropy asymptotic.

E The finite-chain spectral mesh

The Hilbert spectral-counting approach goes back to Widom [66]; see also Fedele [65]. Otte [61, Sections 3–4] uses the Hilbert-to-Carleman reduction for determinant asymptotics. The unit-cell comparison and periodization below give the ordered spectral estimate needed for the finite XX chain.

Lemma 39 (Finite Hilbert sections and the Carleman kernel)**.** *For $m \ge 1$ and fixed $a > 0$ let*

$$H_a^{(m)} = \left(\frac{1}{j+k+a}\right)_{j,k=0}^{m-1}, \qquad L_m = \ln(2m+1).$$

Write $\sigma_j(A)$ for the nonincreasing singular values of A, padding finite lists by zeros. Write $(h_j^{(m)})_{j\ge1}$ for the decreasing rearrangement of $\{\pi \operatorname{sech}(2\pi^2 k/L_m) : k \in \mathbb{Z}\}$. There is a constant C_a independent of m with

$$\sum_{j\ge1} \left|\sigma_j(H_a^{(m)}) - h_j^{(m)}\right| \le C_a.$$

Proof. The difference $H_a^{(m)} - H_2^{(m)}$ is sign definite, since its entries are the moments of $(t^{a-1} - t)\,dt$ on $[0,1]$. Hence

$$\|H_a^{(m)} - H_2^{(m)}\|_1 = \sum_{j=0}^{m-1} \left|\frac{1}{2j+a} - \frac{1}{2j+2}\right| \le C_a'.$$

Consider the integral operator C_m on $L^2([1/2, m+1/2], dx)$ with kernel $(x+y)^{-1}$. The Laplace representation $(x+y)^{-1} = \int_0^\infty e^{-tx}e^{-ty}\,dt$ makes C_m a positive trace-class operator, with trace $\int_{1/2}^{m+1/2}(2x)^{-1}dx = L_m/2$.

Compress C_m to the orthonormal indicators of the unit intervals $[j+1/2, j+3/2)$, $0 \le j < m$. The resulting matrix has entries

$$\int_0^\infty e^{-t(j+k+2)} \left(\frac{2\sinh(t/2)}{t}\right)^2 dt.$$

The factor in parentheses is at least one. The difference from $H_2^{(m)}$ is positive semidefinite, as is seen by evaluating quadratic forms inside the Laplace integral. The min–max principle for compression and positive perturbation therefore gives $\sigma_j(C_m) \ge \sigma_j(H_2^{(m)})$ for every j, with zero padding. Their difference in ℓ^1 is

$$\operatorname{Tr} C_m - \operatorname{Tr} H_2^{(m)} = \frac{1}{2}\left(\ln(2m+1) - \sum_{j=1}^{m}\frac{1}{j}\right) = O(1).$$

The unitary change of variables $x = e^s/2$ transforms C_m to the convolution kernel $K(s-t)$ on $[0, L_m]$, where $K(z) = (2\cosh(z/2))^{-1}$. For $L = L_m$, define the periodized operator P_L by the kernel $\sum_{k\in\mathbb{Z}} K(s-t+kL)$ on the same interval. The Fourier transform with convention $\widehat{K}(\xi) = \int_{\mathbb{R}} K(z)e^{-i\xi z}\,dz$ is $\pi\operatorname{sech}(\pi\xi)$. This follows by setting $u = e^z$ in the integral and using $\int_0^\infty u^{b-1}(1+u)^{-1}du = \pi/\sin(\pi b)$ at $b = 1/2 - i\xi$. The Fourier modes on $[0, L]$ diagonalize P_L, with eigenvalues $\widehat{K}(2\pi k/L)$ for $k \in \mathbb{Z}$. They are nonnegative and summable for each L.

The trace norm of the difference between P_L and the original convolution operator is bounded uniformly for $L \ge 1$. For $k \ge 1$, the image kernel $K(s-t+kL)$ is, after reflection of the second variable, $K(s+t+(k-1)L)$. Negative images are transposes. For $k = 1$, expand for $z > 0$

$$K(z) = \sum_{j\ge 0}(-1)^j e^{-(j+1/2)z}.$$

Group consecutive terms into rank-two kernels $f_b \otimes f_b - f_{b+1} \otimes f_{b+1}$ with $f_b(x) = e^{-bx}$ and $b = 2j + 1/2$. On $[0, L]$, $\|f_b\|_2 \le (2b)^{-1/2}$ and $\|f_b - f_{b+1}\|_2 \le \|xe^{-bx}\|_{L^2(0,\infty)} = O(b^{-3/2})$. Each paired trace norm is thus $O(b^{-2})$, uniformly in L. The series converges in trace norm and its kernel agrees almost everywhere with $K(s+t)$. For $k \ge 2$, there is a positive offset $d = (k-1)L \ge 1$. The trace norms of the unpaired rank-one kernels are bounded in sum by

$$\sum_{j\ge 0}\frac{e^{-(j+1/2)d}}{2j+1} \le Ce^{-d/2}.$$

These bounds sum over the images. The trace-norm perturbation inequality for singular values now bounds the ℓ^1 distance between the spectra of C_m and P_{L_m} by an absolute constant. Combining with the compression comparison proves the claim. □

Corollary 40 (Quantitative active-mode mesh of the finite open XX chain)**.** *Let $N = 2n \ge 4$ be even, $m_1 = \lceil n/2\rceil$, $m_2 = \lfloor n/2\rfloor$. For the cross block B_N of the half-filled finite open XX covariance, the singular values are at bounded ℓ^1 distance from the decreasing rearrangement of*

$$\left\{\frac{1}{2}\operatorname{sech}(2\pi^2 k/L_{m_i}) : i = 1, 2,\ k \in \mathbb{Z}\right\}. \tag{66}$$

For each fixed $0 < a < b < \infty$, among the mesh entries whose energy $4\pi^2|k|/L_{m_i}$ belongs to $[a, b]$, all but $O_{a,b}(1)$ can be matched to finite-chain modes with energies in $[a/2, 2b]$. The sum of the absolute energy errors over those matches is $O_{a,b}(1)$.

Proof. Write $M = N + 1$. In reversed left and forward right coordinates $1 \le u, v \le n$ the covariance cross block, up to a global sign from choosing the other occupied half of the standing waves, has entries

$$\frac{\sin(\pi(u+v-1)/2)}{2M\sin(\pi(u+v-1)/(2M))} - \frac{\sin(\pi(M+v-u)/2)}{2M\cos(\pi(v-u)/(2M))}.$$

This follows from the product-to-sum identity for the sine waves and the finite cosine sum. Subtract the principal matrix with entries $\sin(\pi(u+v-1)/2)/[\pi(u+v-1)]$. The two remaining reciprocal factors have the form $M^{-1}f(a_u + b_v)$, with $f(z) = (2\sin(\pi z/2))^{-1} - (\pi z)^{-1}$ and $|a_u|, |b_v| \le 1/2$ for the first term. For the

second use $f(z) = (2\cos(\pi z/2))^{-1}$ with centered coordinates $|a_u|, |b_v| \le 1/4$. The functions are analytic on the disks of radii 2 and 1, respectively, with a removable singularity at zero for the first. If $|a_u|, |b_v| \le r$, the trace norm of the degree-k monomial matrix is at most $(n/M)(2r)^k$ by the binomial expansion into rank-one matrices. Each sine is a sum of two products of phases in u and v. Their diagonal-unitary multipliers preserve these bounds. Absolute summability of the Taylor coefficients on the relevant smaller radii gives a uniformly bounded trace-norm error.

Separating the parities and removing the alternating signs makes the principal matrix equivalent in singular values to the direct sum of $(2\pi)^{-1}H_{1/2}^{(m_1)}$ and $(2\pi)^{-1}H_{3/2}^{(m_2)}$. Apply Lemma 39 to the blocks, together with the trace-norm perturbation inequality. Rearranging the union in decreasing order only decreases the ℓ^1 distance, which proves (66).

A true singular value s equals $\sqrt{\lambda(1-\lambda)}$ for an eigenvalue λ of the restricted covariance, since the full covariance is a projection. The one-mode energy $|\ln(\lambda/(1-\lambda))|$ is therefore $e(s) = 2\,\mathrm{arcosh}(1/(2s))$, with $e(0) = \infty$. Match the ordered singular-value lists from (66). Mesh values of energy in $[a, b]$ have positive distance from the complement of the singular-value interval corresponding to $[a/2, 2b]$. Thus at most a bounded number of their matches are outside that interval. On the remaining matches $e(s)$ has bounded derivative, which converts the ℓ^1 singular-value bound to the stated energy bound. □

Corollary 41 (Fourier damping for tilted energy laws)**.** *Let e_j denote the finite open XX mode energies and $W = \ln N$. Under a tilt β in a fixed compact subinterval of $(0, \infty)$, let X_j be independent Bernoulli variables of parameters $(1 + e^{\beta e_j})^{-1}$, with deterministic zero contribution from any infinite-energy mode. For $\varphi_\beta(t) = \mathbb{E}_\beta \exp(it \sum_j e_j X_j)$ there are constants $c, \delta, \tau > 0$ such that for all sufficiently large even N,*

$$|\varphi_\beta(t)| \le \begin{cases} \exp(-cWt^2), & |t| \le \delta, \\ \exp(-cW), & \delta \le |t| \le \tau W. \end{cases}$$

The constants and large-N threshold can be chosen uniformly in the given compact set of tilts.

Proof. Use just one mesh branch from (66) with $k \ge 1$ and mesh energies in a fixed interval $[a, b] \subset (0, \infty)$. The matched true energies lie in $[a/2, 2b]$ with $O(1)$ deletions and total absolute energy discrepancy $O(1)$. On these modes, the Bernoulli variances are uniformly bounded below. The squared modulus of one characteristic factor is $1 - 4q(1-q)\sin^2(te_j/2)$, where $q = (1 + e^{\beta e_j})^{-1}$. It follows that $|\varphi_\beta(t)|$ is bounded by $\exp(-c_0 \sum_j \sin^2(te_j/2))$ using the matched modes only. For $|t| \le \delta$ with δb sufficiently small, each sine square is bounded below by a constant times t^2 and the number of matches is of order W.

For $|t| \ge \delta$ the integral $\int_a^b \sin^2(tx/2)\,dx$ is bounded below by a strictly positive constant. It is positive on compact subsets away from zero and tends to $(b-a)/2$ at infinity. The mesh spacing is $4\pi^2/L_{m_i}$ with $L_{m_i} = W + O(1)$. The corresponding Riemann sum for the sine squares differs from this integral divided by the mesh spacing by $O(1 + |t|)$, since the derivative in x is bounded by $|t|/2$. Deleting $O(1)$ terms and replacing mesh energies by matched true energies changes the sum by at most another $O(1 + |t|)$ by the same derivative bound. The resulting lower bound is $c_1 W - C_1(1 + |t|)$ with $c_1, C_1 > 0$ independent of N. Choose $\tau \le c_1/(4C_1)$ and take N large enough that $C_1 \le c_1 W/4$. The sum is then at least $c_1 W/2$ throughout $\delta \le |t| \le \tau W$, proving the second estimate. □

Corollary 42 (Transfer to the finite open critical Ising chain)**.** *Let L be even. Across its middle cut, the entanglement-mode energies of the finite open critical Ising chain on L spins occur twice in the finite open XX chain on $2L$ spins across its middle cut. In particular, the XX bounds in Corollary 41 apply to the Ising characteristic function with adjusted constants.*

Proof. This is the covariance decomposition behind the even-block identity of Iglói and Juhász [50, eqs. (19)–(23)]. Write $a_j = c_j + c_j^\dagger$ and $b_j = i(c_j^\dagger - c_j)$ for the two Majoranas of the XX site j. Its hopping term is $\frac{i}{2}(a_j b_{j+1} - b_j a_{j+1})$. There are two disjoint Majorana paths, one using a_j on odd sites and b_j on even sites and the other using the opposite assignment. In each path, the signs of the hopping terms can be made uniform by changing signs of individual Majoranas. Each path is then, up to a positive scale of its Hamiltonian, the critical Ising chain of L spins in its Majorana representation. Since L is even, the first L XX sites restrict each path to its first L Majoranas, which are precisely the first $L/2$ Ising spins. The restricted Gaussian covariance is the direct sum of the two path covariances. The mode-energy lists therefore add by multiset union. The tilted characteristic function of the XX energy is the square of the Ising characteristic function. Taking its modulus square root gives the desired damping. □

E.1 Uniformity over cuts of open chains

Corollary 43 (Arbitrary end cuts of the finite chains)**.** *For the half-filled open XX chain of even length N, take the site bipartition $\{1,\dots,\ell\} \mid \{\ell+1,\dots,N\}$ and put $m = \min(\ell, N-\ell)$. As $m \to \infty$, uniformly in N and ℓ,*

$$S = \tfrac{1}{6}\log m + O(1), \qquad V = \tfrac{1}{6}\log m + O(1), \qquad \mathcal{E}_d = \frac{3(\log d)^2}{2\log m} + O_d((\log m)^{-2}).$$

For the open critical Ising chain of any length L cut after k spins, put $m = \min(k, L-k)$. The same statement holds with $S, V = \frac{1}{12}\log m + O(1)$ and $\mathcal{E}_d = 3(\log d)^2/\log m + O_d((\log m)^{-2})$.

Proof. By reflection it suffices to treat the XX prefix cut with $\ell = m$. Write $M = N+1$. For $1 \le x \le m < y \le N$, the exact cross-covariance entry, up to the sign of the filled standing-wave half, is

$$\frac{\sin(\pi(x-y)/2)}{2M\sin(\pi(x-y)/(2M))} - \frac{\sin(\pi(x+y)/2)}{2M\sin(\pi(x+y)/(2M))}.$$

This follows by expanding the product of standing waves and summing the first $N/2$ cosines. In both terms replace the reciprocal factor $(2M\sin(\pi(x\pm y)/(2M)))^{-1}$ by $(\pi(x\pm y))^{-1}$. The error in trace norm is $O(1)$ uniformly in m, N. Indeed the difference is $M^{-1}f((x\pm y)/M)$ with $f(z) = (2\sin(\pi z/2))^{-1} - (\pi z)^{-1}$ analytic for $|z| < 2$. For a matrix $M^{-1}(a_x + b_y)^q$, the binomial rank-one expansion bounds its trace norm by $M^{-1}\sqrt{m(N-m)}(\max_x |a_x| + \max_y |b_y|)^q$. Here the sum of maxima is at most $3/2$ and the prefactor is at most $1/2$. The Taylor series of f is absolutely summable on that smaller radius. Multiplication by either sine gives a linear combination of two diagonal-unitary left and right multipliers, so the same bound applies to the covariance remainder.

We use a uniform bound for the rectangular matrix with entries $(j+k+a)^{-1}$ for $0 \le j < p$ and $p \le k < q$, with fixed $a > 0$ and $1 \le p \le q$. Its Laplace rank-one representation bounds the trace norm by

$$\int_0^\infty e^{-(p+a)t}\,\frac{\sqrt{\min\{p, (1-e^{-2t})^{-1}\}}}{\sqrt{1-e^{-2t}}}\,dt.$$

On $0 < t \le 1/p$ the integrand is bounded by $C\sqrt{p/t}$. On $t \ge 1/p$ it is bounded by $Ce^{-pt}(1+t^{-1})$. Both integrals are bounded independently of p, q. Taking $a = 2$ shows that the reflected principal matrix with entries $\sin(\pi(x+y)/2)/(\pi(x+y))$ has trace norm $O(1)$, using the same sine phase decomposition.

Reverse the order on the left by setting $u = m+1-x$ and set $v = y-m$ on the right. The remaining principal matrix has entries $\sin(\pi(u+v-1)/2)/(\pi(u+v-1))$. Its parity blocks, after removing alternating signs, are rectangular Hilbert matrices scaled by $(2\pi)^{-1}$, with $a = 1/2, 3/2$ and row counts $m_1 = \lceil m/2\rceil$, $m_2 = \lfloor m/2\rfloor$. Each column count is at least its row count. The rectangular bound just proved gives trace-norm error $O(1)$ when the columns beyond the leading square are deleted with zero padding. Lemma 39 and the singular-value perturbation inequality therefore give the comparison (15) with $L_i = \log(2m_i+1) = \log m + O(1)$. All constants in the comparison are independent of the cut. The proof of Theorem 1 applies with $W = \log m$.

For Ising, reflect so that $k \le L/2$ and take the XX chain on $2L$ sites with a cut after $2k$ sites. In the Majorana decomposition of Corollary 42, each path then restricts to its first $2k$ Majoranas, the first k Ising spins. The XX mode list is again the multiset union of two Ising lists. The XX result at $W = \log(2k)$ and exact doubling prove the claimed Ising estimates uniformly in L, k. □

F Microscopic energy counts and the smooth rank profile

We use the assumptions of Proposition 8 and the notation in (12). All constants below are uniform in the real tilt interval of that proposition.

Lemma 44 (Cumulative counts with their left limits)**.** *For $|E-\mu| \le 2B$, the exact cumulative counts satisfy (13), including energies at which the count jumps.*

Proof. Put $\Delta = (\tau W)^{-1}$ and let $K(u) = c_K(\sin(u/4)/(u/4))^4$, where c_K normalizes the integral to one. This even, nonnegative kernel has finite second moment, a positive minimum on $[-1,1]$, and Fourier transform

supported on $[-1,1]$. The support statement follows by writing the sinc function as the Fourier transform of a uniform density on $[-1/4,1/4]$ and taking the fourfold convolution. By (10),

$$\int_{|t|\le\Delta^{-1}} |\varphi_\beta(t)|\,dt = O(W^{-1/2}).$$

Consequently the convolution of the tilted law of A with $K_\Delta(u)=\Delta^{-1}K(u/\Delta)$ has density bounded by $C/\sqrt{W}$. At the midpoint of a closed interval of length Δ, this density is at least c/Δ times the probability of the interval. Covering longer intervals by such intervals gives, for every closed interval I,

$$\Pr_\beta(A\in I)\le C(|I|+\Delta)/\sqrt{W}. \tag{67}$$

The same bound applies to singletons by enclosure.

For $k_\beta(u)=e^{-\beta u}\mathbf{1}_{\{u\ge 0\}}$ and a real shift v, the change $|k_\beta(E-A-v)-k_\beta(E-A)|$ is bounded by the variation measure of k_β on the closed interval swept out by the shift. This measure has total mass two. Integrating first over the law of A and using (67) bounds the expectation of the change by $2C(|v|+\Delta)/\sqrt{W}$. Thus smoothing by an independent ΔZ with Z of density K changes the expectation by $O(\Delta/\sqrt{W})$.

At $E=m_\beta$, Fourier inversion gives the smoothed expectation as

$$\mathbb{E}_\beta k_\beta(E-A-\Delta Z)=\frac{1}{2\pi}\int_{-\Delta^{-1}}^{\Delta^{-1}}\frac{\mathbb{E}_\beta e^{it(A-m_\beta)}}{\beta-it}\widehat{K}(\Delta t)\,dt. \tag{68}$$

The integral outside $|t|\le W^{-5/12}$ is $O(e^{-c'W^{1/6}})$ by (10). On the inner interval put $u=t\sqrt{v_\beta}$. The analytic derivative bounds give

$$\log\mathbb{E}_\beta e^{it(A-m_\beta)}=-u^2/2+i\alpha_\beta u^3+O(W^{-1}u^4),\qquad \alpha_\beta\in\mathbb{R},\quad |\alpha_\beta|\le CW^{-1/2}.$$

Here $|u|\le CW^{1/12}$, so after factoring out $e^{-u^2/2}$, the exponential equals $1+i\alpha_\beta u^3+O(W^{-1}(u^4+u^6))$. Expanding $(\beta-it)^{-1}=\beta^{-1}+it\beta^{-2}+O(t^2)$ and $\widehat{K}(\Delta t)=1+O(\Delta^2t^2)$, the two first-order corrections integrate to zero by oddness. The remaining terms are bounded in the rescaled integral by $CW^{-1}e^{-u^2/2}(1+|u|^8)$. The Gaussian tail beyond the inner interval is exponentially small. Therefore

$$\mathbb{E}_\beta k_\beta(E-A-\Delta Z)=\frac{1}{\beta\sqrt{2\pi v_\beta}}\big(1+O(W^{-1})\big).$$

The $O(\Delta/\sqrt{W})$ smoothing error has this same relative order. For the strict count, dropping the event $A=E$ changes the unsmoothed expectation by $O(\Delta/\sqrt{W})$ by (67). Multiplying both expectations by $e^{L(\beta)+\beta E}$ and taking logarithms gives (13). □

Lemma 45 (Smooth profile on the logarithmic rank axis)**.** *With the exact square-root profile g and smooth cutoff profile G defined in the proof of Proposition 8, the three estimates in (14) hold.*

Proof. Throughout this proof $U=A-\mu$ under the tilt $\beta=1$. The derivative bounds give

$$\mathbb{E}_1U=0,\quad \mathbb{E}_1U^2=V,\quad \mathbb{E}_1U^3=-L'''(1)=O(W),\quad \mathbb{E}_1U^4=L^{(4)}(1)+3V^2=O(W^2). \tag{69}$$

Also $\log\mathbb{E}_1e^{\theta U}\le CW\theta^2$ for small real θ by Taylor expansion of $L(1-\theta)$. Taking θ of order B/W with either sign gives

$$\Pr_1(|U|>B/2)\le 2e^{-c(\log W)^2}. \tag{70}$$

The first three moments restricted to this tail are smaller than every power of W^{-1} by Hölder's inequality and the fourth moment bound.

From $\beta'(E)=-1/v_{\beta(E)}$ we obtain

$$F'(E)=\beta(E)+\frac{1}{\beta(E)v_{\beta(E)}}+\frac{L'''(\beta(E))}{2v_{\beta(E)}^2}. \tag{71}$$

In particular F' is bounded above and away from zero. Each E-derivative applies $-v_\beta^{-1}\partial_\beta$. Thus, for $0\le j\le 3$, $\partial_E^j(F'-\beta)=O(W^{-j-1})$, and for $1\le j\le 3$, $\partial_E^j\beta=O(W^{-j})$. These bounds follow term by term from $v_\beta\asymp W$ and the uniform bounds on the fixed-order derivatives of L. For $a(E)=1-1/F'(E)$ we have

$$a(\mu)=O(W^{-1}),\qquad a'(\mu)=-1/V+O(W^{-2}),\qquad \partial_E^j a(E)=O(W^{-j})\quad(1\le j\le 3).$$

When $y \in F([\mu - 3B/2, \mu + 3B/2])$, the lower bound on F' and (13) put e^y between $R(\mu - 2B)$ and $R(\mu + 2B)$. Hence the exact energy E_{e^y} lies in the counting window. Conversely, when $|E_{e^y} - \mu| \le B/2$, the two bounds on the exact rank give $y \in F([\mu - B, \mu + B])$. In either case

$$E_{e^y} = E_*(y) + O(W^{-1}). \tag{72}$$

Indeed, $R(E_{e^y}^-) < e^y \le R(E_{e^y})$ away from the immaterial endpoints of the rank intervals, and the lower bound on F' converts the two logarithmic count errors into the energy error. Therefore, on the support of G, the uncut profile $G_0(y) = \exp((y - L(1) - E_*(y))/2)$ satisfies $G_0^2/g^2 = 1 + O(W^{-1})$. On the ranks with $|E_{e^y} - \mu| \le B/2$, the cutoff is one. The law of E_{e^y} under $g(y)^2\,dy$ equals the law of A under the original Gibbs law. Thus (70) and the ratio bound give $\|g - G\|_2 = O(W^{-1})$.

The cutoff and its first two y-derivatives are bounded. Its derivatives are supported where the exact energy lies in the tail of (70). On the support of G,

$$G_0' = \tfrac{1}{2} a(E_*) G_0, \qquad G_0'' = \left(\tfrac{1}{4} a(E_*)^2 + \frac{a'(E_*)}{2F'(E_*)} \right) G_0.$$

The bound $|a(E_*)| \le C(1 + |E_* - \mu|)/W$, the ratio bound and the fourth moment imply $\|G''\|_2 = O(W^{-1})$. Contributions from the cutoff derivatives are smaller than every power of W^{-1}.

To compute $\|G'\|_2^2$, we can drop cutoff derivative terms and restrict to $|E_{e^y} - \mu| \le B/2$ at negligible cost. On this set, replacing G_0^2 by g^2 costs $O(W^{-2})$ because the squared score has integral $O(W^{-1})$. Replacing $a(E_*)$ by $a(E_{e^y})$ changes the score pointwise by $O(W^{-2})$ by (72). It remains to compute $\mathbb{E}_1[a(\mu + U)^2 \mathbf{1}_{\{|U| \le B/2\}}]$. Write

$$a(\mu + U) = a_0 + a_1 U + a_2 U^2 + \rho(U), \qquad |\rho(U)| \le C|U|^3/W^3,$$

with $a_0 = O(W^{-1})$, $a_1 = -1/V + O(W^{-2})$, $a_2 = O(W^{-2})$. On the truncated set the squared sum of the constant, quadratic and remainder terms has expectation $O(W^{-2})$, since $|U| \le W$ there and $\mathbb{E}_1 U^4 = O(W^2)$. Its cross term with $a_1 U$ is also $O(W^{-2})$. The constant cross term has zero full mean by (69), the quadratic cross term uses $\mathbb{E}_1 U^3 = O(W)$, and the remainder cross term is bounded by $CW^{-4}\mathbb{E}_1 U^4$. Reinstating the first three full moments has negligible cost by (70). The linear term alone contributes $a_1^2 V = 1/V + O(W^{-2})$. This proves $\|G'\|_2^2 = 1/(4V) + O(W^{-2})$. □

Trace sums for the free chains. For finite XX the mesh gives $\sum_j s_j = O(W)$. For $e \ge 1$, a fixed-order z-derivative of $\log(1 + e^{-ze})$ is bounded by $C_k e^k e^{-(1-2\eta)e}$ on a small fixed complex neighborhood of the tilt interval. Take $\eta < 1/4$ so this bound is $O_k(s)$. For $e \le 1$, shrinking the imaginary neighborhood bounds $|1 + e^{-ze}|$ away from zero, and the number of such modes is $O(W)$. This proves the analytic derivative bounds needed in Proposition 8. The analytic logarithms can be chosen real on the real tilt interval. Zero modes contribute a constant $\log 2$ and infinite-energy modes may be omitted.

The entropy and variance summands in the main text are smooth even functions of e at zero, and e^2 is smooth in s at $s = 1/2$. At $s = 0$ the derivatives with respect to s are $O(s(1 + |\log s|^2))$. Thus the summands are Lipschitz in s on $[0, 1/2]$. For each function $f(e)$ used in these trace sums, the integral comparison on the branch $e_k = 4\pi^2|k|/L_i$ has error $O(1)$ because f' is integrable. The alternating exponential series gives $\int_0^\infty \log(1 + e^{-e})\,de = \pi^2/12$ and $\int_0^\infty e/(1 + e^e)\,de = \pi^2/12$. Integration by parts gives $\int_0^\infty e^2 e^e/(1 + e^e)^2\,de = \pi^2/6$.

The largest-probability summand, as a function of s, is $F_b(s) = \log 2 - \log(1 + \sqrt{1 - 4s^2})$. Let s_j^0 denote the mesh values in decreasing order. For mesh energy $e(s_j^0) \ge \epsilon$, where $0 < \epsilon \le 1$, the square-root difference identity gives $|F_b(s_j) - F_b(s_j^0)| \le C\epsilon^{-1}|s_j - s_j^0|$. For mesh energies below ϵ, the global bound is $C|s_j - s_j^0|^{1/2}$ and there are $O(1 + W\epsilon)$ entries. The ℓ^1 comparison and Cauchy–Schwarz give a trace-sum error of order $\epsilon^{-1} + (1 + W\epsilon)^{1/2}$. Taking $\epsilon = W^{-1/3}$ and evaluating the mesh integral therefore yields

$$b_N = \frac{\log N}{12} + O((\log N)^{1/3}).$$

Exact doubling gives the Ising counterpart $b_L = (\log L)/24 + O((\log L)^{1/3})$.

G Rank variance and information variance

Lemma 46 (Rank variance and information variance)**.** *Let $(p_j)_{j\ge1}$ be decreasing probabilities with finite support, and let T have density $p(t) = p_{\lceil t\rceil}$ on $(0,\infty)$. Put $X = \ln T$, $Y = -\ln p(T)$ and $S = \mathbb{E}Y$. Then*

$$|\operatorname{sd}(X) - \operatorname{sd}(Y)| \le 2\sqrt{2} + \ln 2 + 2\ln(e+S)$$

and, for every integer $d \ge 2$,

$$\mathcal{E}_d \ge \frac{(\ln d)^2}{4\operatorname{Var}(X) + (\ln d)^2}.$$

Proof. Monotonicity gives $tp(t) \le \int_0^t p(u)\,du \le 1$, so $X \le Y$. The function

$$q(t) = \frac{1}{(1+t)(1+\ln(1+t))^2}$$

is a probability density. For $Z = \ln(q(T)/p(T))$ we have $\Pr[Z > z] \le e^{-z}$ for $z \ge 0$, since $\mathbb{E}e^Z = \int_{\{p>0\}} q(t)\,dt \le 1$. Hence $\|Z_+\|_2 \le \sqrt{2}$. Also $\|X_-\|_2^2 = 2p_1 \le 2$. Now

$$0 \le Y - X = Z + \ln((1+T)/T) + 2\ln(1+\ln(1+T)) \le Z_+ + \ln 2 + X_- + 2\ln(e+Y).$$

The squared logarithm $y \mapsto \ln^2(e+y)$ is concave for $y \ge 0$. Jensen's inequality and the triangle inequality give the stated bound on $\|Y - X\|_2$. Subtracting means is an orthogonal projection in L^2, so the same bound holds for the difference of the standard deviations.

Write $a = \ln d$ and $f(x) = e^x p(e^x)$, the density of X. By aligning the Schmidt coefficients of the resource and target states,

$$\mathcal{E}_d = \int_0^\infty \left(\sqrt{p(t)} - d^{-1/2}\sqrt{p(t/d)}\right)^2 dt = \int_{\mathbb{R}} (\sqrt{f(x)} - \sqrt{f(x-a)})^2\,dx.$$

With $m = \mathbb{E}X + a/2$, the means of $f(x-a)$ and $f(x)$ differ by a. Factoring their difference and applying Cauchy–Schwarz gives

$$a \le \sqrt{\mathcal{E}_d}\left(\|(x-m)\sqrt{f(x-a)}\|_2 + \|(x-m)\sqrt{f(x)}\|_2\right) = 2\sqrt{\mathcal{E}_d}\sqrt{\operatorname{Var}(X) + a^2/4}.$$

□

For XX, $S_N, V_N \sim (\ln N)/6$ while the standard-deviation error in the lemma is $O(\ln\ln N)$. Hence the logarithmic-rank variance is $V_N(1+o(1))$, and the lemma gives $\liminf S_N\mathcal{E}_d \ge (\ln d)^2/4$ directly. The cumulative-counting argument in Appendix F supplies the matching upper limit.

H Proof of the near-critical law for the XXZ CTM spectrum

Proof of Theorem 24. Put $a = \pi^2/12$ and $\ell(t) = \log \mathcal{Z}(t)$. Writing η_{D} for the Dedekind eta function, the product in (38) gives $\mathcal{Z}(t) = e^{t/24}\eta_{\mathrm{D}}(it/\pi)/\eta_{\mathrm{D}}(it/(2\pi))$. The eta transformation $\eta_{\mathrm{D}}(i/y) = \sqrt{y}\,\eta_{\mathrm{D}}(iy)$ for $y > 0$ yields

$$\ell(t) = a/t - \tfrac{1}{2}\log 2 + t/24 - \ell(2\pi^2/t).$$

Writing $w = 2\pi^2/t$, the entropy $S = \ell(t) - t\ell'(t)$ equals $2a/t - \frac{1}{2}\log 2 - \ell(w) - w\ell'(w)$. The product gives $\ell^{(k)}(w) = O(e^{-w})$ for $0 \le k \le 2$, so the last two terms are $O(we^{-w})$. Differentiating twice also gives the information variance $V = t^2\ell''(t) = 2a/t + O(w^2e^{-w})$.

Apply Proposition 8 with $W = t^{-1}$, one-mode energies tj, and $L(z) = \ell(tz)$. Define the analytic logarithm by summing $\log(1+e^{-tjz})$ over $j \ge 1$, with each term given by its power series in e^{-tjz}. For $\operatorname{Re} z > 1/2$,

$$|L(z)| \le \sum_{k\ge1} \frac{1}{k(e^{kt/2}-1)} \le \frac{2}{t}\sum_{k\ge1}\frac{1}{k^2}.$$

Cauchy's estimate gives the required derivative bounds on a fixed neighborhood of $[3/4, 5/4]$. The tilted occupations are independent Bernoulli variables with probabilities $(1+e^{\beta tj})^{-1}$. For $M = \lfloor t^{-1}\rfloor$, modes

$j \le M$ have $tj \le 1$, so their Bernoulli variances are bounded below. Thus $v_\beta \ge c\sum_{j\le M}(tj)^2 \ge c'W$. Their characteristic factors also give

$$|\varphi_\beta(u)| \le \exp\left[-c\sum_{j=1}^{M}\sin^2(jtu/2)\right].$$

Put $\theta = tu$. For $|\theta| \le 4\pi/M$, the first $\lfloor M/4\rfloor$ terms give $\sum_{j\le M}\sin^2(j\theta/2) \ge cM^3\theta^2$. For $4\pi/M \le |\theta| \le \pi$, the geometric sum of $e^{ij\theta}$ has modulus at most $\pi/|\theta| \le M/4$, so the sine-square sum is at least $3M/8$. It follows that $|\varphi_\beta(u)| \le \exp[-c'W\min\{1, u^2\}]$ for $|u| \le \pi W$. This is (10), uniformly in β. The criterion therefore gives

$$\mathcal{E}_d = \frac{(\log d)^2}{4V} + O_d(t^2) = \frac{t(\log d)^2}{8a} + O_d(t^2).$$

Using $t = 2\eta$ and multiplying by S proves the three estimates.

In the symmetric CTM spectrum, the binary mode $j = 0$ contributes two copies of every weight $p_j/2$ [49, Sec. 4.1]. Thus its rank density is $p(x/2)/2$ and its logarithmic-rank square root is $T_{\log 2}g_p$. Since translations are unitary and commute, the error in (47) is identical. The entropy of the duplicated spectrum is $S + \log 2$. □

I Local deformations and the quartic impurity

I.1 Finite-depth and finite-time deformations

Proposition 47 (Preservation of the represented half-chain pair)**.** *Suppose α is an automorphism of $\mathfrak{A}$ of the form*

$$\alpha = \mathrm{Ad}(u)\circ(\beta_L \otimes \beta_R), \qquad u \in \mathcal{U}(\mathfrak{A}), \quad \beta_s \in \mathrm{Aut}(\mathfrak{A}_s).$$

The half-chain von Neumann algebras for $\omega\circ\alpha$ are simultaneously unitarily conjugate to M_L, M_R. In particular their types and the equality $M_L = M_R'$ are preserved. Under this identification the new state vector is $\pi(u)^*\Omega$. If α preserves each $\mathfrak{A}_s$ setwise, one may take $u = 1$, and all state-specific game values and attainment properties are preserved.

The displayed factorization holds for every circuit of finite depth whose layers consist of disjoint gates of uniformly bounded diameter. It also holds for a fixed-time dynamics generated by an interaction whose two restrictions to the halves generate dynamics and whose terms meeting both halves are absolutely summable in operator norm.

Proof. The GNS triple for $\omega\circ\alpha$ can be realized as $(\pi\circ\alpha, \mathcal{H}, \Omega)$ because α is onto. In this representation the two algebras are $\pi(u)M_s\pi(u)^*$. Applying $\pi(u)^*$ to the Hilbert space proves the first claims. If α preserves each half algebra, its restrictions determine α on the tensor product, with $u = 1$.

In a circuit layer the gates wholly in each half define the two half-chain automorphisms. All gates meeting both halves lie in a finite interval around the cut, so their product is one local unitary. Disjoint gates commute, giving the displayed form for each layer. Products retain the form by $\beta\circ\mathrm{Ad}(v) = \mathrm{Ad}(\beta(v))\circ\beta$. The crossing unitary for the product is local because a finite-depth circuit maps each local observable into another finite interval. For the dynamics, remove the boundary sum v to get the product dynamics $\gamma_t = \gamma_{L,t}\otimes\gamma_{R,t}$. The full dynamics is a bounded perturbation of γ_t by $v = v^* \in \mathfrak{A}$. Its Dyson cocycle $u_t \in \mathcal{U}(\mathfrak{A})$ gives $\alpha_t = \mathrm{Ad}(u_t)\circ\gamma_t$. □

The factorization is related to the local automorphism factorizations of Refs. [70, 71]. The explicit bulk applications are proved in Theorem 27.

I.2 The quartic impurity

Proof of Theorem 28. Uniform finite-volume estimate. On $I_m = [-m+1, m]$ for $m \ge 2$, use open boundaries and write h_m for the one-particle compression. Its eigenvalues $e_k = 2\cos(k\pi/(2m+1))$ are nonzero. In the finite Jordan–Wigner representation the unperturbed Hamiltonian minus its lowest energy is $\sum_k |e_k| b_k^* b_k$, where b_k annihilates a particle if $e_k > 0$ and a hole if $e_k < 0$. The free ground vector Ω_m is the common vacuum of the b_k. Denote by $K_{\lambda,m}$ this vacuum-normalized Hamiltonian plus λV, by E_m its least eigenvalue, and by Ψ_m a normalized ground vector. The local even identification of V with the same spin operator V_{s} is independent of m.

The pull-through identity gives

$$(K_{\lambda,m} - E_m + |e_k|)b_k\Psi_m = -\lambda[b_k, V]\Psi_m.$$

Since the operator on the left is bounded below by $|e_k|$, this implies $\|b_k\Psi_m\| \le |\lambda|\|[b_k, V]\|/|e_k|$. For normalized eigenvectors u_k of h_m, commuting b_k across the four factors of V by the anticommutation relations gives the bound

$$\|[b_k, V]\| \le 2\big(|\langle u_k, f\rangle| + |\langle u_k, g\rangle|\big).$$

Here each remaining factor is a contraction. On these volumes $f = h_m\delta_0/\sqrt{2}$ and $g = h_m\delta_1/\sqrt{2}$, so Parseval yields

$$\sum_k \|b_k\Psi_m\|^2 \le 8\lambda^2 \sum_k \frac{|\langle u_k, f\rangle|^2 + |\langle u_k, g\rangle|^2}{|e_k|^2} = 8\lambda^2.$$

The complement of the vacuum projection is bounded by the excitation-number operator. Consequently $|\langle\Omega_m, \Psi_m\rangle|^2 \ge 1 - 8\lambda^2$. The estimate holds for every ground vector and also implies finite-volume nondegeneracy, since an eigenspace of dimension at least two would contain a vector orthogonal to Ω_m.

A normal ground component. Let ω_0 be the infinite XX ground state and ψ any weak-$*$ limit of the finite perturbed ground states, considered on local spin observables. The limit is a ground state for the perturbed dynamics. Indeed the finite-volume ground-state inequalities pass to the limit on the local algebra, a generator core for a finite-range spin interaction [40]. The free finite-volume states converge locally to ω_0: the compressions h_m, extended by zero, converge strongly to h, and the negative spectral projections converge strongly because h has no spectral atom at zero. Wick's rule then gives convergence on the even spin local algebra, while both states vanish on odd observables. The finite-volume overlap bound therefore gives

$$\|\psi - \omega_0\| \le 2\sqrt{8}\,|\lambda| < 2,$$

first on local observables and then by norm density.

Let $z \in \mathfrak{A}^{**}$ be the central support of the XX GNS representation $(\pi_0, \mathcal{H}_0, \Omega_0)$. Since ω_0 is pure, the corner $z\mathfrak{A}^{**}$ is normally isomorphic to $B(\mathcal{H}_0)$. The normal extensions of the states satisfy $\psi(z) > 0$: if it were zero, evaluation on $2z - 1$ would give $\|\psi - \omega_0\| = 2$. The projection z is invariant under the perturbed dynamics. To see this, write the unperturbed dynamics in the XX representation as $\mathrm{Ad}(e^{itK_0})$. Bounded perturbation by λV_{s} is implemented there by $K = K_0 + \lambda\pi_0(V_{\mathrm{s}})$, using the same Dyson cocycle on $\mathfrak{A}$. The representation after each perturbed automorphism is therefore unitarily equivalent to π_0 and has the same central support.

The normalized component $a \mapsto \psi(za)/\psi(z)$ is a normal ground state in $B(\mathcal{H}_0)$ for $\mathrm{Ad}(e^{itK})$. Here is the positive-energy argument for this step. Extend the GNS representation of ψ normally to $\mathfrak{A}^{**}$ and let p be the image of z. The ground-state implementing unitaries fix its cyclic vector and commute with p by invariance of z. They thus have nonnegative generator on $p\mathcal{H}_\psi$, with invariant normalized vector $p\Omega_\psi/\sqrt{\psi(z)}$. This is the GNS space of the component, and the normal representation there factors through $z\mathfrak{A}^{**}$. It is also the GNS space of the corresponding normal state on $B(\mathcal{H}_0)$, because $\pi_0(\mathfrak{A})$ is strongly dense and the representation is normal.

Write this normal state as a trace-class density D. Its invariance implies D commutes with e^{itK}. Each positive-eigenvalue eigenspace of D is finite-dimensional and invariant under e^{itK}. It therefore has a basis of eigenvectors of K. If u is one such unit vector with $Ku = eu$ and $Du = du$, $d > 0$, then in the Hilbert–Schmidt model of the component GNS space the subspace spanned by $|v\rangle\langle u|\sqrt{D}$ for $v \in \mathcal{H}_0$ carries the implementing group $e^{it(K-e)}$. Its generator must be nonnegative, so $e = \inf\mathrm{spec}(K)$. Thus K has a ground eigenvector Ψ. Every such vector is cyclic for the irreducible spin representation π_0, so its spin state is pure and has the original half-chain pair.

The interaction changes the state and leaves gapless excitations. The operator V is a two-body term in the original fermions, with a nonzero two-particle matrix element and zero action on the bare zero- and one-particle spaces. It also acts nontrivially on the free Fermi sea. If P is the negative spectral projection of h, then $(1-P)f, (1-P)g$ are linearly independent, as are Pf, Pg, since in Fourier space the two form factors are proportional to $\cos k$ and $e^{ik}\cos k$ on each Fermi arc. The two-particle, two-hole component of $\pi_0(V_{\mathrm{s}})\Omega_0$ is the wedge product of these two pairs (conjugating the hole pair). It is nonzero. This computation takes place in the common even cyclic subspace of the spin and Fock representations under the even Jordan–Wigner identification. The even cyclic subspaces are identified by GNS uniqueness because the states agree on the even algebra [72, Sec. 6.2]. It follows that Ω_0 cannot be a ground eigenvector of K for $\lambda \neq 0$. The purity and Haag duality assertions follow from π_0 and the XX theorem [11].

For gaplessness, let τ_x be spin translation. For any fixed local observable a, $\pi_0(\tau_x(a))$ converges weakly to $\omega_0(a)1$ as $x \to \infty$. All its weak cluster points commute with the full spin algebra by locality, and purity and translation invariance identify the scalar. Let $\mathcal{L}_0$ denote the unperturbed local commutator derivation without the factor i. For any $e \in (0,2)$ and any $\eta > 0$, choose a local even b with $\|\pi_0(b)\Omega_0\| \geq 1/2$ and $\|\pi_0(\mathcal{L}_0 b - eb)\Omega_0\| < \eta$. Such b comes from $a^*(f_+)a(g_-)$, with normalized wave packets f_+, g_- supported in spectral bands of h near $e/2, -e/2$, followed by finite spatial truncation. The free energy of this particle–hole vector is within the chosen band width of e. Truncation controls the observable and the commutator in norm because h is bounded. Translate b away from the impurity, so $[V_s, \tau_x(b)] = 0$. Local observables are differentiable for the quasi-local dynamics, so their representatives preserve the domain of K and satisfy the local commutator identity on it. Writing $E = \inf \operatorname{spec}(K)$, one has

$$(K - E - e)\pi_0(\tau_x(b))\Psi = \pi_0\big(\tau_x(\mathcal{L}_0 b - eb)\big)\Psi.$$

The squared norms of the translated vectors on both sides of this estimate converge to the corresponding squared norms at the free vacuum, by the weak limit for translates applied to the appropriate local products. Taking $\eta \to 0$ and normalizing gives approximate eigenvectors of $K - E$ at e. Hence all $e \in (0,2)$ belong to its spectrum. The game claim follows from the half-chain pair and the proof of Theorem 4. □

Proof of Corollary 29. Let Q be the ground spectral projection of K and let p_0 be the rank-one projection onto Ω_0, viewed in the central corner $z\mathfrak{A}^{**} \cong B(\mathcal{H}_0)$. The norm bound in the theorem, applied to $2p_0 - 1$, gives $\psi(p_0) \geq 1 - \sqrt{8}|\lambda|$. The density of the normal component is supported on Q, so $\psi(p_0) \leq \langle \Omega_0, Q\Omega_0 \rangle$. Hence $\Psi_\lambda = Q\Omega_0/\|Q\Omega_0\|$ is well defined and converges in norm to Ω_0 as $\lambda \to 0$. The gauge rotations fix Ω_0 and commute with K_0 and the density interaction, so they fix Ψ_λ. In particular Ψ_λ is in the even spin subspace. In the corresponding even Fock subspace its vector state extends to the full CAR algebra. The extension is even and gauge invariant and is pure, because every nonzero vector is cyclic in the irreducible free Fock representation.

Suppose this fermionic state were quasifree. Its covariance P' is then a projection by purity. On $\operatorname{span}\{f, g\}$ the free covariance has diagonal entries $1/2$ and off-diagonal entry of modulus $4/(3\pi)$, as follows by integrating $2\cos^2 k$ and $2e^{ik}\cos^2 k$ on the occupied arc with measure $dk/(2\pi)$. For a normalized linear combination r of f, g, the expectation of $a^*(r)a(r)$ changes by at most $2\sqrt{\sqrt{8}|\lambda|}$. This is the vector-state norm bound from the squared overlap with the free vacuum. The stated range makes that change smaller than $1/2 - 4/(3\pi)$. Both P' and $1 - P'$ are therefore strictly positive on $\operatorname{span}\{f, g\}$, so the two particle and hole form-factor pairs remain linearly independent.

Use the particle–hole Fock realization of this hypothetical pure quasifree state and denote its vacuum by Φ. The operator

$$B = a^*((1 - P')f)a^*((1 - P')g)a(P'f)a(P'g)$$

is a product of four excitation creators, and $B\Phi \neq 0$. The four-creation component of $V\Phi$ is $\pm B\Phi$. Therefore the vacuum expectation of $[V, B^*]$ is $\mp\|B\Phi\|^2 \neq 0$. The vacuum expectation of $\mathcal{L}_0(B^*)$ is zero because the quadratic derivation changes only one annihilation factor at a time to a linear combination of excitation creators and annihilators, while the other three factors still annihilate excitations. Every such term changes excitation number and has zero vacuum expectation. The even CAR observable B^* is in the domain of the spin commutator derivation after Jordan–Wigner. Finite spatial truncations of its smearing vectors approximate both B^* and its unperturbed commutator in norm since h is bounded, and the impurity commutator is bounded. Stationarity of the ground state requires the expectation of $\mathcal{L}_0(B^*) + \lambda[V, B^*]$ to vanish, a contradiction. □

Proof of Corollary 30. Use a common ambient Fock space over $\mathfrak{k}_* = \ell^2(\mathbb{Z}) \oplus \overline{\ell^2(\mathbb{Z})}$. Let $P_m^{\pm}$ be the projections onto the positive and negative spectral subspaces of the open h_m, extended by zero outside I_m. The finite quasiparticle space is $\mathfrak{k}_m = P_m^+\ell^2 \oplus \overline{P_m^-\ell^2}$. For the spin finite-volume ground vectors Ψ_m use the finite Jordan–Wigner identification with their quasiparticle Fock vectors and embed these into the ambient Fock space. The free finite vacua all map to the same ambient vacuum. The infinite free particle–hole space is $\mathfrak{k} = (1 - P)\ell^2 \oplus \overline{P\ell^2}$. On fixed local even spin observables the finite Fock realizations converge in operator norm to the infinite one when viewed as polynomials in ambient annihilators and creators, since $P_m^{\pm}$ converge strongly. The finite vectors are even in excitation parity. The Hamiltonian commutes with parity, its ground vector is unique by the overlap bound, and the overlap with the even free vacuum is nonzero.

We show precompactness of these vectors. Write v_k for the excitation one-particle basis, $\epsilon_k = |e_k|$, and $T_m = K_{\lambda,m} - E_m \geq 0$. The CAR commutator used above has the form

$$[b_k, V] = \sum_{\ell=1}^{4} \langle v_k, w_{m,\ell} \rangle A_{m,\ell}, \qquad \|A_{m,\ell}\| \leq 1,$$

where the $w_{m,\ell}$ are the particle and hole components of f, g in $\mathfrak{k}_m$. Each $A_{m,\ell}$ is the remaining signed product of three operators and is independent of k. Thus the vector of annihilations in $\mathfrak{k}_m \otimes \Gamma(\mathfrak{k}_m)$ is

$$\sum_k v_k \otimes b_k \Psi_m = -\lambda \sum_{k,\ell} v_k \langle v_k, w_{m,\ell} \rangle \otimes (T_m + \epsilon_k)^{-1} A_{m,\ell} \Psi_m.$$

For $\epsilon_k < \delta$ its squared norm is bounded by $8\lambda^2$ times

$$\frac{1}{2} \sum_{j=0}^{1} \big\langle \delta_j, \mathbf{1}_{(-\delta,\delta)}(h_m) \delta_j \big\rangle.$$

This is the same estimate as for the total excitation number, restricted to these energies. The displayed bound tends to zero as $\delta \downarrow 0$ after taking $\limsup_m$, by strong convergence of these spectral projections and absence of a free zero-energy atom.

For $\epsilon_k \geq \delta$, partition $[\delta, 2]$ into finitely many energy intervals J of diameter at most η, and replace the resolvent by $(T_m + c_J)^{-1}$ with $c_J \in J$. Its norm error is at most η/δ^2 by the resolvent identity. The resulting vector lies in $F_m \otimes \Gamma(\mathfrak{k}_m)$, where F_m is the span of all energy-band projections of the four $w_{m,\ell}$. Its error in norm on the high-energy part is at most $4|\lambda|\eta/\delta^2$, using the orthogonality of the v_k and $\|w_{m,\ell}\| \leq 1$. With the low-energy bound this gives arbitrarily small $\limsup_m$ distance to $F_m \otimes \Gamma(\mathfrak{k}_m)$.

In Fock space the expectation of the number operator outside F_m is the squared norm of the part of the annihilation vector orthogonal to F_m in the first tensor factor. That number operator bounds the complement of the Fock projection onto $\Gamma(F_m)$. Hence Ψ_m has arbitrarily small $\limsup_m$ distance to $\Gamma(F_m)$. The finitely many vectors spanning F_m converge in the ambient one-particle space to the corresponding infinite-volume band projections. They are linearly independent in the limit: different bands and particle and hole sectors are orthogonal, and within a band the two Fourier form factors $\cos k$ and $e^{ik} \cos k$ are independent on a set of positive measure. The partition can be chosen with each interval of positive length, and the free dispersion has no atoms at the endpoints. Consequently their span projections converge in norm to a finite-rank projection onto $F \subset \mathfrak{k}$. The projections onto the finite Fock spaces $\Gamma(F_m)$ also converge in norm to that onto $\Gamma(F)$, as can be seen by choosing convergent orthonormal bases and taking wedge products. For any tolerance, the vectors thus lie within that tolerance of a fixed finite-dimensional subspace for all sufficiently large m. This proves norm precompactness, with every limit vector a unit even vector in $\Gamma(\mathfrak{k})$.

Each such limit determines the spin limit state using the common even GNS subspace and vanishing odd expectations. It is a pure vector state in the XX representation and a ground state by the local ground-state inequalities. If the finite-chain game value did not tend to one, a subsequence with a uniform positive loss floor would have a further subsequence converging locally to one of these limit states. That state has value one by its half-chain pair. By local POVM approximation as in Lemma 21, it admits a finite-support strategy with loss smaller than the putative floor. Its loss in the convergent finite states tends to that same value, a contradiction. The same argument with local embezzling unitaries, as in the finite-chain proof of Theorem 4, proves convergence of the embezzlement error. □

J Transferring bounded continuum strategies

Let $\mathfrak{A}$, $\mathfrak{A}_L$, $\mathfrak{A}_R$ denote the quasi-local spin algebra and its two halves, with pure state ω and GNS triple $(\pi, \mathcal{H}, \Omega)$. Write $M_s = \pi(\mathfrak{A}_s)''$, $s = L, R$.

Call a bounded sequence (x_n) of elements of $\mathfrak{A}_s$ supported asymptotically far from the cut if, for each finite $F \subset \mathbb{Z}$,

$$\operatorname{dist}\big(x_n, \mathfrak{A}_{s \setminus F}\big) \longrightarrow 0.$$

For matrices the same condition is taken in matrix norm. This condition holds for uniformly bounded observables supported in scaled intervals $[an, bn]$ or $[-bn, -an]$ with fixed $0 < a < b < \infty$.

Proposition 48 (Transfer within the lattice GNS representation)**.** *Suppose that for each d_0 and each unit vector $\zeta \in \mathbb{C}^{d_0} \otimes \mathbb{C}^{d_0}$ there are unitary sequences $U_{s,n} \in \mathfrak{A}_s \otimes M_{d_0}$ supported asymptotically far from the cut and satisfying*

$$\left\| U_{L,n} U_{R,n} (\Omega \otimes |00\rangle) - \Omega \otimes \zeta \right\| \longrightarrow 0.$$

Here the two unitaries use separate ancillas and are represented in the original GNS space. The same sequences work for every GNS vector. Both half-chain algebras have type III_1 *if ω is pure. For the physical pair M_L, M_R, the value at every GNS vector is at least the supremal finite-dimensional tensor-product value of any finite game.*

Proof. Use the isometries $V_n\xi = U_{L,n}U_{R,n}(\xi \otimes |00\rangle)$ and $W\xi = \xi \otimes \zeta$ and the positive errors $D_n = (V_n - W)^*(V_n - W)$ on $\mathcal{H}$. They are uniformly bounded by 4, asymptotically commute in norm with each $\pi(a)$ for local a, and satisfy $\langle \Omega, D_n \Omega\rangle \to 0$. Thus $\|D_n\Omega\|^2 \le 4\langle \Omega, D_n\Omega\rangle \to 0$. Commuting D_n through $\pi(a)$ and using cyclicity gives $D_n \to 0$ strongly. This proves the vector claim and $\widetilde{\varphi}(D_n) \to 0$ for every normal state $\widetilde{\varphi}$ on $B(\mathcal{H})$.

Let φ be a normal state on M_R and choose a normal state extension $\widetilde{\varphi}$ to $B(\mathcal{H})$, equivalently a trace-class density operator representing φ in this representation. For every contraction $x \in M_R \overline{\otimes} M_{d_0}$, acting trivially on the left ancilla, x commutes with $U_{L,n}$. The expectations of $V_n^* x V_n$ and $W^* x W$ in $\widetilde{\varphi}$ are therefore those of x in $\mathrm{Ad}(U_{R,n})_*(\varphi \otimes |0\rangle\langle 0|)$ and $\varphi \otimes \rho_R(\zeta)$, respectively. Their difference in norm is bounded by $2\sqrt{\widetilde{\varphi}(D_n)}$, by Cauchy–Schwarz applied to $V_n - W$ and the two isometries. Every ancilla density operator is a marginal of a pure ζ. Composing two such transformations gives approximate transitions between arbitrary ancilla states. Thus every normal state of M_R is monopartite embezzling in the sense of [13, Theorem E]. The same argument applies to M_L. Purity implies each is a factor since the halves generate $B(\mathcal{H})$ and commute. The classification in [13, Theorem E] then gives type III_1.

For the game claim choose any finite-dimensional tensor-product strategy. Embezzle its state into the two ancillas, then perform the strategy there. Conjugating the measurements by the corresponding $U_{s,n}$ and compressing to the initial ancilla vectors gives POVMs in the physical halves. Their success converges to that of the target strategy. Taking its supremum proves the claim, and gives value one for the game G. □

Theorem 49 (A bounded-observable scaling hypothesis)**.** *Let $\mathcal{D}_s$ be a unital C^*-algebra of uniformly bounded families in $\prod_n \mathfrak{A}_s$ supported asymptotically far from the cut as above. Let $\widehat{\omega}((x_n)) = \lim_{n\to\mathcal{U}} \omega(x_n)$ on $C^*(\mathcal{D}_L, \mathcal{D}_R)$ for a free ultrafilter $\mathcal{U}$. Write $(\widehat{\pi}, \widehat{\mathcal{H}}, \widehat{\Omega})$ for its GNS triple. Suppose the pair $\widehat{\pi}(\mathcal{D}_L)'', \widehat{\pi}(\mathcal{D}_R)''$ contains, on this vacuum vector, a pair of continuum algebras whose state is bipartite embezzling. Then the conclusions of Proposition 48 hold. In particular the hypothesis holds if these closures contain the two complementary wedge algebras of a local net in a separable vacuum representation with wedge duality and their joint state at $\widehat{\Omega}$ agrees with the vacuum state [13, Theorem 97].*

Proof. Fix an ancilla dimension, a target vector, and an error tolerance. The assumed continuum strategy supplies a unitary in each side's von Neumann algebra amplified by the local ancilla. Each such unitary is e^{iH} for a bounded self-adjoint H in that algebra. Kaplansky density approximates H strongly by bounded self-adjoints in the represented C^*-algebra. Exponentiation preserves strong convergence on bounded sets, and hence the two unitaries can be approximated on the vectors relevant to the error norm. Lift those self-adjoints to self-adjoint elements of $M_{d_0}(\mathcal{D}_s)$ and exponentiate there. At each coordinate one has genuine lattice unitaries. Their squared errors at Ω have ultralimit equal to the squared error of the represented approximants at $\widehat{\Omega}$, as follows by expanding the squared norm in the finite ancilla matrix entries. Select a coordinate far enough out to make the desired error small and to make the chosen unitaries close in norm to the halves outside an arbitrarily large interval about the cut. Taking the tolerance to zero and the interval to infinity yields the sequences in Proposition 48. □

Theorem 49 requires bounded representatives preserving joint vacuum expectations and localization on both sides. For a continuum net generated by bounded observables in regions strictly inside each wedge, the corresponding lattice representatives must be approximable in norm outside every fixed cut neighbourhood. A continuum factor type, finite-size energy levels, and a formula for Rényi entropies do not supply those representatives. The theorem proves type and game value under this hypothesis, with no assumption or conclusion about microscopic Haag duality or attainment.

K Numerical methods and error control

The source archive includes a portable Python implementation and the data used in every figure. The critical XX model is the open hopping chain at half filling. The critical Ising model is $-\sum_j X_j X_{j+1} - \sum_j Z_j$. The gapped comparison adds the staggered potential $\delta \sum_j (-1)^j a_j^* a_j$ to XX, with $\xi = 1/\operatorname{arsinh}(\delta/2)$. All numerical chain cuts are central.

For a free-fermion reduced state, the Schmidt probabilities are products of the one-mode probabilities $\zeta_k, 1-\zeta_k$, obtained from the restricted covariance [53]. Dense diagonalization checks the small sizes. At large sizes, sine transforms implement the covariance without storing a dense matrix, and the eigensolver retains the modes furthest from zero and one. The critical-chain sweep retains up to 60 Ising modes and uses two copies of them for XX according to the exact correspondence [50]. Direct XX calculations check the doubled spectra. The code enumerates and sorts the Schmidt probabilities formed from the retained modes that exceed 10^{-11} times the largest one (a run that would keep more than 12,000,000 of them stops with an error), and records the retained mass, $\mathcal{E}$, and the explicit strategy losses with and without Φ_9. The largest sizes in this sweep are 131,072 Ising spins and 262,144 XX sites. The gapped curves retain the original data, with selected values independently recomputed; they are estimates of the long-chain limit.

If a sorted Schmidt tail of probability τ is discarded and the prefix renormalized to $\widehat{p}$, then

$$\|\sqrt{p} - \sqrt{\widehat{p}}\|_2^2 = 2 - 2\sqrt{1-\tau}, \qquad \left|\sqrt{\mathcal{E}_d(p)} - \sqrt{\mathcal{E}_d(\widehat{p})}\right| \le 2\sqrt{2 - 2\sqrt{1-\tau}}. \tag{73}$$

The code propagates this bound and estimates the mass of omitted one-body modes by bounding their minority occupation probabilities by the smallest retained one, conditional on retention of the leading exact modes. Dense, matrix-free, doubled-mode and archived-data comparisons assess floating-point eigensolver error. For the game score itself the block-diagonal fallback of Corollary 15 gives the sharper linear tail cost.

Tower probabilities are counted with exact integer degeneracies from $\prod_{\ell \ge 0}(1+q^{2\ell+1})^r$. This avoids expanding exponentially large lists at small ε. The smooth spectrum uses scaled Bessel functions and stable inverse-rank quadrature. Tables interpolate the computed tower and smooth curves at the physical value $b = -\ln p_1$. The code also evaluates finite tensor powers through exact binomial multiplicities and checks the finite sine-profile witness by an independent compressed-rank calculation.

Two implementations verify the 19×19 matrix inequality using exact arithmetic. Direct construction of all POVMs and evaluation of all question pairs checks (28) independently of its quadratic-form implementation. The archive's README gives the commands for the checks and for regenerating each figure.

For the XXZ data in Figure 4, the coefficients of $\prod_{j \ge 1}(1+q^j)$ are computed as exact integers using Euler's pentagonal recurrence. The squared error is summed over intersections of exact integer rank intervals, without expanding the Schmidt list. An exponential-moment estimate bounds the omitted tail. The partition function and entropy are evaluated through their modular transformation. The four-spin tilted CHSH strategy is checked independently with exact algebraic arithmetic on the full 16×16 spin Hamiltonian, its ground vector, the local basis changes and the physical Bell observables.

The assisted certificates extend the XX calculation to $N = 2^{19}$ and 2^{20} by compressing the exact covariance to dimensions 320 and 344. Directed interval arithmetic, exact residual bounds and integer rank counts give losses below 0.000703 and 0.000663 on $\psi_N \otimes \Phi_9$. An independent implementation uses rational mode ratios and a different retained prefix to check both separations. Table 6 gives the certified intervals, including the control at $N = 2^{18}$. These certificates account for arithmetic and spectral errors as well as the discarded Schmidt mass.

The chain-only certificate for the original game at $N = 2^{84}$ uses a separate calculation with rigorous arithmetic and spectral error bounds. A dyadic Gaussian quadrature compresses one parity block of the cross covariance to dimension 1418. Residual intervals and a Frobenius-norm tail estimate control its singular values. Exact integer multiplicities and directed arithmetic bound the model strategy loss below 0.000410720762. Transferring the strategy to the physical ground state gives the certified loss bound 0.000461. The complete parameters and error propagation appear in Appendix C.

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